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Heldman et al.

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(54) **MOVEMENT DISORDER THERAPY SYSTEM, DEVICES AND METHODS, AND INTELLIGENT METHODS OF TUNING**

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Oct. 14, 2019, now Pat. No. 11,383,087, which is a
(Continued)

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A61B 5/00 (2006.01)
A61B 5/11 (2006.01)
A61N 1/05 (2006.01)
A61N 1/372 (2006.01)

(52) **U.S. Cl.**
CPC **A61N 1/36135** (2013.01); **A61B 5/11**
(2013.01); **A61B 5/4836** (2013.01); **A61N**
1/36067 (2013.01); **A61N 1/36082** (2013.01);
A61N 1/37247 (2013.01); **A61N 1/0534**
(2013.01)

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1/36082; **A61N 1/36135**; **A61N 1/37247**;
A61B 5/11; **A61B 5/4836**

See application file for complete search history.

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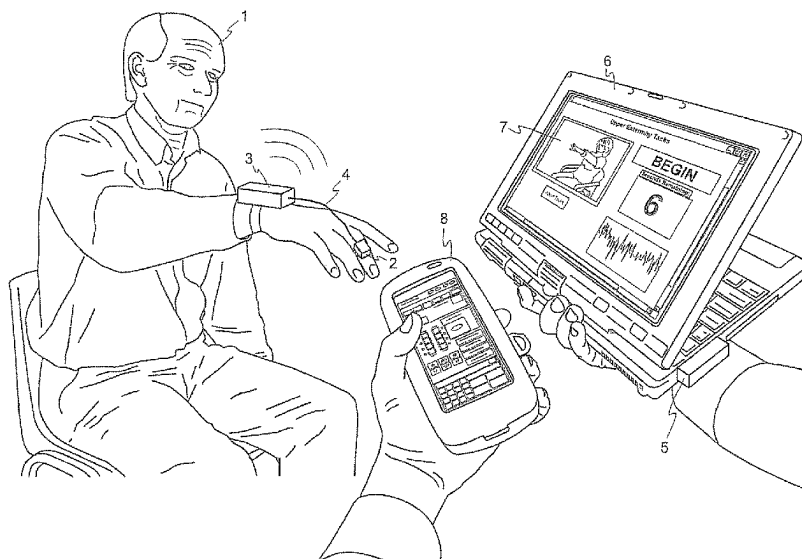
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(57) **ABSTRACT**

The present invention relates to methods for tuning treatment parameters in movement disorder therapy systems. The present invention further relates to a system for screening patients to determine viability as candidates for certain therapy modalities, such as deep brain stimulation (DBS). The present invention still further provides methods of quantifying movement disorders for the treatment of patients who exhibit symptoms of such movement disorders including, but not limited to, Parkinson's disease and Parkinsonism, Dystonia, Chorea, and Huntington's disease, Ataxia, Tremor and Essential Tremor, Tourette syndrome, stroke, and the like. The present invention yet further relates to methods of tuning a therapy device using objective quantified movement disorder symptom data acquired by a movement disorder diagnostic device to determine the therapy setting or parameters to be provided to the subject via his or her therapy device. The present invention also provides treatment and tuning remotely, allowing for home monitoring of subjects.

20 Claims, 15 Drawing Sheets



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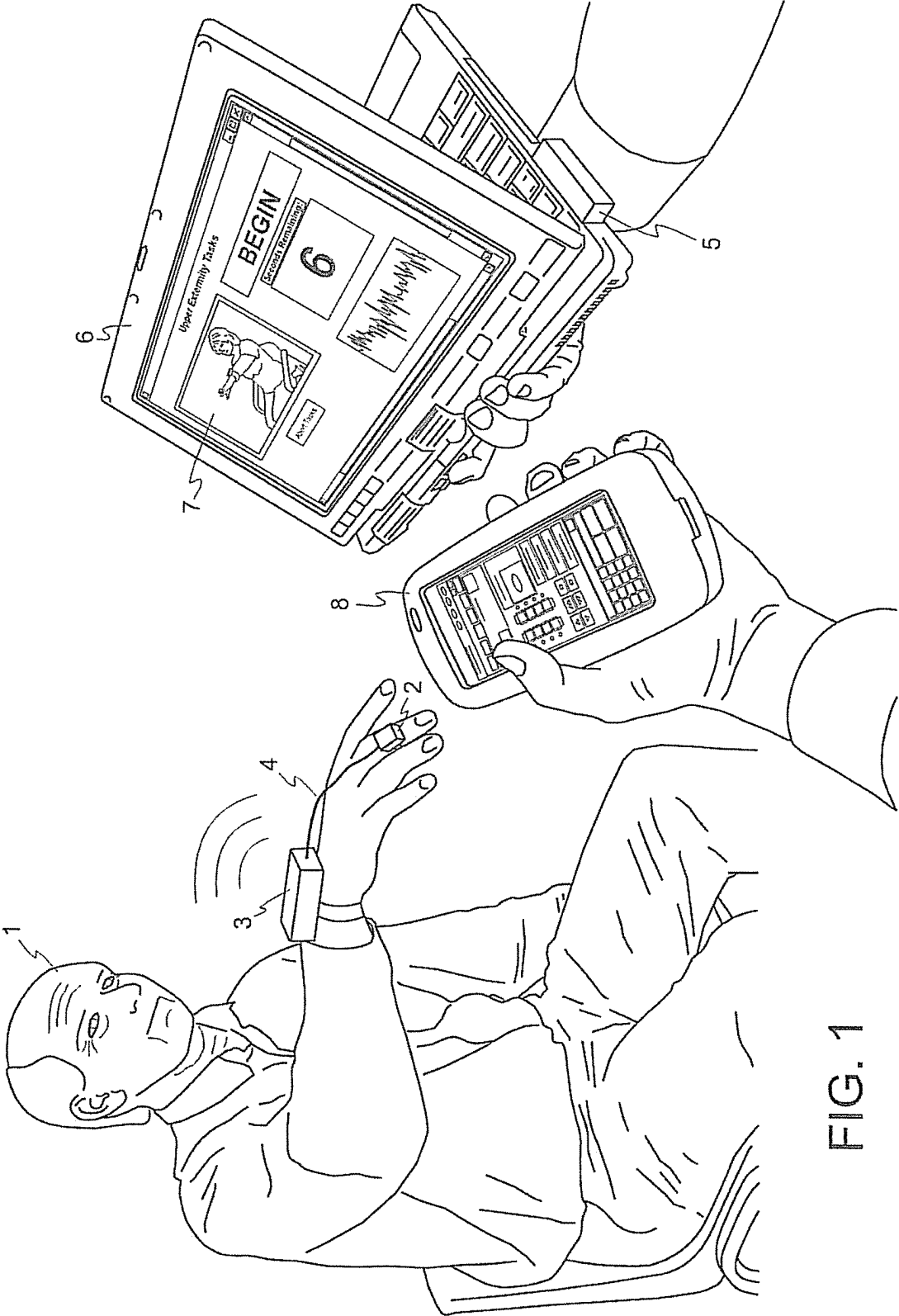


FIG. 1

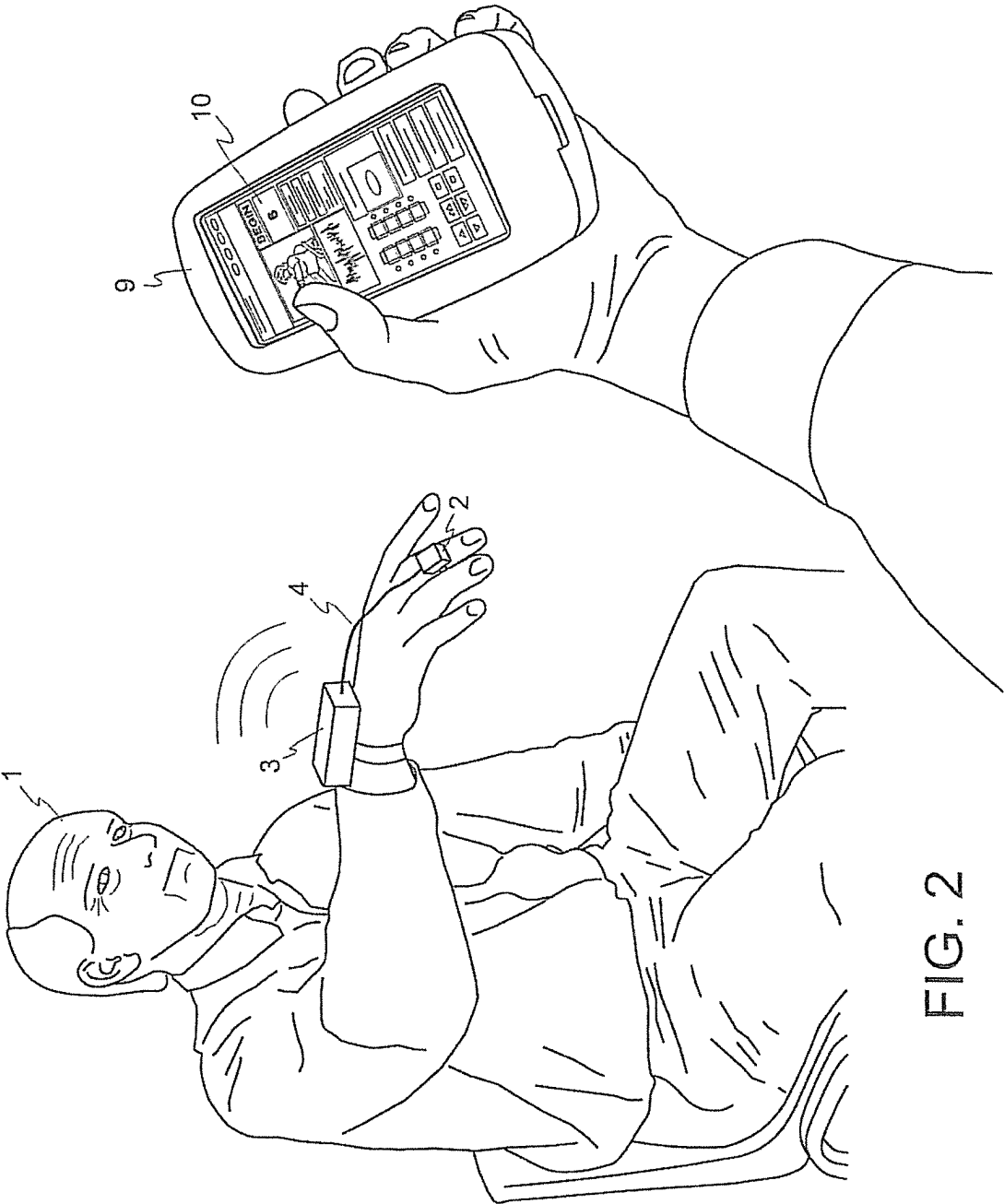


FIG. 2

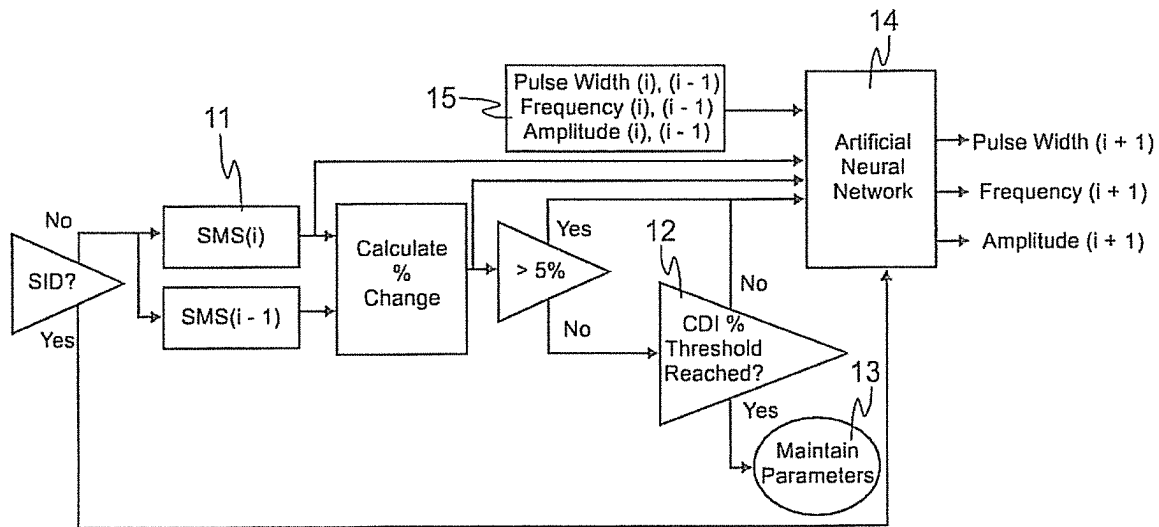


FIG. 3

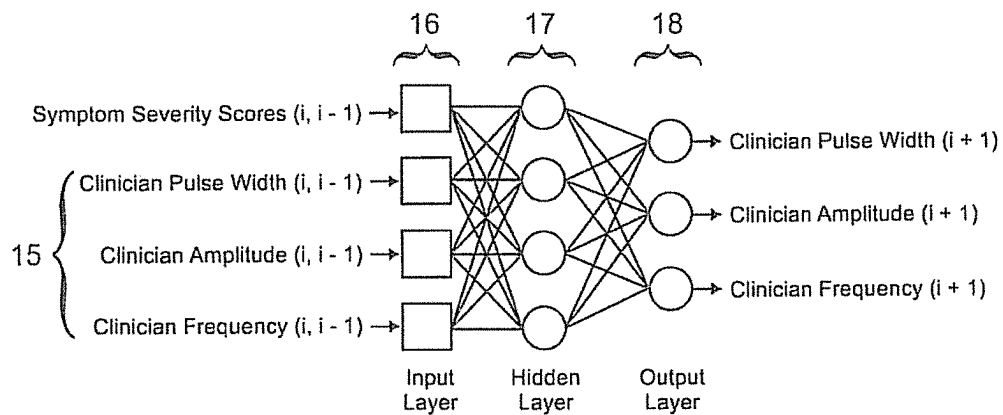


FIG. 4

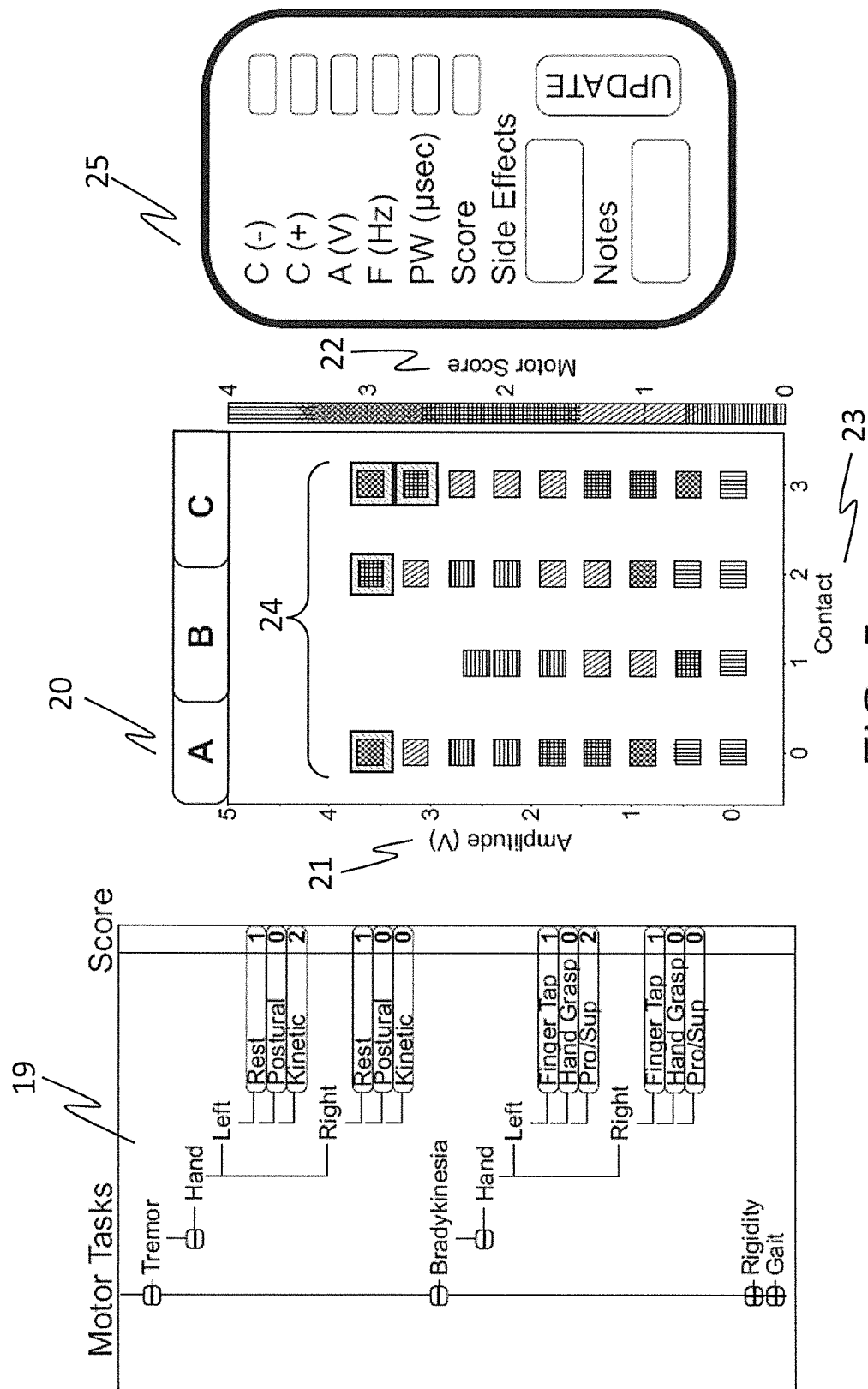
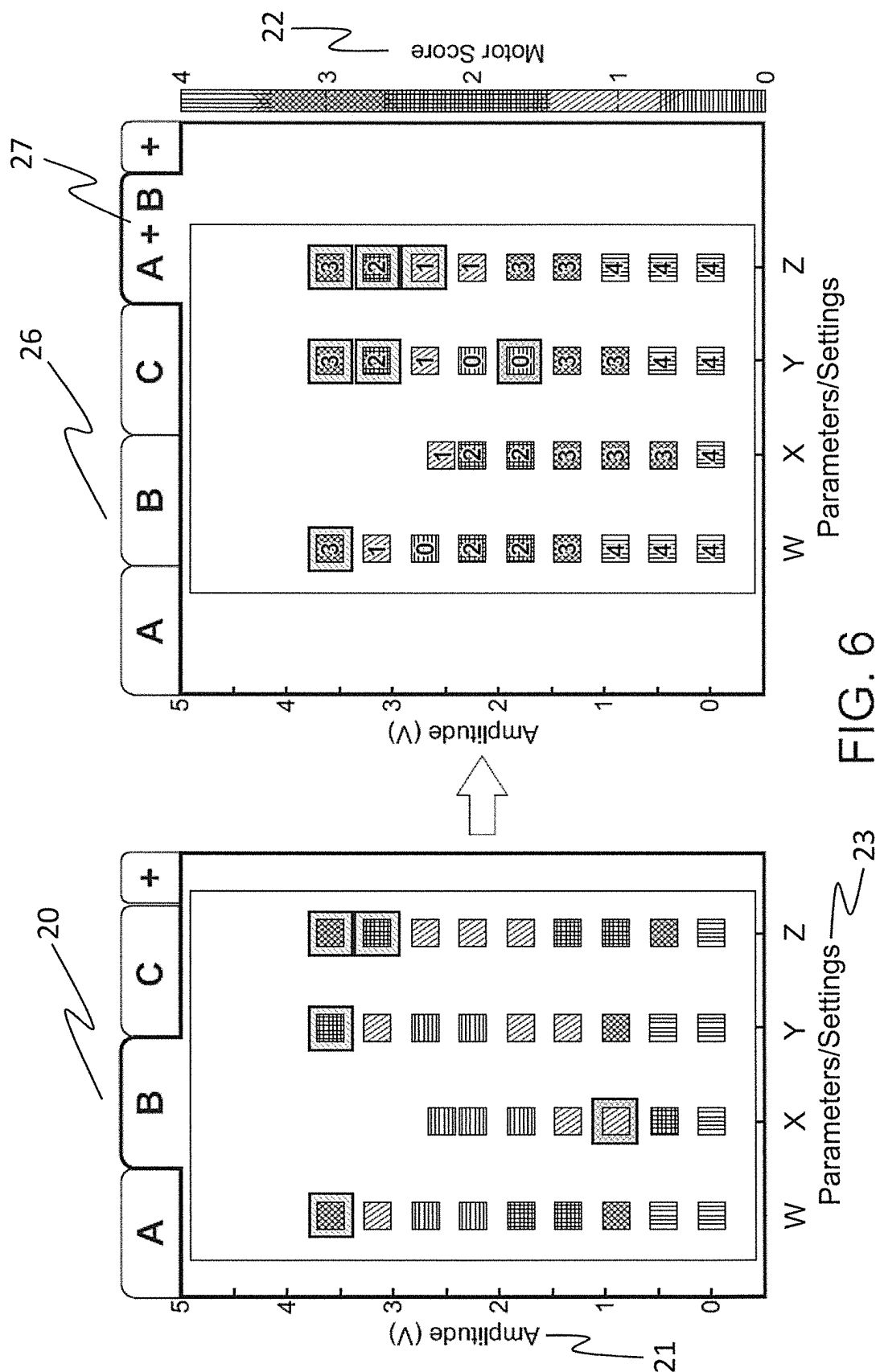


FIG. 5



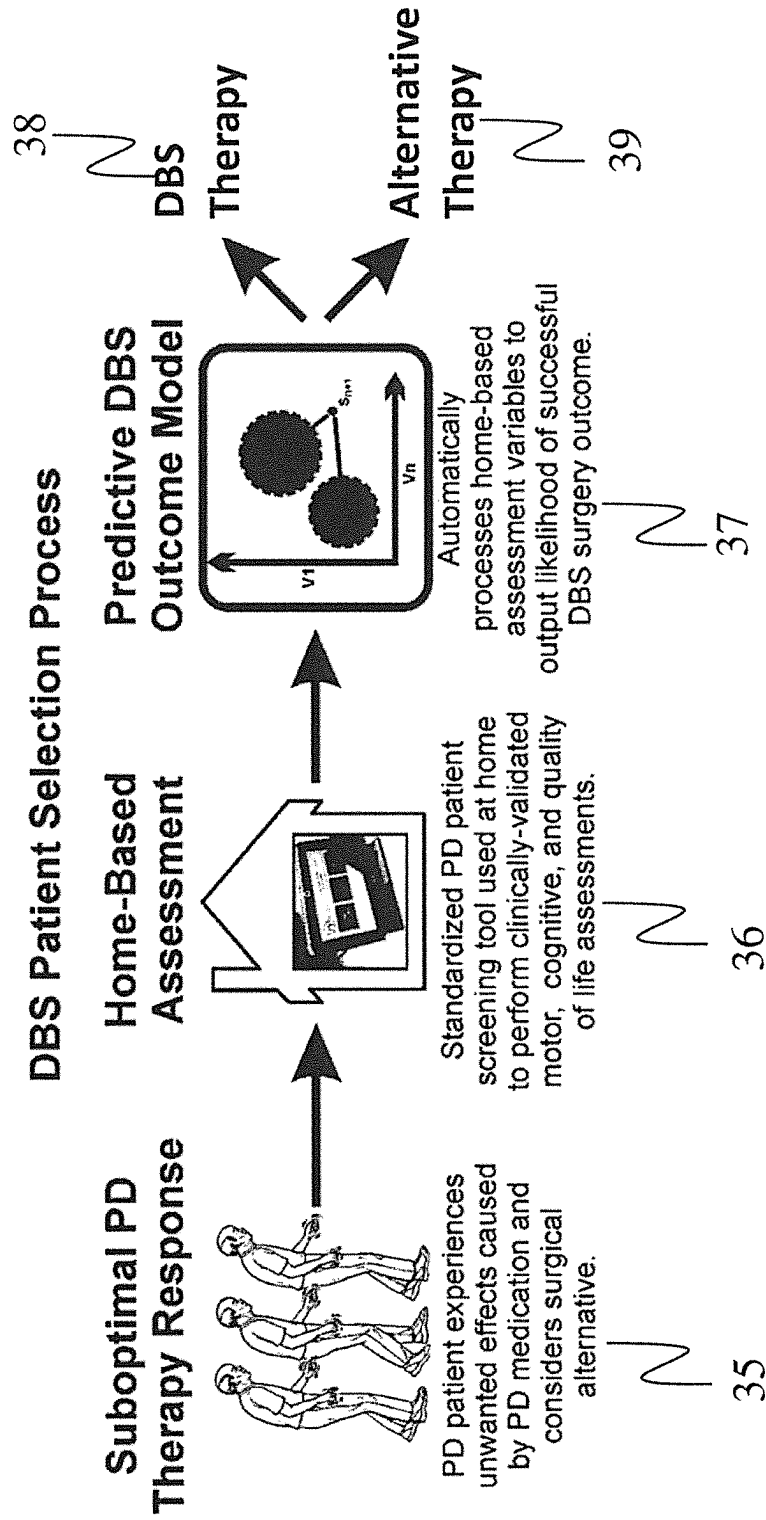


FIG. 7

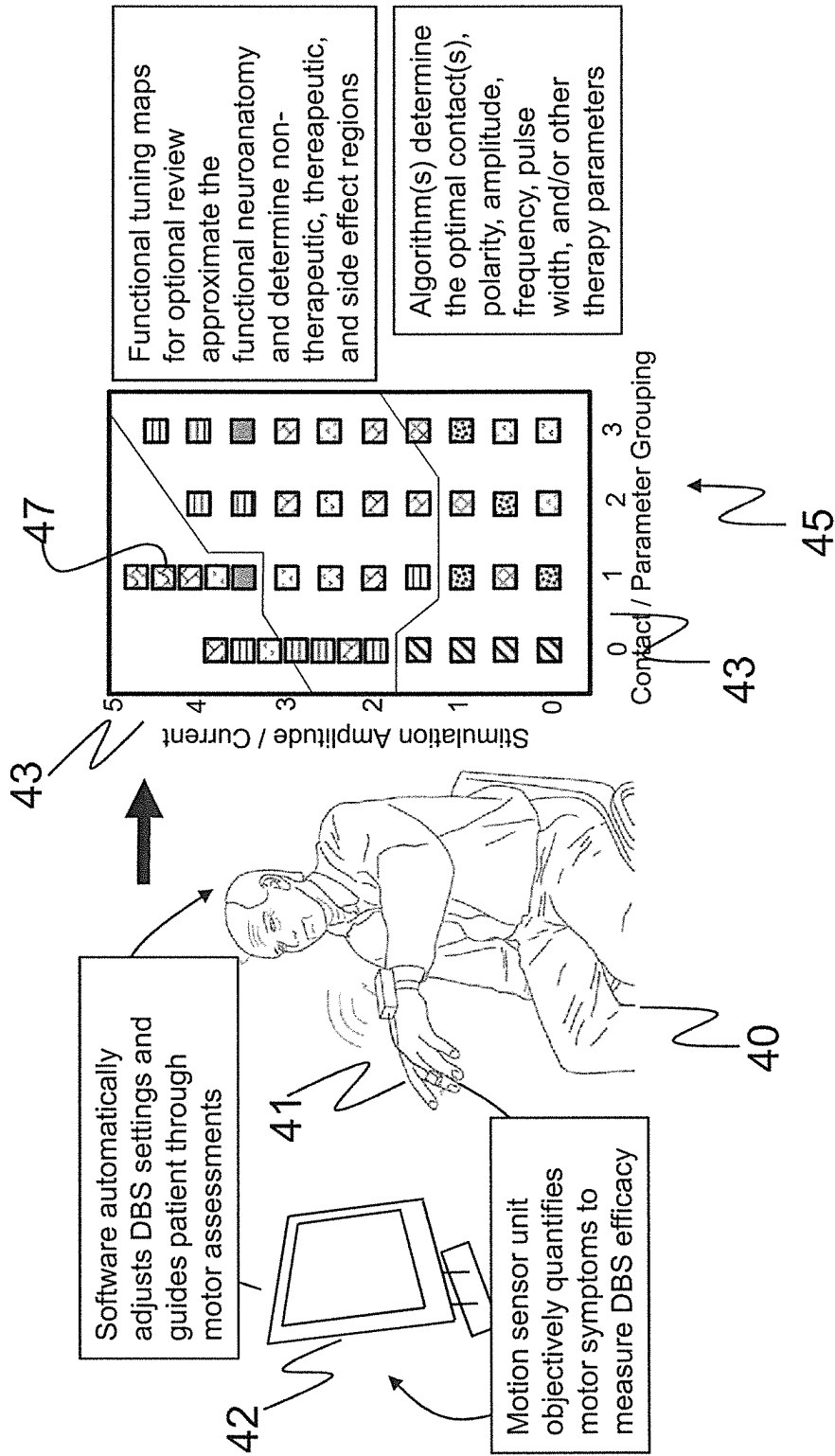


FIG. 8

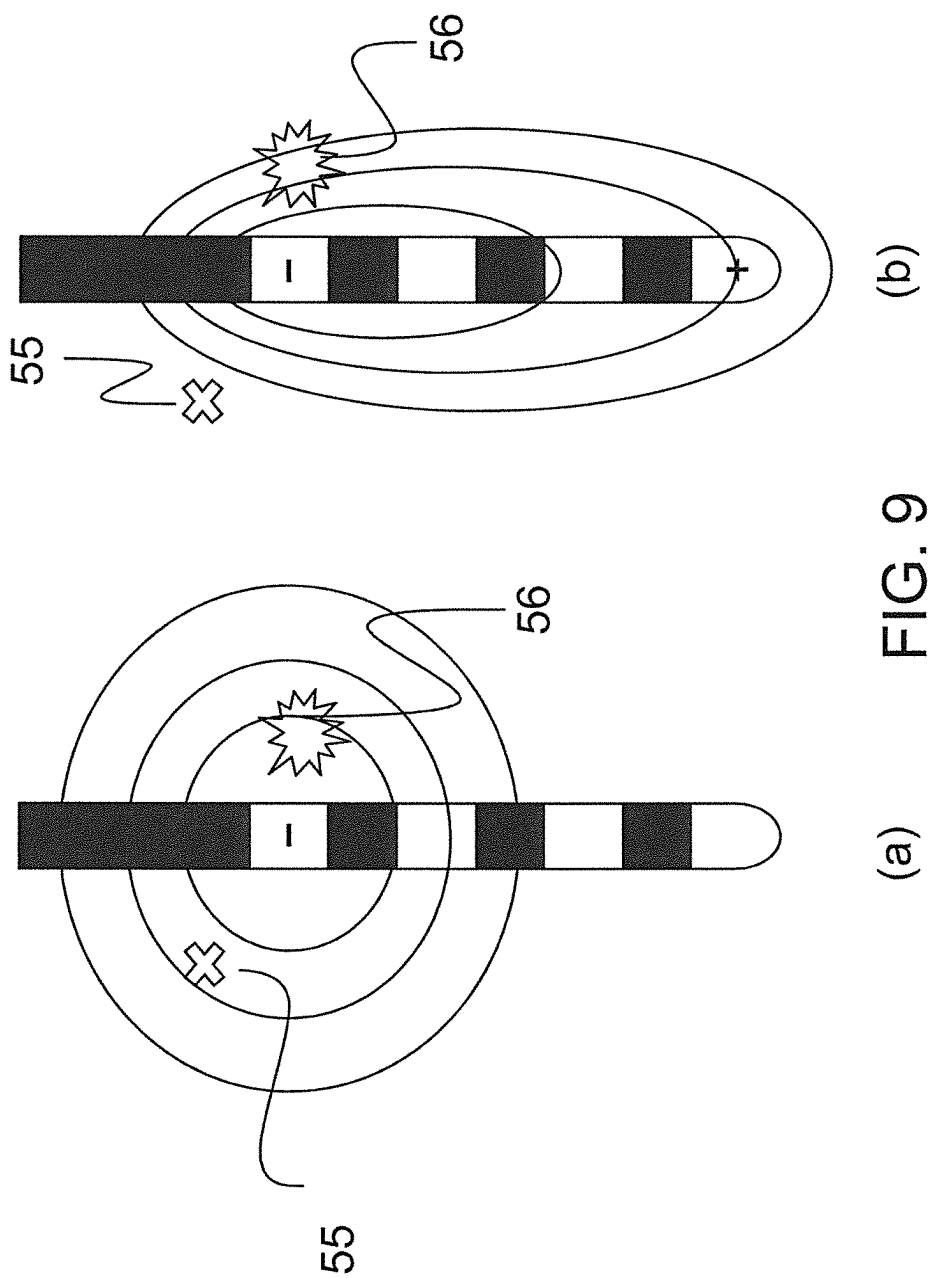


FIG. 9

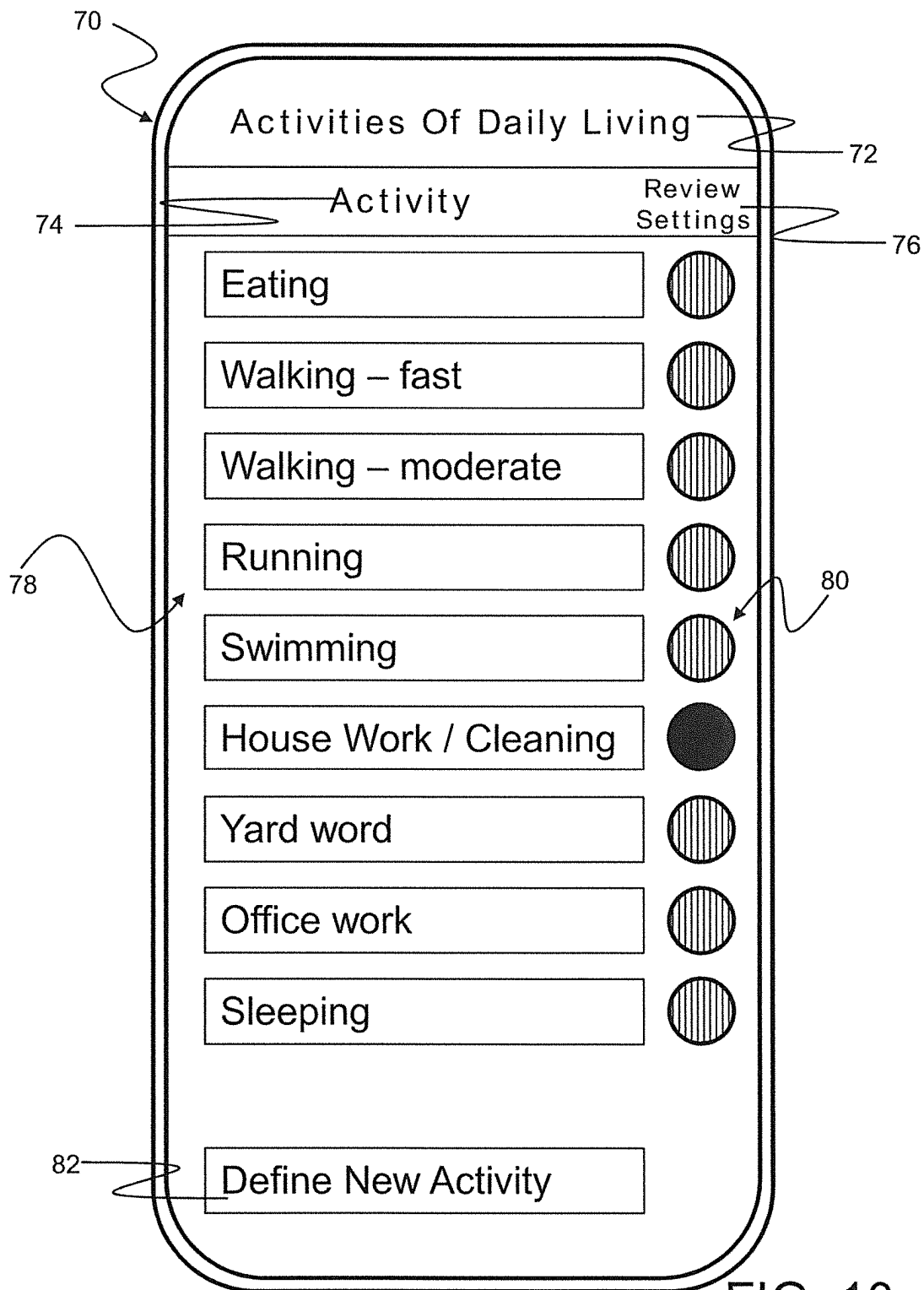


FIG. 10

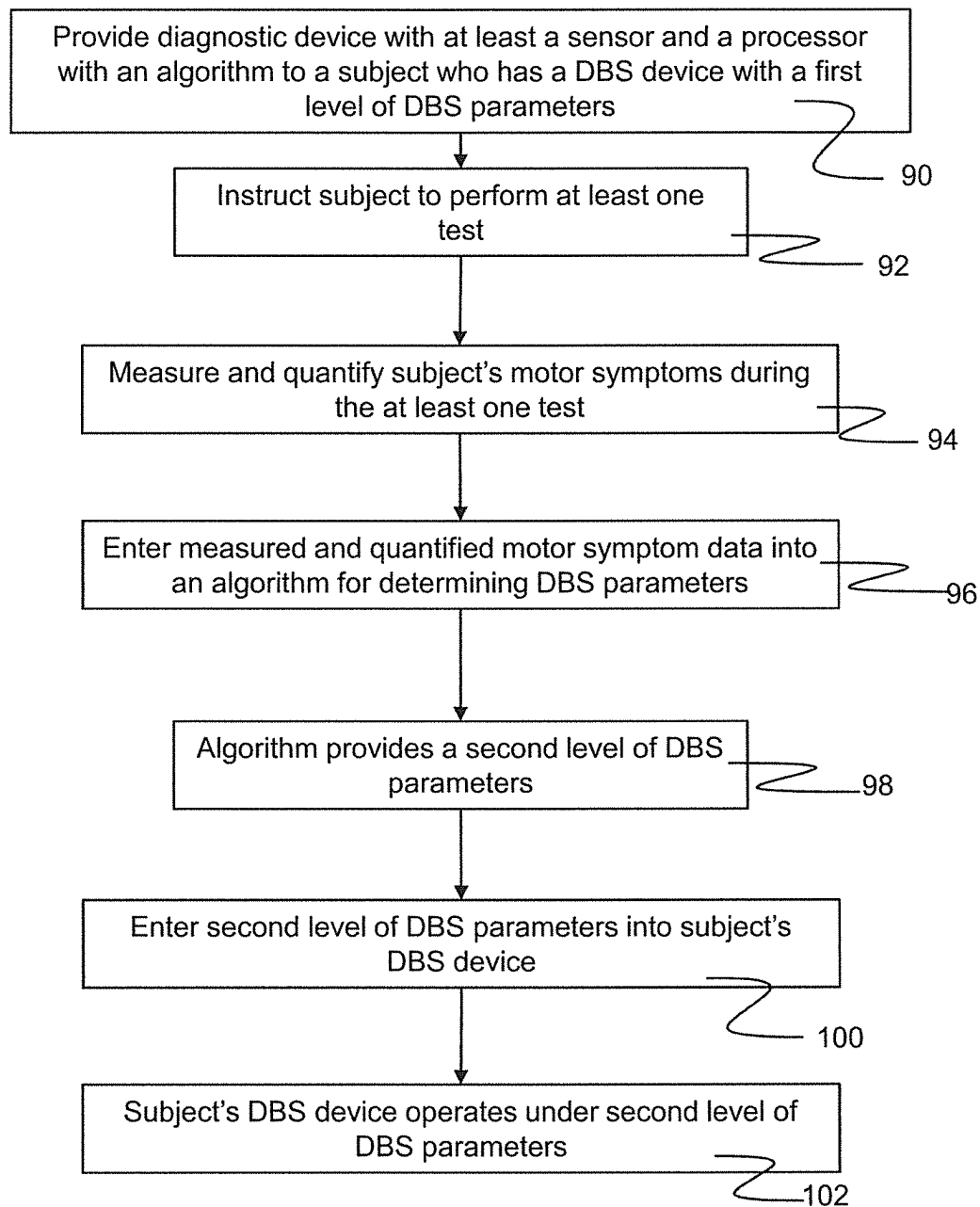


FIG. 11

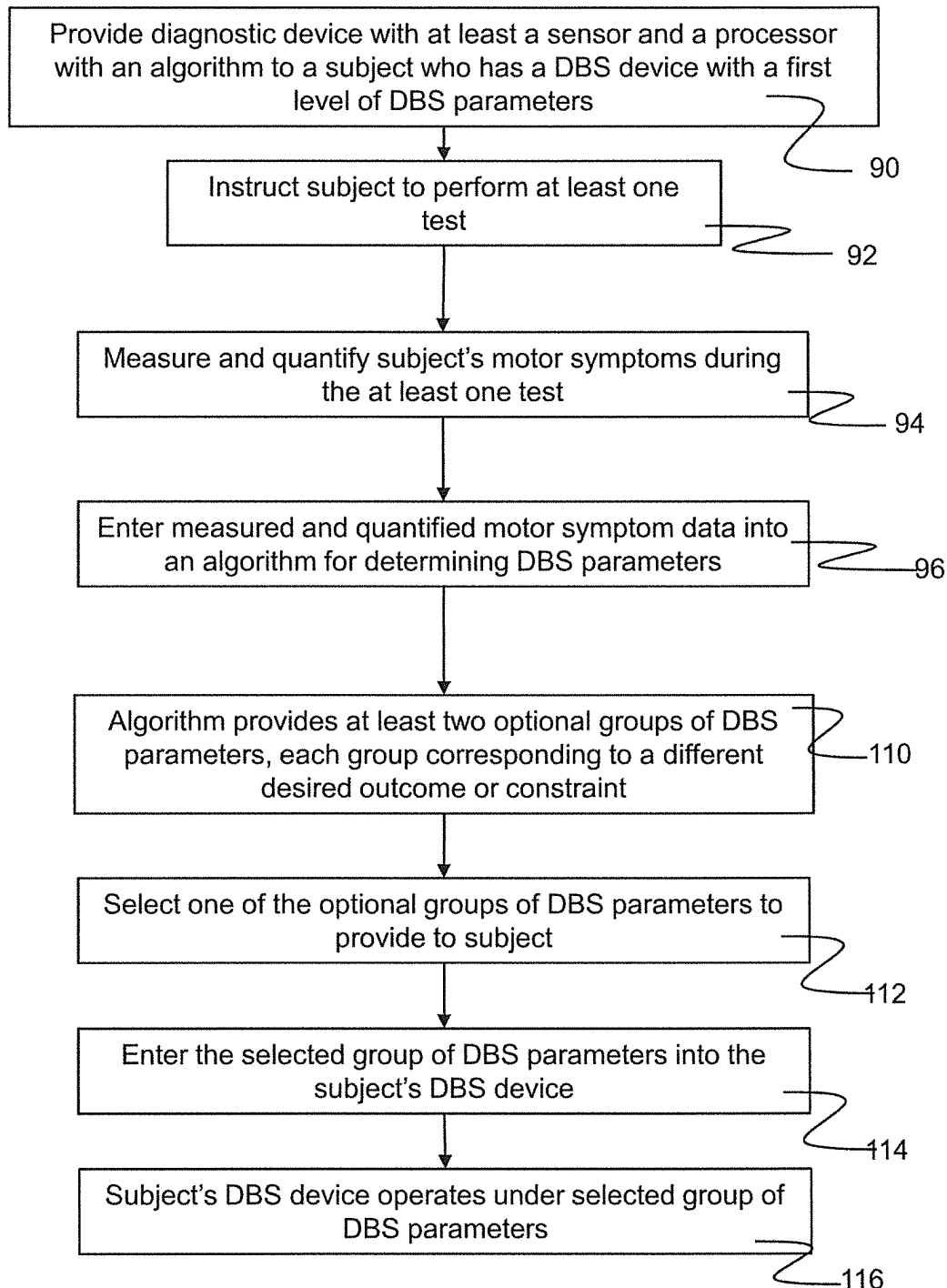


FIG. 12

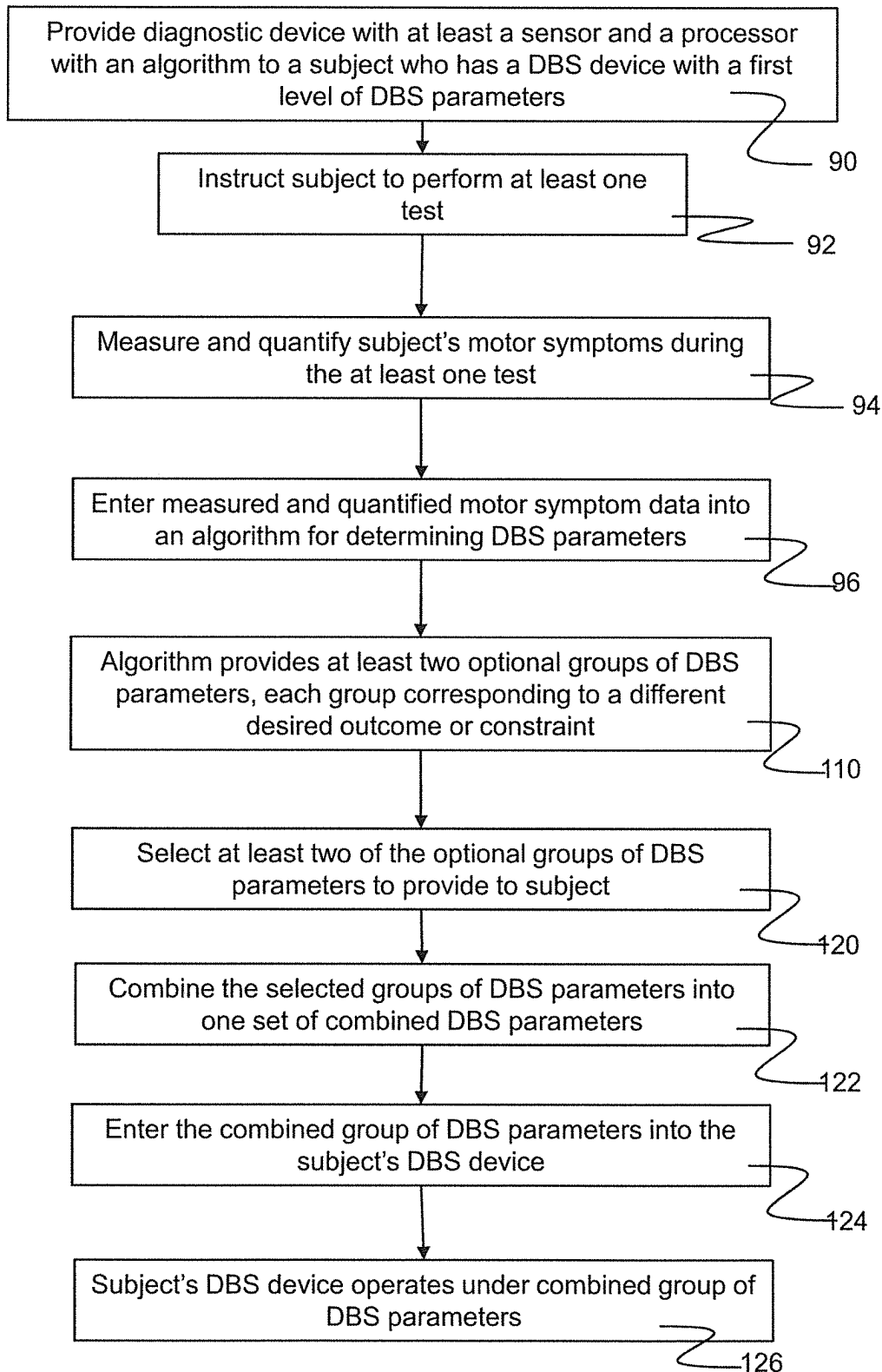


FIG. 13

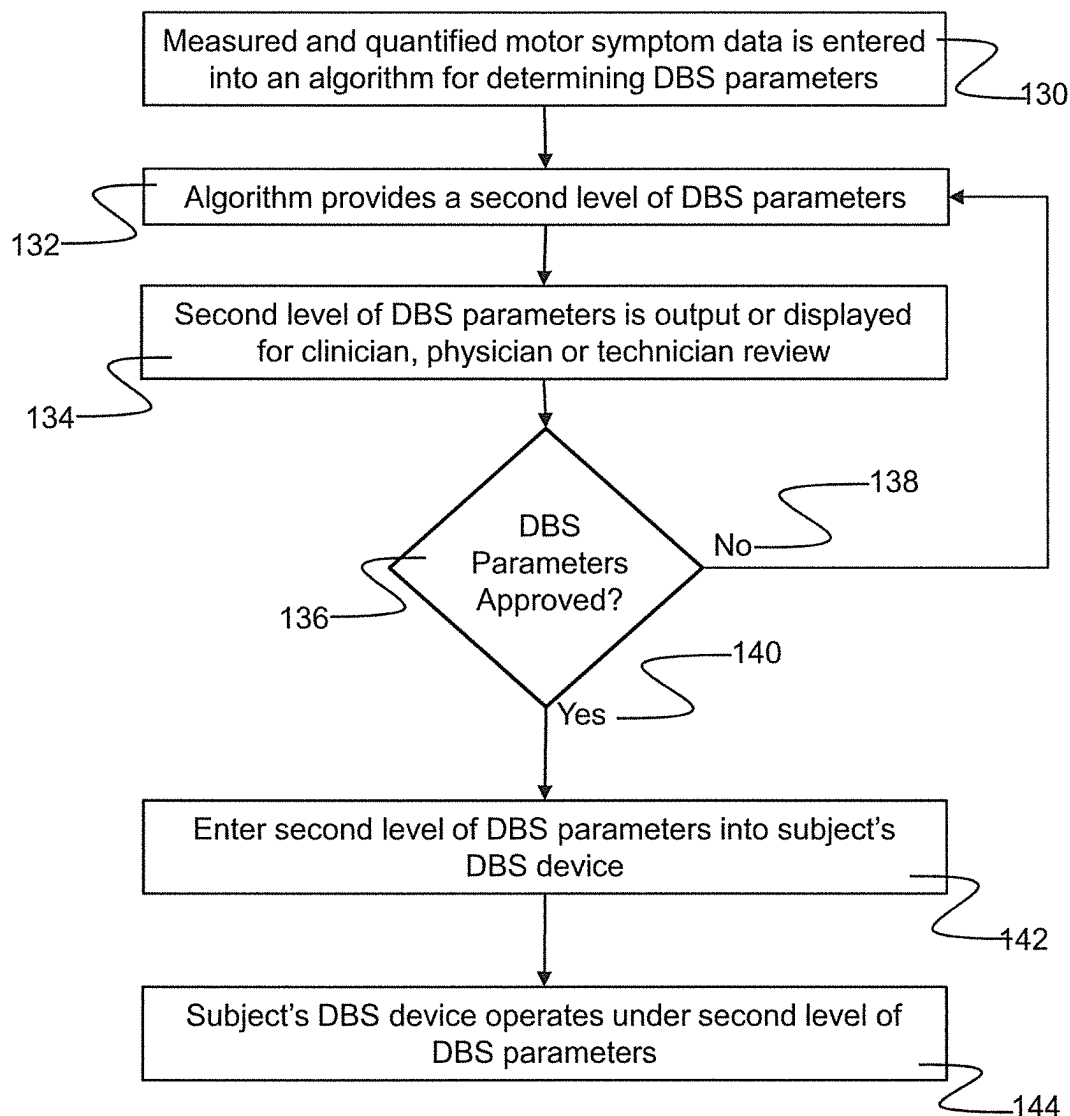
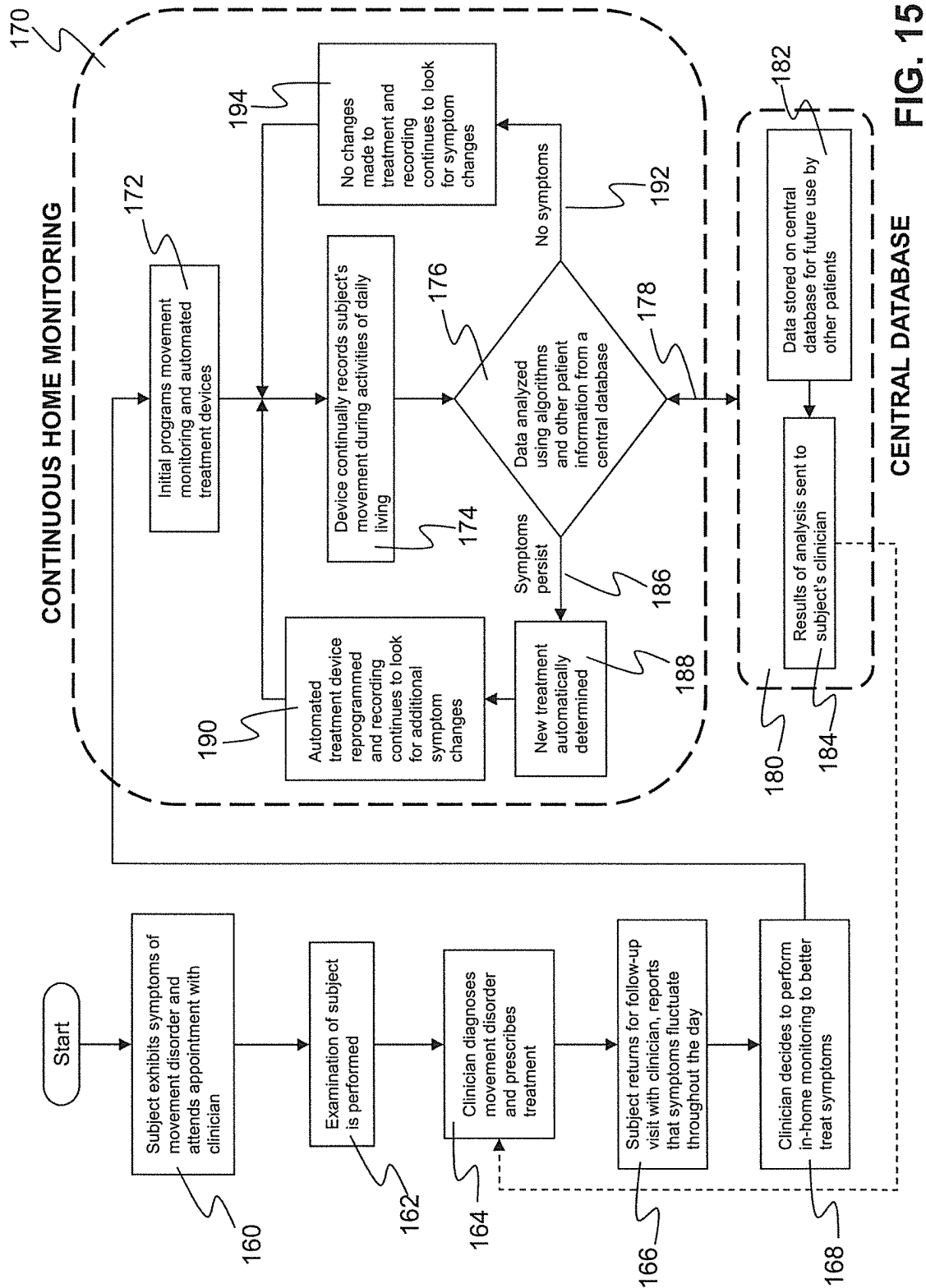
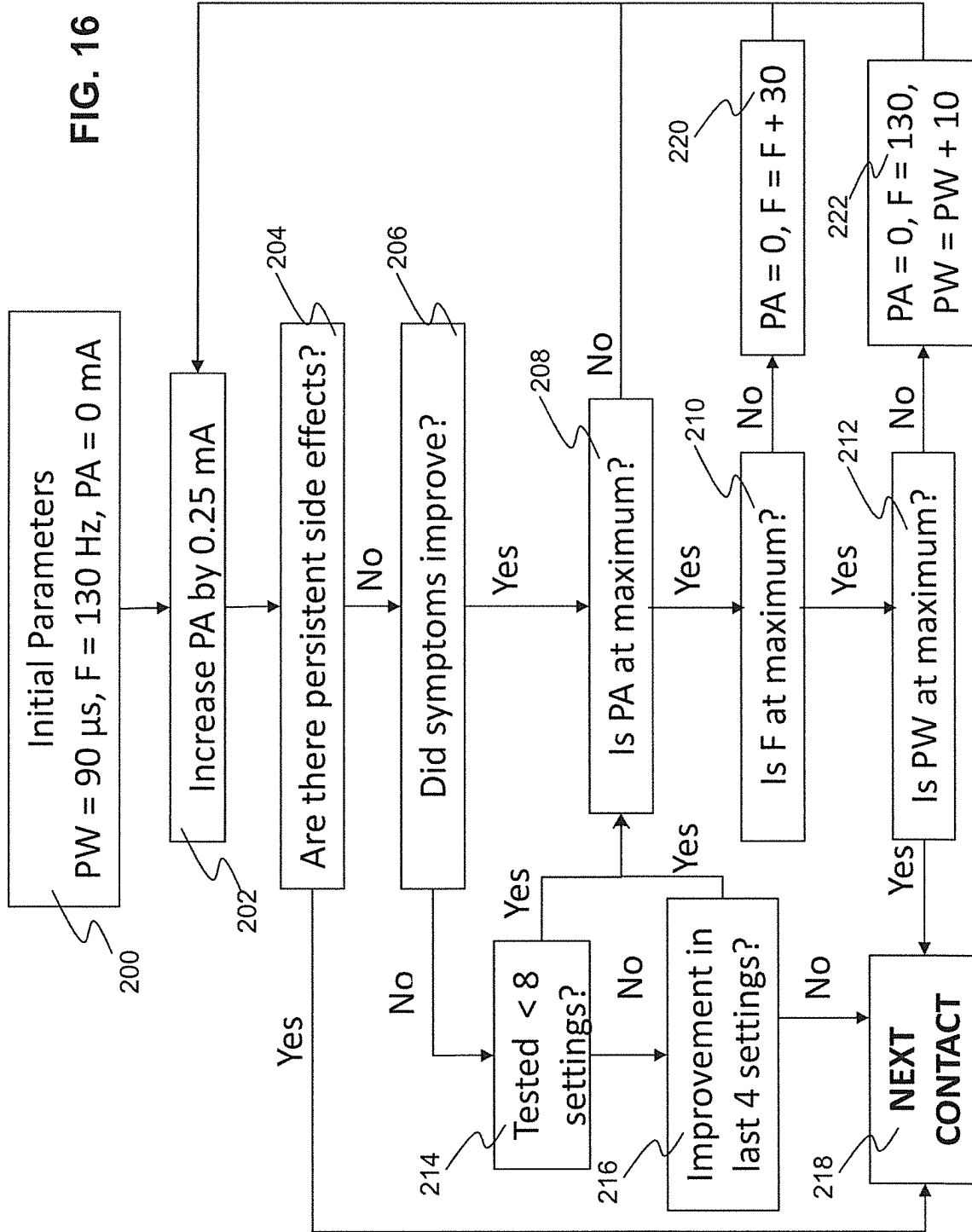


FIG. 14





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MOVEMENT DISORDER THERAPY SYSTEM, DEVICES AND METHODS, AND INTELLIGENT METHODS OF TUNING

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation of U.S. patent application Ser. No. 16/600,834 which was filed on Oct. 14, 2019, and which is a continuation of U.S. patent application Ser. No. 15/632,540, which was filed on Jun. 26, 2017, issued as U.S. Pat. No. 10,478,626 on Nov. 19, 2019, and is a continuation of U.S. patent application Ser. No. 14/933,078, which was filed on Nov. 5, 2015, issued as U.S. Pat. No. 9,717,920 on Aug. 1, 2017, and which is a continuation of U.S. patent application Ser. No. 13/918,948, filed on Jun. 15, 2013, issued as U.S. Pat. No. 9,211,417 on Dec. 15, 2015, and which is a non-provisional application that claims priority to provisional U.S. Patent Application Ser. No. 61/698,890, which was filed on Sep. 10, 2012.

BACKGROUND OF THE INVENTION

(1) Field of the Invention

The present invention relates to therapeutic medical apparatus systems, delivery systems, devices and/or methods, and to apparatus and methods for using neural stimulation to alleviate the symptoms of movement disorders, such as those associated with Parkinson's disease, essential tremor, dystonia, and Tourette's syndrome, including tremor, bradykinesia, rigidity, gait/balance disturbances, and dyskinesia, and also for treating mental health disorders such as major depression, bipolar disorder, and obsessive compulsive disorder for example. The present invention further relates to the use of a movement disorder diagnostic device for automatically adjusting therapeutic systems, devices, delivery systems, as well as methods thereof.

(2) Technology Review

A current trend in the treatment of diseases identified as being associated with the central nervous system is the stimulation of target areas of the central nervous system to affect therapeutic benefit. Such stimulation has been accomplished with, for example, implanted electrodes that deliver electrical stimulation to target brain regions; one class of electrical neural stimulation devices has been categorized under the name "deep brain stimulation" (DBS). Although the exact neurological mechanisms by which DBS therapies succeed are complex and are not yet fully understood, such therapies have proven effective in treating Parkinson's disease motor symptoms (such as tremor, bradykinesia, rigidity, and gait disturbances), and investigation into the use of DBS for the treatment of this and other neurological and mental health disorders, including major depression, obsessive-compulsive disorder, tinnitus, obesity, criminal tendencies, and antisocial disorders, is ongoing.

Parkinson's disease (PD) affects the motor system and can be characterized by motor symptoms including tremor, bradykinesia, and impaired gait. When diagnosed, dopamine replacement medication is prescribed. However, over time drug effectiveness decreases, requiring increased dosage. Frequent and stronger side effects such as dyskinesias (uncontrolled, irregular movements) and unpredictable "on"/"off" episodes are cause for more invasive intervention. Deep brain stimulation (DBS) surgery is performed when

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medication no longer adequately treats symptoms. Significant costs are not only associated with the initial implant surgeries (~\$40,000), but also subsequent stimulator battery replacements (~\$10,000-20,000), outpatient programming sessions (~\$1,000), and geographic disparities can put a significant financial and emotional burden on patients over many years, especially if the expected therapeutic improvement is not achieved. Although DBS has become a standard of care for many advanced stage PD patients, post-surgical outcomes are not equal across patients. A University of Florida study followed over 100 DBS patients, primarily PD, seeking referral to their movement disorder specialist after experiencing unsatisfactory improvement. Of those patients, 28% had been misdiagnosed presurgery and 90% had unsatisfactory symptomatic benefit including 40% for tremor, 37% gait, 11% for motor fluctuations and dyskinesias, and 14% for bradykinesia. The study also strongly argues for the need of preoperative education to ensure appropriate referral and selection of DBS candidates.

As a result, there are no set criteria for surgical patient selection. Subjective screening questionnaires have been developed to determine appropriate candidates based on a significant levodopa response (greater than 25%), the presence of motor fluctuations and dyskinesias, and minimal to no cognitive decline. However, methods to accurately evaluate these criteria are severely lacking for several reasons. First, the pattern and severity of motor symptom and dyskinesias vary greatly throughout the day. Therefore, an in-clinic assessment over multiple medication dose cycles is not feasible and is conducted under artificial conditions. Patients are typically asked to come in OFF PD medications from the previous night. Long travel days and pain and stress associated with being in this therapy state may induce fatigue and increased symptom severity. Second, clinical rating scales, most commonly the Unified Parkinson's Disease Rating Scale (UPDRS), are used to evaluate ON and OFF PD symptom states. Under the UPDRS, symptoms are subjectively rated on a 0-4 scale corresponding to normal, slight, mild, moderate, and severe, which can suffer from poor inter- and intra-rater reliability. An alternative, using home diaries, relies entirely on the patient's perception of their medication state and their level of compliance. Patients may also underestimate dyskinesia and motor symptom severity and have difficulty distinguishing between dyskinesia, tremor, and normal voluntary movements, a situation that can make evaluating medication response particularly challenging. Third, limited work has been conducted to directly compare pre-surgery medication response with post-surgery outcome in order to better select DBS candidates.

Access to movement disorder specialists to undergo the subjective screening questionnaires may require many clinical visits and can be financially burdensome for the geographically disparate subset of the PD population or those unable to travel. Movement disorder center locations can limit access to well-trained clinicians and effective symptom management. Rural patients in one study had a significantly worse quality of life score than their urban counterparts. Telehealth technologies such as home monitoring and online patient data management can have a significant impact on the equity, accessibility, and management of PD for patients who live in rural and remote communities or those unable to travel. In particular, one study showed that over a three-year period, telemedicine used for 100 follow-up visits for 34 PD patients left patients and providers satisfied with use of the technology and their savings amounted to approximately 1500 attendant travel hours, 100,000 travel kilometers, and \$37,000 in travel and lodging costs. Typically, medication

for Parkinson's disease (PD) consists of Levodopa to alleviate symptoms. Over time, however, the medication has reduced efficacy and shows increased occurrence of side effects such as dyskinesias. Once side effects outweigh benefits, patients consider deep brain stimulation (DBS). An electrode/wire lead is implanted in a specific location in the brain which shows hyperactivity in PD patients and is sensitive to electrical stimulation. PD target sites are typically the subthalamic nucleus (STN) or globus pallidus internus (GPI). The tremor-specific target site is generally the ventral intermedius nucleus of the thalamus (VIM). Electrical pulses characterized by amplitude (volts), current (amps), frequency (Hz), and pulse width (microseconds) are regulated by an implantable pulse generator (IPG) placed beneath the skin on the chest. Stimulation affects motor symptoms on the contralateral side, i.e., right side tremor will be treated on the left brain. After a patient has been implanted and recovered, programming sessions will fine tune stimulation settings described above in order to minimize symptom severity, minimize side effects, and maximize IPG battery life span. Although medication is not eliminated, it is typically reduced significantly. DBS efficacy decreases over time as the body adjusts to stimulation and protein buildup around electrode lead attenuates electrical field. Programming sessions are required throughout the patient's lifetime, though the frequency of adjustments is typically greater at first.

A typical implanted DBS stimulation lead consists of a thin insulated needle comprising four platinum/iridium electrodes spaced 0.5 or 1.5 mm apart along the length of the lead. One or multiple leads may be implanted in a target brain region or regions to provide symptom-inhibiting high-frequency stimulation, although some research suggests that excellent results can be achieved even when the lead is implanted distant from a target region. A DBS lead is connected to an implantable pulse generator (IPG), which serves as a controller and power source, via an extension cable tunneled subcutaneously to a subcutaneous pocket in the chest or abdominal cavity. The IPG typically includes a battery and circuitry for telemetered communication with an external programming device used to adjust, or "tune," DBS stimulation parameters, which may include but are not limited to stimulation frequency, amplitude, pulse width (or wavelength), waveform type, and contact configuration (that is, the selection of which electrodes are utilized from among the electrodes available on a lead, and, if two or more electrodes are active, the relative polarity of each), and the like. These parameters are initially set during implantation surgery separately and independently for each DBS lead that is implanted, and are then further fined-tuned in the outpatient clinic or in a doctor's office following surgery to maximize therapeutic benefit and minimize undesirable stimulation-induced side effects. The first such programming session usually takes place several weeks following implantation surgery, after the patient has recovered and inflammation at the lead placement site has subsided.

DBS programming may be performed by movement disorder neurologists, neurosurgeons, fellows, occupational and physical therapists, nurses, or employees of the DBS manufacturer. However, many patients have inadequate access to DBS programming due to physicians and patients relocating as well as implantations occurring at facilities far from a patient's home. Additionally, there is a shortage of health care professionals highly trained in DBS programming. This can partially be explained by a reluctance to participate in DBS management due to a lack of familiarity with electrophysiology or possibly the costs associated with

postoperative DBS management. Retrospective studies found that DBS programming sessions take more than twice as long as typical evaluations by movement disorder neurologists. Furthermore, programming sessions must be limited to 1-3 hours since longer sessions result in patient fatigue or lightheadedness. Multiple visits lead to additional travel costs and can be particularly difficult for those traveling from rural areas.

The approaches to programming can vary greatly across institutions. Strict iterative procedures whereby initial subjective test results based on human observation are used to determine the effect the parameters have of the patient and new parameters are determined based on those results by clinician calculation and observation are quite time consuming and therefore rarely followed. Many programmers make educated guesses as to the best settings based on their prior experience; however, this experience can vary across institutions and may not take into account varied lead positioning. Many programmers simply ignore bipolar or tripolar configurations whereby stimulation is provided from two or three contacts on a single DBS lead simultaneously, and do not adjust frequency or pulse width in an attempt to speed the programming process; however, neglecting these options can lead to suboptimal patient outcomes. Many clinician programmers do not fully appreciate the different programming parameters or modes of stimulation. In constant-voltage IPGs, the voltage of each pulse is set, but the current will automatically change based on the electrode impedance. This leads to variable amounts of current being delivered the stimulation target as impedances change. Additionally, since impedances will vary across electrode contacts, applying the same voltage on two different contacts will likely lead to different therapeutic currents being delivered. On the other hand, constant-current IPGs specify the current to be delivered and adjust the voltage accordingly based on the impedance. Since the therapeutic effects of DBS are based on current delivered at a given target, constant-current IPGs are preferable to constant-voltage IPGs.

While the above-described equipment and procedures are typical as of the filing of this application, variations and refinements may become commonplace as neural implant technology advances. Conceivably, uses of a multiplicity of DBS leads or networks of DBS leads may provide greater coverage, enabling the stimulation of larger and more varied target areas, and miniaturization and improved telemetry may obviate the need for the extension cable and/or the IPG altogether as leads become self-powering and/or self-controlling or permit for built-in telemetry. Advances in nanotechnology and materials may also allow DBS leads in the future to become self-repositioning, self-cleaning, or resistant to biological rejection for improved long-term therapeutic operation and more precisely targeted implantation.

The current standard in evaluating the severity of movement disorder symptoms in Parkinson's disease is the manually human scored Unified Parkinson's Disease Rating Scale (UPDRS) used to score motor tests, many of which involve repetitive movement tasks such as touching the nose and drawing the hand away repeatedly, or rapidly tapping the fingers together. A battery of exercises, typically a subset of the upper extremity motor section of the UPDRS, is normally completed during DBS lead placement surgery and subsequent programming sessions to evaluate performance while a clinician qualitatively assesses symptoms. Each test is evaluated by a clinician based solely on visual observation and graded on a scale that ranges from 0 (minor) to 4 (severe).

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During DBS implantation surgery, various lead placement strategies are used, including inversion recovery imaging, reformatted anatomical atlases, and formula coordinates based on known landmarks. Implantation location is verified and adjusted based on electrophysiological mapping using techniques such as microelectrode recording and micro and macro stimulation. Currently, lead placement and stimulation parameters are modified based on subjective motor examinations such as clinical observation such as the UPDRS motor tasks during the implantation procedure. After lead placement, patient motor symptoms are evaluated in response to a set of stimulation parameters. Stimulation parameters are then adjusted, and motor exam repeated. This trial-and-error process of adjusting parameters and monitoring patient response is continued until an optimal electrode position and stimulation set are established. During this programming or "tuning" process, the clinician subjectively assesses motor symptom improvement.

Postoperatively, assessing DBS response and reprogramming stimulation parameters require a significant time commitment. Several stimulation parameters can be modified, including electrode polarity, amplitude, current, pulse width, waveform type, and frequency. DBS programming and patient assessment may be performed by a variety of healthcare professionals, including movement disorder neurologists, neurosurgeons, fellows, occupational and physical therapists, nurses, and employees of the DBS manufacturer. Stimulation optimization is typically performed based on results of an exam such as the UPDRS, with the patient in four states (off medication/off DBS, off medication/on DBS, on medication/off DBS, and on medication/on DBS). The process of DBS adjustment is iterative and largely involves trial-and-error. Programming and patient assessment from preoperatively to one year after surgery requires approximately 30 hours of nursing time per patient.

Clinicians presently lack tools that combine physiological, electrical, and behavioral data to optimize electrode placement and stimulator programming. Optimizing electrode placement and stimulation parameters improves patient outcome by alleviating motor symptoms and minimizing complications. The present invention addresses this need for improved electrode placement and adjustment of deep brain stimulation parameters by providing a repeatable, automated or semi-automated tool that can assist stimulation parameter tuning during surgical electrode placement and outpatient programming sessions. In particular, the present invention aims to provide methods for the collection and transmission of objective biokinetic data during these procedures, which data is then processed to output objective movement disorder symptom severity measures on a continuous scale in real-time to guide clinician decision making. The improved resolution and repeatable results of the present invention should reduce time and costs of DBS procedures as well as improve patient outcomes.

It is therefore an object of the present invention to provide a system for screening patients for viability of DBS therapy prior to extensive, repetitive travel and expense, and prior to requiring surgical implantation of DBS leads. It is further an object of the present inventions to provide such a screening system to help minimize healthcare costs and to prevent adverse effects in patient quality of life associated with ineffective or unnecessary surgery, and to help clinicians to better select courses of treatment for patients.

It is further an object of the present invention to couple at-home patient viability screening and automatically-assigned quantitative motor assessments with procedures and practices for DBS implantation and parameter tuning and

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programming in semi-automatic and automatic ways to provide improved and less costly movement disorder patient therapy.

It is further an object of the present invention to provide automated functional mapping based on objective motor assessments and algorithms for resolving an optimal set of programming parameters out of the thousands of possibilities to provide an expert system to enable programming at a local medical facility. The system is designed for use by a general practitioner or nurse rather than by a neurologist or neurophysiologist with years of experience in DBS programming and disease management to increase access to high-quality postoperative DBS management. The system will minimize the required expertise of the clinician by requiring little or no advanced knowledge of complex neurophysiology or MRI imaging.

Existing systems for quantifying Parkinson's disease motor symptoms are described in this application's parent application, U.S. patent application Ser. No. 12/250,792, which is herein incorporated by reference, and which describes a novel system for measuring motor dysfunction symptoms and computing measures based on UPDRS scores therefrom. Preferably, the system and methods described therein are incorporated, in whole or in part, into the present invention as a means of automatic symptom quantification. The resultant scores objectively quantify movement disorder symptoms advantageously using a scale that is familiar to clinicians.

SUMMARY OF THE INVENTION

The present invention relates to methods for semi-automatically and automatically adjusting, or tuning, treatment parameters in movement disorder therapy systems. Semi-automatic adjustment includes providing the clinician, physician or technician with objective, quantitative or semi-quantitative data or measurements related to a subject's movement disorder symptoms, determining desired parameters, and then entering those parameters either semi-automatically or automatically into the subject's therapy device. Semi-automatic or automatic adjustment includes providing data including but not limited to objective, quantitative or semi-quantitative data and/or measurements related to a subject's movement disorder symptoms to an algorithm, using the data or measurements for determining desired parameters using the algorithm, and then entering those parameters either semi-automatically (i.e., allowing clinician, physician or technician to review and/or approve/adjust) or to automatically into the subject's therapy device. The present invention further relates to a system for screening patients to determine if they are viable candidates for certain therapy modalities. The present invention still further provides methods of quantifying movement disorders for the treatment of patients who exhibit symptoms of such movement disorders including, but not limited to, Parkinson's disease and Parkinsonism, Dystonia, Chorea, and Huntington's disease, Ataxia, Tremor and Essential Tremor, Tourette syndrome, and the like. The present invention yet further relates to methods of automatically and intelligently tuning a therapy device using objective quantified movement disorder symptom data acquired by a movement disorder diagnostic device with the therapy settings or parameters to be provided to the subject via his or her therapy device.

Objective measurement and quantification of a subject's movement disorder symptoms, including tremor, bradykinesia, dyskinesia, gait and/or balance disturbances, and the like requires, as a first step, a measurement of the movement.

This measurement can be performed by measuring a single movement metric, different movement metrics, or a combination of a number of movement metrics; and the movement metric or metrics being measured may include linear or rotational displacement, velocity, or acceleration, or any other metric that could give a quantitative indication of motion; and the part of the body being measured for motion may be a limb (as at a wrist, ankle, or finger) or may be the trunk of the body (as at a shoulder or torso), and the head. Sensors used for measuring body movement or motion include gyroscopes and accelerometers, preferably miniaturized, electromagnets, video, a multitude of sensors or system disclosed herein, or other sensors known to those skilled in the art. Additionally, sensors for measuring physiological signals such as electromyogram (EMG), electrooculogram (EOG), electroencephalogram (EEG), electrocardiogram (EKG), or other physiological signals which can directly or indirectly measure movement metrics in the subject may be included if such sensors and signals may be used to sense, detect, measure, and/or quantify the subject's external body motion, or related aspects. Other systems that can be used to detect and measure body motion include motion capture systems, machine vision systems, sonic or laser Doppler velocity transducers, infrared systems, GPS, or any other system known to those skilled in the art. The movement disorder diagnostic device used in the present invention may incorporate one or more of any of the above sensors or systems. Currently used movement data acquisition and diagnostic systems, such as the one described in U.S. Pat. No. 8,187,209, herein incorporated by reference, may similarly be used. In the present disclosure, "movement data" is construed as including, but not being limited to, any signal or set of signals, analog or digital, corresponding to movement of any part of the body or multiple parts of the body, independently or in conjunction with each other. This includes physiological signals from which movement data or symptoms can be derived. Preferably, this movement data is generated with a movement sensor such as for example a gyroscope and/or an accelerometer, and additionally or optionally a video sensor.

Movement may be continuously measured over long time spans, or may be measured only over a short time span, for example, as during the period of one or more tests taken from or based on the UPDRS motor exam. A measurement time period comprises two separate time periods: (i) a sensing time during which the movement disorder diagnostic device and its included sensors are used to sense and measure the subject's external physical motion; and (ii) a processing or calculation time wherein the measured motion data is used to calculate objective scores and/or other kinematic data that quantify the severity of the subject's movement disorder symptoms and side effects, and wherein the scores. The measurement time required to adequately and accurately sense, measure and quantify the subject's movement can depend on the particular movement test or task being performed, which typically corresponds to a particular symptom of a movement disorder. Generally, however, the system aims to minimize the amount of measurement time required to obtain sufficient movement data and provide quantitative scores and/or kinematic data to continue the evaluative process. Preferably, the measurement time required to provide objective scores and/or kinematic data is less than about 120 minutes. More preferably, the measurement time required to provide objective scores and/or kinematic data is less than about 90 minutes. Still more preferably, the measurement time required to provide objective scores and/or kinematic data is less than about 60

minutes. Yet more preferably, the measurement time required to provide objective scores and/or kinematic data is less than about 45 minutes. Even more preferably, the measurement time required to provide objective scores and/or kinematic data is less than about 30 minutes. Still yet more preferably, the measurement time required to provide objective scores and/or kinematic data is less than about 15 minutes. Still even more preferably, the measurement time required to provide objective scores and/or kinematic data is less than about 10 minutes. Yet still more preferably, the measurement time required to provide objective scores and/or kinematic data is less than about 5 minutes. Yet even more preferably, the measurement time required to provide objective scores and/or kinematic data is less than about 60 seconds. Even still more preferably, the measurement time required to provide objective scores and/or kinematic data is less than about 30 seconds. Even yet more preferably, the measurement time required to provide objective scores and/or kinematic data is less than about 15 seconds. Still even yet more preferably, the measurement time required to provide objective scores and/or kinematic data is less than about 1 second.

In some embodiments, a periodic system may be employed wherein the subject's external body motions are sensed, measured, and quantified repeatedly but at predefined or altering intervals. In such periodic embodiments, the periodic measurements preferably conform to the above described measurement time period standards. Embodiments utilizing periodic measurement may begin when the subject attaches or dons the movement disorder diagnostic device, and may involve a step of instructing the subject to attach or don the device to begin a measurement period. Protocols for periodic measurement may be envisioned wherein a subject follows a particular schedule for measurement and quantification of movement disorder data, and wherein the schedule may change throughout the course of treatment and/or therapy. In still other embodiments, a continuous monitoring system may be employed wherein the movement disorder diagnostic device continuously senses, measures and quantifies the subject's external body movements over extended periods of time, such as hours, days, weeks or months. Preferably, in the continuous measurement embodiments, the diagnostic device senses, measures and/or quantifies the subject's external body movements substantially continuously, with no breaks or stoppages in its operation. However, the limits of continuous operation may be defined by characteristics of the device, such as battery life, form factor and construction (e.g., if it needs to be removed to shower), and other such concerns.

The movement disorder diagnostic device contains at least one electronic component that further may contain internal or onboard memory for storage of the movement data such that the data may be transferred at a later time. More preferably, the movement disorder diagnostic device further may contain communications electronics, which transmit the movement data to an external device for storage and/or analysis. The communication electronics preferably is/are wireless, and most preferably is/are radio frequency wireless. The external device may be a centralized storage database, parallel databases, a cloud-based database, a computer, tablet, cell phone including for example smartphone, personal data assistant (PDA) or similar device, or a combination of database and computer or communication devices. Preferably, such transmission of data occurs substantially in real-time. By real-time, it is meant that preferably, data is transmitted within 30 minutes of being acquired, measured, or calculated. More preferably, data is transmitted

within 20 minutes of being acquired, measured, or calculated. Still more preferably, data is transmitted within 10 minutes of being acquired, measured, or calculated. Yet more preferably, data is transmitted within 5 minutes of being acquired, measured, or calculated. Even more preferably, data is transmitted within 5 minutes of being acquired, measured, or calculated. Still yet more preferably, data is transmitted within 3 minutes of being acquired, measured, or calculated. Even yet more preferably, data is transmitted within 60 seconds of being acquired, measured, or calculated. Yet still more preferably, data is transmitted within 45 seconds of being acquired, measured, or calculated. Yet even more preferably, data is transmitted within 30 seconds of being acquired, measured, or calculated. Even still more preferably, data is transmitted within 15 seconds of being acquired, measured, or calculated. Even yet more preferably, data is transmitted within 5 seconds of being acquired, measured, or calculated. Still even yet more preferably, data is transmitted within 1 second of being acquired, measured, or calculated. Yet even still more preferably, data is transmitted substantially simultaneously within milliseconds of being acquired, measured, or calculated.

Following measurement of symptomatic movement, the next step in objective quantification of a subject's movement disorder symptoms is the extraction of statistical kinematic features from the acquired movement data via processing. This processing may take place during or following data acquisition and may occur within a movement data acquisition device or within a different processing device, such as a personal computer, PDA, smart phone, tablet computer, touch screen interface, or the like, with which the acquisition device interfaces, either through a cable connection or by wireless transmission. Useful kinematic features that may be extracted from gyroscopic data may include, for example, peak power angular velocity, peak power angle, RMS angular velocity, frequency, maximum amplitude, maximum peak-to-peak amplitude, mean angular velocity, and wavelet parameters, as well as the covariance or standard deviation over time of any of these metrics. Useful kinematic features that may be extracted from accelerometer data may include, for example, peak power acceleration, peak power velocity, peak power position, RMS acceleration, RMS velocity, RMS position, frequency, maximum amplitude, maximum peak-to-peak amplitude, mean acceleration, and wavelet parameters, as well as the covariance or standard deviation over time of any of these metrics. In a movement data acquisition system, or movement disorder diagnostic measuring apparatus, that combines a three-axis accelerometer and a three-axis gyroscope to produce 6 channels of movement data, one or any combination of the above kinematic features can be extracted from any of the 6 kinematic channels to be used as inputs to a trained scoring algorithm in the next step. The listed kinematic features for the sensors above are intended to be exemplary, and not limiting; other types of sensors will produce different data from which different sets of features may be extracted.

The trained scoring algorithm used to process the kinematic features extracted from the movement data may comprise, for example, one or more of a simple or multiple linear regression, an artificial neural network, a Bayesian network, or a genetic algorithm. The output of the trained scoring algorithm may be a single score or multiple scores of any scale; a single score on the same scale as that of the UPDRS may be preferred in certain applications where simplicity or familiarity is the paramount concern, while more sophisticated scores and scales may be preferred for

other advanced applications, such as those that involve recommendations for treatment or closed-loop automated treatment delivery.

In various embodiments, following the step of symptom quantification, a separate tuning algorithm may compute suggested changes to the therapy system parameter settings based on the result of the symptom quantification algorithm and known or predicted current therapy system parameter settings and physiological models.

Depending on the embodiment of the invention, the current therapy system parameter settings changes may be input into the algorithm by a software user interface (integrated tuning), or may be automatically sensed and input from the DBS parameter settings by communicating with a DBS implant or its programmer device or unit (intelligent tuning), or may be known because the DBS parameter settings have been preset to some known baseline settings or restored to a previously saved settings preset. The existing parameter settings might also be predicted or derived based, for example, on observed or measured therapy effectiveness. Suggested therapy system parameter settings changes are then input into the therapy system, and their effectiveness is measured using the above-described method of symptom quantification.

The process of tuning therapy system parameter settings may remain iterative, but the present invention significantly minimizes, or at least greatly reduces the time and expertise required to arrive at optimized stimulation or therapy parameter settings, advantageously allowing clinicians, technicians or physicians with lesser training or experience to adjust parameter settings during patient visits, and to do so in less time than is currently required. Additionally, the present invention increases access to geographically disparate populations by putting the expertise into the system and reducing or eliminating the need for an expert or trained clinician to be present with each subject.

Many embodiments of the present invention include optimization or tuning algorithm(s) which are used to determine or recommend optimum therapy settings or parameters. Such optimization algorithms may include, but are not limited to simplex algorithms, extensions of the simplex algorithm designed for quadratic and/or linear function programming, combinatorial algorithms, and other multi-variant optimization algorithms known to those in the art. In order to determine what a desired or optimal level of therapy parameters might be, the subject's symptoms or side effects must first be measured and quantified. The measurement and quantification preferably take place while the subject is performing at least one movement disorder test as instructed. Once the initial measurement and quantification has been obtained, the system and/or, in some embodiments a clinician, physician or technician, programs a second level of therapy parameters into the subject's therapy device, and the subject repeats the movement disorder test(s) while the symptoms or side effects are again measured and quantified. This process is repeated until the desired result(s), goals or constraints are achieved. These processes and steps are described in greater detail below. Preferably, whether obtaining optimized therapy parameters or settings, or when iteratively testing to determine a second level of therapy parameters, preferably, the subject is instructed to perform, and performs, at least 1 movement disorder test, where the test comprises at least one task related to the subject's external body motion. More preferably, the subject is instructed to perform, and performs, at least 2 movement disorder tests. Still more preferably, the subject is instructed to perform, and performs, at least 3 movement disorder tests.

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tion in the occurrence and/or severity of the subject's side effects while the subject is receiving therapy or is under the effects of recently received therapy. Yet still even more preferably, an optimized or second level of therapy parameters or settings corresponds to at least a 95% reduction in the occurrence and/or severity of the subject's side effects while the subject is receiving therapy or is under the effects of recently received therapy. Most preferably, an optimized or second level of therapy parameters or settings corresponds to substantially eliminating the occurrence and/or severity of the subject's side effects while the subject is receiving therapy or is under the effects of recently received therapy.

Preferably, where the desired result is to reduce or minimize the subject's movement disorder symptoms, the optimized or second level of therapy parameters results in a reduction or minimization of at least 1 movement disorder symptom. More preferably, the optimized or second level of therapy parameters results in a reduction or minimization of at least 2 movement disorder symptoms. Still more preferably, the optimized or second level of therapy parameters results in a reduction or minimization of at least 3 movement disorder symptoms. Yet more preferably, the optimized or second level of therapy parameters results in a reduction or minimization of at least 4 movement disorder symptoms. Even more preferably, the optimized or second level of therapy parameters results in a reduction or minimization of at least 5 movement disorder symptoms.

Preferably, where the desired result is to reduce or minimize the subject's side effects from medication or therapy, the optimized or second level of therapy parameters results in a reduction or minimization of at least 1 side effect. More preferably, the optimized or second level of therapy parameters results in a reduction or minimization of at least 2 side effects. Still more preferably, the optimized or second level of therapy parameters results in a reduction or minimization of at least 3 side effects. Yet more preferably, the optimized or second level of therapy parameters results in a reduction or minimization of at least 4 side effects. Even more preferably, the optimized or second level of therapy parameters results in a reduction or minimization of at least 5 side effects.

Other secondary constraints or desired results may also be considered when optimizing or determining a second level of therapy parameters or settings such as maximizing the battery life of the therapeutic (e.g., DBS) device, maximizing the therapeutic window, and the like. Such constraints or desired results as these are secondary only in that the primary goal of the therapy is to increase the subject's quality of life by reducing or minimizing symptoms or side effects, or balancing both, while also trying to improve the duration and quality of therapy otherwise. For example, maximizing battery life of the therapy device helps to increase the time required between subject's visits to the clinician, physician or technician as well as ensuring that the device has sufficient power and capability to effectively provide the determined levels of therapy. Similarly with maximizing the therapeutic window, which also increases the time between visits, but also maximizes the length of time that the stimulation therapy has a positive effect on the subject and reducing the number of stimulations required to achieve the desired results. Typically, the subject and his or her clinician, physician or technician will agree upon the primary desired result, such as minimizing symptoms, but numerous other such constraints will also be considered, weighed and balanced in determining the optimized or second level of parameters or settings.

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Several embodiments may include a general optimization strategy in which combinations of the above desired results or constraints are used to select the appropriate optimized settings. For example, such embodiments may optimize based on reducing or minimizing both symptoms and side effects. Any combination of type and/or number of desired results or constraints may be used to optimize the system. Preferably, at least two different desired results or constraints are considered when determining an optimized group of therapy settings or parameters. More preferably, at least three different desired results or constraints are considered when determining an optimized group of therapy settings or parameters. Still more preferably, at least four different desired results or constraints are considered when determining an optimized group of therapy settings or parameters. Yet more preferably, at least five different desired results or constraints are considered when determining an optimized group of therapy settings or parameters. Even more preferably, at least six different desired results or constraints are considered when determining an optimized group of therapy settings or parameters. Most preferably, more than seven different desired results or constraints are considered when determining an optimized group of therapy settings or parameters.

Numerous embodiments of the present invention are envisioned in this disclosure. These following embodiments are examples of the many embodiments encompassed by the present invention, but do not in any way limit the many other embodiments covered by this disclosure.

One embodiment of the present invention includes a method of tuning a movement disorder therapy system comprising steps of providing a movement disorder diagnostic device to a subject having a deep brain stimulation (DBS) device with a first level of DBS parameters, the movement disorder diagnostic device comprising at least one physiological or movement sensor having a signal, and a processor comprising an algorithm, instructing the subject to perform at least one movement disorder test(s) while the subject is undergoing DBS therapy or is under the effects of DBS therapy, measuring and quantifying motor symptoms of the subject based at least in part on the signal from the at least one physiological or movement sensor(s) during the at least one movement disorder test(s), entering data corresponding to the subject's measured and quantified motor symptoms into an algorithm, providing, with the algorithm, a second level of DBS parameters based at least in part on the data entered into the algorithm, and entering the second level of DBS parameters into the subject's DBS device such that the subject's DBS device operates under the second level of DBS parameters.

Another embodiment of the present invention includes a method of tuning a movement disorder therapy system comprising steps of providing a movement disorder diagnostic device to a subject having a deep brain stimulation (DBS) device with a first level of DBS parameters, the movement disorder diagnostic device comprising at least one physiological or movement sensor having a signal, and a processor comprising an algorithm, instructing the subject to perform at least one movement disorder test(s) while the subject is undergoing DBS therapy or is under the effects of DBS therapy, measuring and quantifying motor symptoms of the subject based at least in part on the signal from the at least one physiological or movement sensor(s) during the at least one movement disorder test(s), entering data corresponding to the subject's measured and quantified motor symptoms into an algorithm, determining, with the algorithm, at least two optional groups of DBS parameters, each

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optional group of parameters corresponding to a different desired outcome or constraint, selecting one of the at least two optional groups of DBS parameters, and entering the selected optional group of DBS parameters into the subject's DBS device such that the subject's DBS device operates under the second level of DBS parameters.

Still another embodiment of the present invention includes a method of tuning a movement disorder therapy system comprising steps of providing a movement disorder diagnostic device to a subject having a deep brain stimulation (DBS) device with a first level of DBS parameters, the movement disorder diagnostic device comprising at least one physiological or movement sensor having a signal, and a processor comprising an algorithm, instructing the subject to perform at least one movement disorder test(s) while the subject is undergoing DBS therapy or is under the effects of DBS therapy, measuring and quantifying motor symptoms of the subject based at least in part on the signal from the at least one physiological or movement sensor(s) during the at least one movement disorder test(s), entering data corresponding to the subject's measured and quantified motor symptoms into an algorithm, determining, with the algorithm, at least two optional groups of DBS parameters, each optional group of parameters corresponding to a different desired outcome or constraint, selecting at least two of the at least two optional groups of DBS parameters, combining the at least two selected optional groups of DBS parameters into one set of combined DBS parameters, and entering the combined group of DBS parameters into the subject's DBS device such that the subject's DBS device operates under the second level of DBS parameters.

A method of tuning a movement disorder therapy system comprising steps of providing a movement disorder diagnostic device to a subject having a deep brain stimulation (DBS) device with a first level of DBS parameters, the movement disorder diagnostic device comprising at least one physiological or movement sensor having a signal, and a processor comprising an algorithm, instructing the subject to perform at least one movement disorder test(s) while the subject is undergoing DBS therapy or is under the effects of DBS therapy, measuring and quantifying motor symptoms of the subject based at least in part on the signal from the at least one physiological or movement sensor(s) during the at least one movement disorder test(s), entering data corresponding to the subject's measured and quantified motor symptoms into an algorithm, providing, with the algorithm, a second level of DBS parameters based at least in part on the data entered into the algorithm, and uploading, substantially simultaneously with transmitting, with at least one electronic component the second level of DBS parameters and/or measured and quantified motor symptoms to a database for storage and/or review by a clinician, technician or physician.

A method of tuning a movement disorder therapy system comprising steps of providing a movement disorder diagnostic device to a subject having a deep brain stimulation (DBS) device with a first level of DBS parameters, the movement disorder diagnostic device comprising at least one physiological or movement sensor having a signal, and a processor comprising an algorithm, instructing the subject to perform at least one movement disorder test(s) while the subject is undergoing DBS therapy or is under the effects of DBS therapy, measuring and quantifying motor symptoms of the subject based at least in part on the signal from the at least one physiological or movement sensor(s) during the at least one movement disorder test(s), entering data corresponding to the subject's measured and quantified motor

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symptoms into an algorithm, determining, with the algorithm, at least two optional groups of DBS parameters, each optional group of parameters corresponding to a different desired outcome or constraint, selecting one of the at least two optional groups of DBS parameters, and uploading, substantially simultaneously with transmitting, with at least one electronic component the second level of DBS parameters and/or measured and quantified motor symptoms to a database for storage and/or review by a clinician, technician or physician.

A method of tuning a movement disorder therapy system comprising steps of providing a movement disorder diagnostic device to a subject having a deep brain stimulation (DBS) device with a first level of DBS parameters, the movement disorder diagnostic device comprising at least one physiological or movement sensor having a signal, and a processor comprising an algorithm, instructing the subject to perform at least one movement disorder test(s) while the subject is undergoing DBS therapy or is under the effects of DBS therapy, measuring and quantifying motor symptoms of the subject based at least in part on the signal from the at least one physiological or movement sensor(s) during the at least one movement disorder test(s), entering data corresponding to the subject's measured and quantified motor symptoms into an algorithm, determining, with the algorithm, at least two optional groups of DBS parameters, each optional group of parameters corresponding to a different desired outcome or constraint, selecting at least two of the at least two optional groups of DBS parameters, combining the at least two selected optional groups of DBS parameters into one set of combined DBS parameters, and uploading, substantially simultaneously with transmitting, with at least one electronic component the second level of DBS parameters and/or measured and quantified motor symptoms to a database for storage and/or review by a clinician, technician or physician.

A method of tuning a movement disorder therapy system comprising steps of providing a movement disorder diagnostic device to a subject having a deep brain stimulation (DBS) device with a first level of DBS parameters, the movement disorder diagnostic device comprising at least one physiological or movement sensor having a signal, and a processor comprising an algorithm, displaying on a programming device a list of activities, actions or tasks for the subject to select from, having the subject elect at least one activity, action or task from the list on the programming device, selecting with the movement disorder diagnostic device a predetermined set of DBS parameters corresponding to the elected at least one activity, action or task, and entering with the programming device the group of selected DBS parameters corresponding to the at least one elected activity, action or task into the subject's DBS device such that the subject's DBS device operates under the selected group of DBS parameters while the subject performs the at least one elected activity, action, or task.

A method of tuning a movement disorder therapy system comprising steps of providing a movement disorder diagnostic device to a subject having a deep brain stimulation (DBS) device with a first level of DBS parameters, the movement disorder diagnostic device comprising at least one physiological or movement sensor having a signal, and a processor comprising an algorithm, displaying on a programming device a list of activities, actions or tasks for the subject to select from, having the subject elect at least one activity, action or task from the list on the programming device, selecting with the movement disorder diagnostic device a predetermined set of DBS parameters correspond-

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ing to the elected at least one activity, action or task, entering with the programming device the group of selected DBS parameters corresponding to the at least one elected activity, action or task into the subject's DBS device such that the subject's DBS device operates under the selected group of DBS parameters, measuring and quantifying motor symptoms of the subject based at least in part on the signal from the at least one physiological or movement sensor(s) while the subject performs the at least one elected activity, action or task, entering data corresponding to the subject's measured and quantified motor symptoms into an algorithm, providing, with the algorithm, a second level of DBS parameters based at least in part on the data entered into the algorithm, the second level of DBS parameters maximizing the subject's ability to perform the at least one elected activity, action or task, and entering the second level of DBS parameters into the subject's DBS device such that the subject's DBS device operates under the second level of DBS parameters while the subject performs the at least one elected activity, action or task.

It is to be understood that both the foregoing general description and the following detailed description are merely exemplary of the invention, and are intended to provide an overview or framework for understanding the nature and character of the invention as it is claimed. The accompanying drawings are included to provide a further understanding of the invention, and are incorporated in and constitute a part of this specification. The drawings illustrate various embodiments of the invention and together with the description serve to explain the principles and operation of the invention.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1. Schematic view of a subject undergoing post-surgical DBS adjustment with one embodiment of the invention involving automated instruction of the subject and electronic transmission between the system and the subject's DBS device.

FIG. 2. Schematic view of a subject undergoing post-surgical DBS adjustment with another embodiment of the invention involving electronic transmission between the system and the subject's DBS device.

FIG. 3. Flow diagram of a parameter adjustment suggestion algorithm in some embodiments of the present invention.

FIG. 4. Flow diagram of an artificial neural network of the parameter adjustment suggestion algorithm in some embodiments of the present invention.

FIG. 5. Graphic depiction of display pages displaying test results and scores, one embodiment of a tuning map, as well as one embodiment of a parameter input screen.

FIG. 6. Graphic depiction of one embodiment of tuning maps used to display test results and symptom severity objectively measured by the system and displayed as a scatter plot of symptom severity scores.

FIG. 7. Illustration of the subject screening process to determine if DBS is a viable option for a particular subject.

FIG. 8. Illustration of various embodiments of the DBS parameter programming/tuning process.

FIG. 9. Illustration depicting the ability of bipolar stimulation to shape the electrical stimulation field to avoid activating a side effect region of the brain as opposed to monopolar stimulation which does not allow shaping of the electrical field.

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FIG. 10. One embodiment of an interface device allowing a user to select an activity to perform, and which can change the subject's therapy parameters or settings based on the selected activity.

FIG. 11. Flow chart describing one embodiment of the present invention wherein a subject's therapy device is intelligently and automatically programmed with updated parameters or settings, and where the updated parameters or settings are determined by an algorithm(s) based on measured and quantified motor symptom data.

FIG. 12. Flow chart describing one embodiment of the present invention wherein a subject's therapy device is intelligently and automatically programmed with updated parameters or settings, where an algorithm provides at least two optional sets of parameters or settings and the subject selects one of the groups based on a particular activity in which he or she wishes to engage.

FIG. 13. Flow chart describing one embodiment of the present invention wherein a subject's therapy device is intelligently and automatically programmed with updated parameters or settings, where an algorithm provides at least two optional sets of parameters or settings and the subject selects at least two of the groups based on a particular activity in which he or she wishes to engage, and the groups are combined and optimized to best meet the combined needs of the subject.

FIG. 14. Flow chart describing one programming option embodiment of the present invention wherein an algorithm(s) provides a recommended set of parameters or settings to be reviewed and approved before those parameters or settings are programmed into the subject's therapy device.

FIG. 15. Flow chart depicting a start-to-finish description of the process from the subject exhibiting movement disorder symptoms through treatment and therapy of those movement disorder symptoms, and where the subject's therapy device is programmed automatically and intelligently using an algorithm to determine optimal therapy parameters or settings.

FIG. 16. Flow chart of one exemplary embodiment of an intelligent tuning algorithm for determining or selecting a second set or group of therapy parameters or settings based on measured and quantified movement data and/or movement disorder and various constraints.

DETAILED DESCRIPTION OF THE INVENTION

The present invention relates to methods for semi-automatically and automatically adjusting, or tuning, treatment parameters in movement disorder therapy systems. Semi-automatic adjustment includes providing the clinician, physician or technician with objective, quantitative or semi-quantitative data or measurements related to a subject's movement disorder symptoms, determining desired parameters, and then entering those parameters either semi-automatically or automatically into the subject's device. Semi-automatic and automatic adjustment includes providing objective, quantitative or semi-quantitative data or measurements related to a subject's movement disorder symptoms to an algorithm, determining desired parameters using the algorithm, and then entering those parameters either semi-automatically (i.e., allowing clinician, physician or technician review) or automatically into the subject's therapy device. The present invention further relates to a system for screening patients to determine if they are viable candidates for certain therapy modalities. The present invention still further provides methods of quantifying movement disorder

ders for the treatment of patients who exhibit symptoms of such movement disorders including, but not limited to, Parkinson's disease and Parkinsonism, Dystonia, Chorea, and Huntington's disease, Ataxia, Tremor and Essential Tremor, Tourette syndrome, stroke, and the like. The present invention yet further relates to methods of automatically and intelligently tuning a therapy device using objective quantified movement disorder symptom data acquired by a movement disorder diagnostic device with the therapy setting or parameters to be provided to the subject via his or her therapy device.

The movement disorder diagnostic device, systems and/or methods of the various embodiments of the present invention are used to screen patients, analyze, score, and treat various disorders, and especially movement disorders and mental health disorders. Movement disorders for purposes of this application include but are not limited to Parkinson's disease and Parkinsonism, Dystonia, Chorea, and Huntington's disease, Ataxia, Tremor and Essential Tremor, Tourette syndrome, stroke, and the like. Mental health disorders include, but are not limited to major depression, bipolar disorder, obsessive compulsive disorder, and antisocial disorders. Some of the treatments used for these disorders involve pharmaceutical interventions, fetal cell transplants, surgery, or deep brain stimulation. The efficacy of an intervention is often judged by the intervention's ability to alleviate subject symptoms and improve subject quality of life. The subject on which the system or method is used is a human or another form of animal.

The movement disorder diagnostic device the various embodiments of the present invention are preferably portable. By portable it is meant among other things that the device is capable of being transported relatively easily. Relative ease in transport means that the device can be carried by a single person, generally in a carrying case to the point of use or application. Additionally, relative ease in transport means that the device is easily worn, carried by or attached to a subject. Furthermore the device preferably should be relatively light-weight. By relatively lightweight, preferably the device weighs less than about 3 lbs., more preferably less than about 2 lbs., even more preferably less than about 1 lb., still more preferably less than about 0.5 lbs., still preferably less than about 2 ounces and most preferably less than 0.5 ounces. By being lightweight and further compact, the device should gain greater acceptance for use by the subject. The system for measuring and calculating the severity of the symptoms including external computers preferably weighs less than about 15 lbs., more preferably less than about 10 lbs., still more preferably less than about 5 lbs., even more preferably less than about 2 lbs., and most preferably less than 0.5 lbs. This system more preferably can fit in a reasonably sized carrying case so the patient or their caregiver can easily transport the system.

Another advantage of the systems and methods of the present invention is the ability to determine or calculate the severity of a subject's symptoms in real time. Throughout this disclosure, by real time it is meant that within 30 minutes of sensing and measurement the severity of a subject's symptoms can be calculated or determined. Real time, more preferably means that the subject's symptoms can be calculated or determined in less than about 30 seconds, more preferably in less than about 1 second, even more preferably in less than about 0.1 seconds, and most preferably in less than about 0.01 seconds.

The devices of the various embodiments of the present invention can form part of a system for use by a physician, veterinarian, technician or clinician for analysis or evalua-

tion of a subject's movement disorder; for pharmaceutical research, for adjustment of neurostimulation therapy such as for example deep brain stimulation (DBS) or spinal cord stimulation (SCS), or for delivery of pharmaceutical compounds. Other elements of this system may include but are not limited to receivers, routers, communication devices, processors, displays, drug delivery devices and the like, some of which are described further in various embodiments described in more detail below.

The movement disorder diagnostic device, described in greater detail below, worn, carried by or attached to the subject, contains various physiological or movement sensor(s) used to measure the subject's external body motion and/or other physiological signals from the subject's body. The movement disorder diagnostic device may temporarily store the subject's movement or physiological data in onboard memory and/or transmit this data to an external device. In some embodiments, the movement disorder diagnostic device may directly transmit the data to a centralized database, to multiple databases at the same or multiple locations, or to a cloud-based database where the data can be stored and accessed essentially immediately by authorized users who can analyze and/or further process the data, use it to diagnose or assess the subject's symptoms or disorders, or the like. Additionally, or alternatively, the movement disorder diagnostic device can transmit the movement or physiological data to an external computer device. The computer device, though called a tablet herein, is understood to be any type of device known to those skilled in the art usable for the intended purpose(s) or function(s), including, but not limited to, desktop computers, laptop computers, tablet computers, personal digital assistants (PDAs, "smart" cellular telephones, and the like). The computer device, or tablet, may be provided as part of the present invention's system, but in many embodiments the movement disorder diagnostic device is designed to work with and communicate with such devices of any third-party manufacturer or provider who provides such devices for the intended function or purpose of the present invention. In such cases, a software installation providing the user interface, tuning map capabilities, diagnostic and analysis tools, and the like would simply be installed on the third-party computer device or tablet as software or an application (or "app"), or the interaction with the user(s) can be web based through a web portal. The tuning map is a tool that allows the clinician, physician or technician to review and/or determine the next, or preferably best (optimized) therapeutic settings or parameters for the subject's therapy device, such as a DBS device. In the present invention, tuning maps are used primarily as a tool for review and analysis of the automatically, intelligently generated parameters or settings provided by a tuning algorithm, and not for full-time or regular interaction and programming by a clinician, physician or technician. The tuning maps are described in greater detail in U.S. patent application Ser. Nos. 13/861,790 and 13/153,063, both of which are herein incorporated by reference. Once the next therapy parameters or settings have been determined, a programmer device or unit can be used to communicate directly with the therapy device. Again, the programmer device or unit may be integrated into the subject-worn diagnostic device, a separate unit in and of itself, or may be part of the tablet or computer, and automatically communicates the parameters or settings to the programmer device or unit or directly to the subject's therapy device.

As noted, various embodiments of the present invention may include a sensor for measuring a subject's external body motion. The invention may also include at least one

sensor for indirectly measuring movement metrics. Many types of sensors are known by those skilled in the art for measuring external body motion or providing physiological signals through which body movement information may be derived. External body motion sensors include but are not limited to accelerometers, gyroscopes, magnetometers, resistive bend sensors, combinations thereof, and the like. Preferably, a combination using at least an accelerometer and gyroscope is used. Sensors through which body movement information may be derived include, but are not limited to, electromyogram (EMG), electrooculogram (EOG), electroencephalogram (EEG), electrocardiogram (EKG), or other physiological signals which can directly or indirectly measure movement metrics in the subject may be included if such sensors and signals may be used to sense, detect, measure, and/or quantify the subject's external body motion, or related aspects.

In embodiments where a gyroscope is a sensor of the present invention, the gyroscope functions on the principle of the Coriolis Effect and a capacitive-based sensing system. Rotation of the sensor causes a shift in response of an oscillating silicon structure resulting in a change in capacitance. A typical application specific integrated circuit (ASIC), manufactured using a standard complementary metal oxide semiconductor (CMOS) manufacturing process, detects and transforms changes in capacitance into an analog output voltage, which is proportional to angular rate. The sensor element design utilizes differential capacitors and symmetry to significantly reduce errors from acceleration and off-axis rotations.

In embodiments where an accelerometer is a sensor of the present invention, it may optionally be a dual axis acceleration measurement system on a single monolithic integrated circuit (IC). Such embodiments may contain a polysilicon surface-micromachined sensor and signal conditioning circuitry to implement open-loop acceleration measurement architecture. For each axis an output circuit converts the analog signal to a duty cycle modulated (DCM) digital signal that can be decoded with a counter/timer port on a microprocessor. The dual axis accelerometer is capable of measuring both positive and negative accelerations. The sensor may be a surface micromachined polysilicon structure built on top of the silicon wafer. Polysilicon springs suspend the structure over the surface of the wafer and provide a resistance against acceleration forces. Deflection of the structure is measured using a differential capacitor that consists of independent fixed plates and central plates attached to the moving mass. The fixed plates are driven by 180-degree out of phase square waves. Acceleration will deflect the beam and unbalance the differential capacitor, resulting in an output square wave whose amplitude is proportional to acceleration. Phase sensitive demodulation techniques are then used to rectify the signal and determine the direction of the acceleration. The output of the demodulator drives a duty cycle modulator (DCM) stage through a 32 kOhm resistor. At this point a pin is available on each channel to allow the user to set the signal bandwidth of the device by adding a capacitor. This filtering improves measurement resolution and helps prevent aliasing. After being low-pass filtered, the analog signal is converted to a duty cycle modulated signal by the DCM stage. A single resistor sets the period for a complete cycle (T2). A 0 g acceleration produces a nominally 50% duty cycle. The acceleration signal can be determined by measuring the length of the T1 and T2 pulses with a counter/timer or with a polling loop using a low cost microcontroller.

In preferred embodiments, a single sensor unit comprising at least an accelerometer and a gyroscope may be used. More preferably, a single chip containing both a 3-axis accelerometer and a 3-axis gyroscope (e.g., InvenSense MPU-6000), may be used. The sensor unit preferably not only comprises at least an accelerometer and a gyroscope, but also allows for integration of other sensors external to the sensor unit. Preferably, the accelerometer and gyroscope are each three-axis sensors capable of measuring their respective movements (acceleration and orientation) in each of the three dimensions of movement (X, Y and Z). Each of the accelerometer and gyroscope may output a separate signal for their respective measurements in each axis, and these signals are all converted from analog to digital by a bank of analog-to-digital converters (ADC). The separate ADCs for each axis of the accelerometer and gyroscope allow for simultaneous sampling of each sensor and eliminate the need for an external multiplexer. Preferably the sensor unit as a whole, and the accelerometer and gyroscope in particular are capable of operation with low power consumption. Preferably, the accelerometer and gyroscope are user-programmable such that the user may define an operating range in which the sensors will work (e.g., the accelerometer may be programmed to operate from as low as ± 2 g to as high as ± 16 g, and the gyroscope from as low as ± 250 degrees/second to as high as ± 2000 degrees/second). Some embodiments may include other sensors integrated into the sensor unit as well, for example, a temperature sensor which may be used to monitor the temperature of the sensor unit and ensure it is operating properly and under safe conditions.

The sensor unit further preferably comprises a digital motion processor (DMP) which may perform some preprocessing or processing of the sensor signals using motion-related algorithms. The digital motion processor at least preprocesses and/or processes the accelerometer and gyroscope signals to begin the analysis of the signals and to decrease the processing load on the external processor. Many embodiments may include external or additional sensors that are not housed within the sensor unit, but whose signals are transmitted to the sensor unit for integration with the accelerometer and gyroscope signals for further transmission to external components such as a processor. Such external or additional sensors may include, but are not limited to, force sensors, magnetometers, pressure sensors, bend sensors, combinations thereof, and the like. These external or additional sensors communicate with the sensor unit by means of an auxiliary communications interface. The digital motion processor can integrate the signal(s) from these external or additional sensors along with the accelerometer and gyroscope signals and perform preprocessing or processing of all of the signals together, thus further streamlining the data acquisition process and reducing the workload of the external processor (not shown).

In many embodiments, the movement disorder diagnostic device comprises a kinetic sensor board (or subject worn external sensor). The kinetic sensor board is preferably configured with at least an accelerometer and a gyroscope for quantifying the subject's motion. In some embodiments, the kinetic sensor board comprises at least three gyroscopes and three orthogonal accelerometers, but in more preferable embodiments the three of each sensor are replaced by at least one 3-axis accelerometer and at least one 3-axis gyroscope. The kinetic sensor board also includes a microprocessor and a power interface section.

In many embodiments, the electrical components of the movement disorder diagnostic device further include a

power receiver. The power receiver is the component which receives the electrical charge from the external power source (not shown). The external power source can be any device for supplying power to the movement disorder diagnostic device. In some embodiments, the external power source may be a docking station to which the movement disorder diagnostic device can be connected, attached, docked, or placed into whereby a physical connection is made between the docking station and the movement disorder diagnostic device thus allowing power to be transferred via the physical connection. In other embodiments, the external power source may merely involve plugging the movement disorder diagnostic device into a traditional power outlet. In still other embodiments, the external power source may be an inductive charging mat or pad onto which the movement disorder diagnostic device is placed and power may be inductively transferred between induction coils in the charging mat or pad and the inductive coils in the power receiver of the movement disorder diagnostic device, as described herein. As the power receiver, which may be wireless or wired depending on the embodiment, receives power, it transfers said power to a power manager which controls and directs where the incoming power is delivered. If the movement disorder diagnostic device is not being presently used to measure a subject's body movements and is instead being charged, then the power manager directs the incoming power to the device's battery for charging. It might be possible, though not necessarily preferred, for some embodiments to allow charging while the unit is being used to measure a subject's body motions, in which case the power manager would direct the incoming power to either the battery or to the micro-controller for powering the device's operation for testing. However, it is more preferable for the device, during operation for testing, to be untethered and not in charging mode, and thus the battery would provide power to the unit for usage and testing purposes. The micro-controller or microprocessor is the internal processing unit that directs the other components to function. Thus, the micro-controller or microprocessor directs the power manager on where to direct the power it is receiving from either the power receiver or the battery. An electronic clock operates as commonly known in the art to control synchronization and operation of the device to maximize efficiency of power usage. The radio of the device controls and carries out communications between the device components, and between the movement disorder diagnostic device and external devices (not shown). The radio receives power directly from the power manager. As described herein, the radio may be a Bluetooth® communications device to provide wireless communications with external components such as computers or processors, data acquisition circuitry, internet or cloud-based memory banks or databases, and the like, as well as internal components such as the internal movement disorder diagnostic device memory, microprocessor, and the like. Both internal (between electrical components of the subject-worn sensor device) and external (between the subject-worn sensor device and external components or devices) communications may also be transmitted through wireless, wired, or a combination of both methods. The micro-controller comprises algorithms and protocols for coordinating the operation of at least these internal electrical components, and in some embodiments also for preprocessing or processing sensor data.

The movement disorder diagnostic device of the present invention further preferably comprises a transceiver module, or command module. Preferably the sensor unit and transceiver/command module are enclosed in the same housing

constituting a single unit, though they may be separate units. The transceiver module includes communications electronics, such as a Bluetooth® radio (e.g., BlueGiga WT12) to provide wireless communications with the patient PC, on board memory, a microprocessor (e.g., Silicon Labs C8051F930), and a battery power supply (e.g., Kokam Lithium Power battery) that supplies power to both the transceiver module and one or more sensor modules. The transceiver module may also include a USB port to provide battery recharging and serial communications with the patient PC. The transceiver module may also include a push button input.

In many embodiments, the transceiver/command module contains one or more electronic components such as a microprocessor for detecting both the signals from the gyroscopes and accelerometers. Optionally, the one or more electronic components also filter the kinetic motion signals, and more preferably convert these signals, which are in an analog form into a digital signal for transmission to a remote receiving unit, computer or other similar device. Though, more preferably, the device uses the herein described 3-axis accelerometer and 3-axis gyroscope chip which comprises ADC circuitry and thus outputs a digital signal. The one or more electronic components are attached to the subject as part of the movement disorder diagnostic device. Further preferably, the one or more electronic components can receive a signal from the remote receiving unit or other remote transmitters. The one or more electronic components may include circuitry for but are not limited to, for example, electrode amplifiers, signal filters, analog to digital converter, Bluetooth® radio or other receiver, transmitter or transceiver components, a DC power source and combinations thereof. The one or more electronic components may comprise one processing chip, multiple chips, single function components or combinations thereof, which can perform all of the necessary functions of detecting a kinetic or physiological signal from the electrode, storing that data to memory, uploading data to a computer through a serial link, transmitting a signal corresponding to a kinetic or physiological signal to a receiving unit and optionally receiving a signal from a remote transmitter. These one or more electronic components can be assembled on a printed circuit board or by any other means known to those skilled in the art including but not limited to an ASIC chip. Preferably, the one or more electronic components can be assembled on a printed circuit board or by other means so its imprint covers an area less than 4 in², more preferably less than 2 in², even more preferably less than 1 in², still even more preferably less than 0.5 in², and most preferably less than 0.25 in².

Preferably, the circuitry of the one or more electronic components is appropriately modified so as to function with any suitable miniature DC power source. More preferably, the DC power source is a battery. The most preferred battery of the present invention is lithium powered batteries. Lithium ion batteries offer high specific energy (the number of given hours for a specific weight), which is preferable. Additionally, these commercially available batteries are readily available and inexpensive. Other types of batteries include but are not limited to primary and secondary batteries. Primary batteries are not rechargeable since the chemical reaction that produces the electricity is not reversible. Primary batteries include lithium primary batteries (e.g., lithium/thionyl chloride, lithium/manganese dioxide, lithium/carbon monofluoride, lithium/copper oxide, lithium/iodine, lithium/silver vanadium oxide and others), alkaline primary batteries, zinc-carbon, zinc chloride, magnesium/manganese dioxide, alkaline-manganese dioxide, mercuric

oxide, silver oxide as well as zinc/air and others. Rechargeable (secondary) batteries include nickel-cadmium, nickel-zinc, nickel-metal hydride, rechargeable zinc/alkaline/manganese dioxide, lithium/polymer, lithium-ion and others.

In some preferred embodiments, the system is capable of inductive charging whereby an electromagnetic field is used to transfer energy from a charging mat or pad to the device. Preferably in such embodiments, the charging mat or pad comprises an induction coil that is used to create an alternating electromagnetic field. When the device, also comprising an induction coil, is placed on the charging mat or pad, the device's induction coil draws power from the electromagnetic field created by the charging mat's or pad's induction coil. The device then converts this drawn power from electromagnetic field energy into electrical current and uses this electrical current to charge the device's battery.

Optionally, the data acquisition circuitry is designed with the goal of reducing size, lowering (or filtering) the noise, increasing the DC offset rejection and reducing the system's offset voltages. The data acquisition circuitry may be constrained by the requirements for extremely high input impedance, very low noise and rejection of very large DC offset and common-mode voltages, while measuring a very small signal of interest. Additional constraints arise from the need for a "brick-wall" style input protection against ESD and EMI. The exact parameters of the design, such as input impedance, gain and passband, can be adjusted at the time of manufacture to suit a specific application via a table of component values to achieve a specific full-scale range and passband.

Also optionally, a low-noise, lower power instrumentation amplifier is used. The inputs for this circuitry is guarded with preferably, external ESD/EMI protection, and very high-impedance passive filters to reject DC common-mode and normal-mode voltages. Still more preferably, the instrumentation amplifier gain can be adjusted from unity to approximately 100 to suit the requirements of a specific application. If additional gain is required, it preferably is provided in a second-order anti-bias filter, whose cutoff frequency can be adjusted to suit a specific application, with due regard to the sampling rate. Still yet more preferably, the reference input of the instrumentation amplifier is tightly controlled by a DC cancellation integrator servo that uses closed-loop control to cancel all DC offsets in the components in the analog signal chain to within a few analog-to-digital converter (ADC) counts of perfection, to ensure long term stability of the zero reference.

Further optionally, where analog signals are acquired, such signals are converted to a digital form. This can be achieved with an electronic component or processing chip through the use of an ADC. More preferably, the ADC restricts resolution to 16-bits due to the ambient noise environment in such chips. Despite this constraint, the ADC remains the preferable method of choice for size-constrained applications such as with the present invention unless a custom data acquisition chip is used because the integration reduces the total chip count and significantly reduces the number of interconnects required on the printed circuit board.

Preferably, the circuitry of the sensor board comprises a digital section. More preferably, the heart of the digital section of the sensor board is a micro-controller or processor. The microcontroller or processor preferably contains sufficient data and program memory, as well as peripherals which allow the entire digital section to be neatly bundled into a single carefully programmed processing chip. Still

more preferably, the onboard counter/timer sections are used to produce the data acquisition timer.

Preferably, the circuitry for the one or more electronic components is designed to provide for communication with external quality control test equipment prior to sale, and more preferably with automated final test equipment. In order to supply such capability without impacting the final size of the finished unit, one embodiment is to design a communications interface on a separate printed circuit board (PCB) using the SPI bus with an external UART and level-conversion circuitry to implement a standard serial interface for connection to a personal computer or some other form of test equipment. The physical connection to such a device requires significant PCB area, so preferably the physical connection is designed to keep the PCB at minimal imprint area. Optionally, the physical connection is designed with a break-off tab with fingers that mate with an edge connector. This allows all required final testing and calibration, including the programming of the processing chip memory, can be carried out through this connector, with test signals being applied to the analog inputs through the normal connections which remain accessible in the final unit. By using edge fingers on the production unit, and an edge connector in the production testing and calibration adapter, the system can be tested and calibrated without leaving any unnecessary electronic components or too large a PCB imprint area on the final unit. More preferably, no break-off tabs are required where a pogo-pin test pad design is used allowing the PCB to be tested without breaking apart.

Preferably, the circuitry for the one or more electronic components comprises nonvolatile, rewriteable memory for storing kinematic data, as well as RAM used to store operational data such as the pending mode (i.e., sleep or test mode), period and number of seconds to record data, daily alarm time, amount of time to collect data, and the like. Preferably, enough nonvolatile memory is included to record at least 8 hours of kinematic data, though preferably more. Alternatively, if the circuitry for the one or more electronic components doesn't comprise nonvolatile, rewriteable memory then an approach should be used to allow for reprogramming of the final parameters such as radio channelization and data acquisition and scaling. Without nonvolatile, rewriteable memory, the program memory can be programmed only once. Therefore one embodiment of the present invention involves selective programming of a specific area of the program memory without programming the entire memory in one operation. Preferably, this is accomplished by setting aside a specific area of program memory large enough to store several copies of the required parameters. Procedurally, this is accomplished by initially programming the circuitry for the one or more electronic components with default parameters appropriate for the testing and calibration. When the final parameters have been determined, the next area is programmed with these parameters. If the final testing and calibration reveals problems, or some other need arises to change the values, additional variations of the parameters may be programmed. The firmware of various embodiments of the present invention scans for the first blank configuration block and then uses the value from the preceding block as the operational parameters. This arrangement allows for reprogramming of the parameters up to several dozen times, with no size penalty for external EEPROM or other nonvolatile RAM. The circuitry for the one or more electronic components has provisions for in-circuit programming and verification of the program memory, and this is supported by the breakoff test connector, as well as the pop-pin test pad. The operational

parameters can thus be changed up until the time at which the test connector is broken off just before shipping the final unit. Thus the manufacturability and size of the circuitry for the one or more electronic components is optimized. Most preferably, however, the system is designed to allow for over-the-air programming even once the circuit design has been completed and the circuit has been installed into the movement disorder diagnostic device. In such embodiments, the firmware contains a boot-loading program that, once turned on, looks for programming signals. Thus, such programming signals can be delivered and the device updated, even after manufacture and shipment to a clinic, or even when in the possession of a subject.

Preferably the circuitry of the one or more electronic components includes an RF transmitter and/or an RF receiver, or a RF transceiver. Still more preferably the circuitry of the one or more electronic components includes a Bluetooth® radio system requiring an average of about 42 mA of electrical current to operate. Another feature of the circuitry of the one or more electronic components preferably is an antenna. The antenna, preferably, is integrated in the rest of the circuitry. The antenna can be configured in a number of ways, for example as a single loop, dipole, dipole with termination impedance, logarithmic-periodic, dielectric, strip conduction or reflector antenna. The antenna is designed to include but not be limited to the best combination of usable range, production efficiency and end-system usability. Preferably, the antenna consists of one or more conductive wires or strips, which are arranged in a pattern to maximize surface area. The large surface area will allow for lower transmission outputs for the data transmission. The large surface area will also be helpful in receiving high frequency energy from an external power source for storage. Optionally, the radio transmissions of the present invention may use frequency-selective antennas for separating the transmission and receiving bands, if a RF transmitter and receiver are used on the electrode patch, and polarization-sensitive antennas in connection with directional transmission. Polarization-sensitive antennas consist of, for example, thin metal strips arranged in parallel on an insulating carrier material. Such a structure is insensitive to or permeable to electromagnetic waves with vertical polarization; waves with parallel polarization are reflected or absorbed depending on the design. It is possible to obtain in this way, for example good cross polarization decoupling in connection with linear polarization. It is further possible to integrate the antenna into the frame of a processing chip or into one or more of the other electronic components, whereby the antenna is preferably realized by means of thin film technology. The antenna can serve to just transfer data or for both transferring data to and for receiving control data received from a computer device and/or receiving unit which can include but is not limited to a wireless relay, a computer or a processor system. Optionally, the antenna can also serve to receive high-frequency energy (for energy supply or supplement). In any scenario, only one antenna is required for transmitting data, receiving data and optionally receiving energy. Optionally, directional couplers can be arranged on the transmitter outputs of the electrode patch and/or the computer device and/or receiving unit. The couplers being used to measure the radiated or reflected radio wave transmission output. Any damage to the antenna (or also any faulty adaptation) thus can be registered, because it is expressed by increased reflection values.

An additional feature of the present invention is an optional identification unit. By allocating identification codes-a patient code, the computer device and/or receiving

unit is capable of receiving and transmitting data to several subjects, and for evaluating the data if the computer device and/or receiving unit is capable of doing so. This is realized in a way such that the identification unit has control logic, as well as a memory for storing the identification codes. The identification unit is preferably programmed by radio transmission of the control characters and of the respective identification code from the programming unit of the computer device and/or receiving unit to the patient worn unit. More preferably, the unit comprises switches as programming lockouts, particularly for preventing unintentional reprogramming.

In any RF link, errors are an unfortunate and unavoidable problem. Analog systems can often tolerate a certain level of error. Digital systems, however, while being inherently much more resistant to errors, also suffer a much greater impact when errors occur. Thus the present invention, when used as a digital system, preferably includes error control sub architecture. Preferably, the RF link of the present invention is digital. RF links can be one-way or two-way. One-way links are used to just transmit data. Two-way links are used for both sending and receiving data.

If the RF link is one-way error control, then this is preferably accomplished at two distinct levels, above and beyond the effort to establish a reliable radio link to minimize errors from the beginning. At the first level, there is the redundancy in the transmitted data. This redundancy is performed by adding extra data that can be used at the computer device and/or receiving unit or at some station to detect and correct any errors that occurred during transit across the airwaves. This mechanism known as Forward Error Correction (FEC) because the errors are corrected actively as the signal continues forward through the chain, rather than by going back to the transmitter and asking for retransmission. FEC systems include but are not limited to Hamming Code, Reed-Solomon and Golay codes. Preferably, a Hamming Code scheme is used. While the Hamming Code scheme is sometimes maligned as being outdated and underpowered, the implementation in certain embodiments of the present invention provides considerable robustness and extremely low computation and power burden for the error correction mechanism. FEC alone is sufficient to ensure that the vast majority of the data is transferred correctly across the radio link. Certain parts of the packet must be received correctly for the receiver to even begin accepting the packet, and the error correction mechanism in the computer device and/or receiving unit reports various signal quality parameters including the number of bit errors which are being corrected, so suspicious data packets can be readily identified and removed from the data stream.

Preferably, at a second, optional level, an additional line of defense is provided by residual error detection through the use of a cyclic redundancy check (CRC). The algorithm for this error detection is similar to that used for many years in disk drives, tape drives, and even deep-space communications, and is implemented by highly optimized firmware within the electrode patch processing circuitry. During transmission, the CRC is first applied to a data packet, and then the FEC data is added covering the data packet and CRC as well. During reception, the FEC data is first used to apply corrections to the data and/or CRC as needed, and the CRC is checked against the message. If no errors occurred, or the FEC mechanism was able to properly correct such errors as did occur, the CRC will check correctly against the message and the data will be accepted. If the data contains residual errors (which can only occur if the FEC mechanism was overwhelmed by the number of errors), the CRC will not

match the packet and the data will be rejected. Because the radio link in this implementation is strictly one-way, rejected data is simply lost and there is no possibility of retransmission.

More preferably, the RF link utilizes a two-way (bi-directional) data transmission. By using a two-way data transmission the data safety is significantly increased. By transmitting redundant information in the data emitted by the electrodes, the computer device and/or receiving unit is capable of recognizing errors and request a renewed transmission of the data. In the presence of excessive transmission problems such as, for example transmission over excessively great distances, or due to obstacles absorbing the signals, the computer device and/or receiving unit is capable of controlling the data transmission, or to manipulate on its own the data. With control of data transmission it is also possible to control or re-set the parameters of the system, e.g., changing the transmission channel. This would be applicable for example if the signal transmitted is superimposed by other sources of interference then by changing the channel the computer device and/or receiving unit could secure a flawless and interference free transmission. Another example would be if the signal transmitted is too weak, the computer device and/or receiving unit can transmit a command to increase its transmitting power. Still another example would be the computer device and/or receiving unit to change the data format for the transmission, e.g., in order to increase the redundant information in the data flow. Increased redundancy allows transmission errors to be detected and corrected more easily. In this way, safe data transmissions are possible even with the poorest transmission qualities. This technique opens in a simple way the possibility of reducing the transmission power requirements. This also reduces the energy requirements, thereby providing longer battery life. Another advantage of a two-way, bi-directional digital data transmission lays in the possibility of transmitting test codes in order to filter out external interferences such as, for example, refraction or scatter from the transmission current. In this way, it is possible to reconstruct falsely transmitted data.

The computer device and/or receiving unit of various embodiments of the present invention can be any device known to receive RF transmissions used by those skilled in the art to receive transmissions of data. The computer device and/or receiving unit, by way of example but not limitation, can include a communications device for relaying the transmission, a communications device for re-processing the transmission, a communications device for re-processing the transmission then relaying it to another computer device and/or receiving unit, a computer with wireless capabilities, a PDA with wireless capabilities, a processor, a processor with display capabilities, a desktop computer, laptop computer, tablet computer, smart phone, and combinations of these or like devices. Optionally, the computer device and/or receiving unit can further transmit data both to another device and/or back. Further optionally, two different computer devices and/or receiving units can be used, one for receiving transmitted data and another for sending data. For example, with the movement disorder diagnostic system of the present invention, the computer device and/or receiving unit of the present invention can be a wireless router, which establishes a broadband Internet connection and transmits the physiological signal to a remote Internet site for analysis, preferably by the subject's physician. Another example is where the computer device and/or receiving unit is a PDA, computer, tablet or cell phone, which receives the physiological data transmission, optionally re-processes the infor-

mation, and re-transmits the information via cell towers, land phone lines or cable to a remote site for analysis. Another example is where the computer device and/or receiving unit is a computer or processor, which receives the data transmission and displays the data or records it on some recording medium, which can be displayed or transferred for analysis at a later time.

The digitized kinetic or physiological signal is then transmitted wirelessly to a computer device and/or receiving unit. This computer device and/or receiving unit allows the subject wide movement. Preferably, the computer device and/or receiving unit can pick up and transmit signals from distances of greater than about 5 feet from the subject, more preferably greater than about 10 feet from the subject, even more preferably greater than about 20 feet from the subject, still even more preferably greater than about 50 feet from the subject, still even more preferably greater than about 200 feet from the subject, and most preferably greater than about 500 feet from the subject. The computer device and/or receiving unit is used to re-transmit the signal based in part from the movement or physiological signal from the movement disorder diagnostic device wirelessly or via the internet to another monitor, computer or processor system. This allows the clinician or monitoring service to review the subject's movement or physiological signals and if necessary to make a determination, which could include modifying the patients treatment protocols.

Optionally, the system of the present invention includes some form of instruction, which can be in written form on paper or on a computer monitor, or on a video. Optionally, a video is used which instructs the subjects to perform a series of tasks during which their kinetic motion and/or EMG and/or other physiological signals related to their motion can be measured. Since the system of the present invention is preferably used in the subject's home, a video giving directions and/or describing various tasks to be performed by the subject is included with the system. The video may be accessed or viewed for example but not by way of limitation through use of video tape, DVD, podcast as part of computer software provided, through the internet, or the like. The directions could include but are not limited to instructions on how to don the device, how to turn the device on, and the like. The description of various tasks could include but is not limited to exercises which are typically used by a technician, clinician or physician to evaluate a subject with a movement disorder including but not limited to hand grasps, finger tapping exercises, other movements and the like. One embodiment of a video includes the technician, clinician or physician looking into the camera, as they would a patient, and instructing them on device setup, instructing the patients through each of the tasks to be performed, providing verbal encouragement via video after a task, and asking subject's to repeat a task if it was not completed. Preferably, these video clips are edited and converted to MPEG or other similar file types either automatically or using editing software. For movement disorders such as Parkinson's disease preferably the technician, clinician or physician instructs the user through multiple tasks as per the UPDRS, TRS, or similar scale guidelines including but not limited to rest tremor, postural tremor, action tremor, all bradykinesia tasks (including but not limited to finger taps, hand grasps, and pronation/supination tasks), and/or rigidity tasks. More preferably, if the video is linked to the user interface software, the software will automatically detect if a subject has performed the requested task and provide feedback through the video to either repeat the task or continue to the next task. Still more

preferably, once the user has setup the device, it will continually record the subject's movement data (including before and after any directed video tasks), be able to quantify the severity of the subject's symptoms during activities of daily living, and communicate that information with the clinician and subject through interface software, video, or the like.

The present invention includes various methods of measuring and scoring the severity of a subject's movement disorder. These methods include a number of steps which may include but are not limited to measuring a subject's external body motion; transmitting wirelessly a signal based in part on the subject's measured external body motion; receiving the wirelessly transmitted signal; downloading data from memory; and scoring the severity of a subject's movement disorder based in part on the wirelessly transmitted or downloaded signal. Optionally, an electromyogram of the subject's muscle activity and/or other physiological signals may be obtained and used in part to score the severity of the subject's movement disorder.

Several preferred embodiments of the present invention include a trained scoring algorithm to determine and provide objective scoring from movement data acquired by the movement disorder diagnostic device. The trained scoring algorithm in part comprises a mathematical model or quantitative representation, used to process kinematic features computed from the movement data and may include some of those steps known to those skilled in the art. In some embodiments of the present invention, the scoring may be done on a continuously variable scale of 0-4 with scores substantially similar to or predictive of scores that would be given on the Unified Parkinson's Disease Ratings Scale by an expert clinician. ("Expert clinician" for the purposes of this application is taken to mean a doctor, nurse, researcher, or other medical or scientific professional trained for and sufficiently experienced in the task of interest, e.g., motor function assessment using the UPDRS, or DBS programming.)

The present invention also preferably includes DBS parameter control methods and tuning algorithms for determining and setting the DBS parameters used to deliver DBS therapy to the subject. In many embodiments, the parameter control methods and algorithms utilize a system of tuning maps, or other parameter display or visualization methods or tools, for DBS programming. Although the term 'tuning map' is used throughout this application, it is intended that tuning maps include any such method or tools that may be used for displaying therapy parameters or settings for human review and/or analysis. As noted herein, in the present invention, the tuning maps are preferably only utilized for optional clinician, physician or technician review of the second levels or optimized levels of therapy parameters or settings. The tuning maps utilized by the present invention are a tool used for recording DBS parameters, the subject's response to stimulation at those parameters, and allowing a clinician, physician or technician to optionally or periodically review parameters and settings. Preferably, when utilized, the tuning map is a two-dimensional representation of a three-dimensional graph or display of data.

The subject is first screened and determined to be a viable candidate for DBS therapy, and then has at least one DBS lead surgically implanted into his or her brain. The screening process preferably involves providing the subject with a diagnostic device for monitoring and assessing the subject's movement disorder systems. Generally, a DBS therapy system will include one or more implanted leads with each lead having one or more electrodes. These leads are con-

nected to a pulse generator, which generally is also implanted with the leads. The pulse generator can be implanted in the cranium or in some cases wiring from the leads will be threaded down the subject's neck and the pulse generator will be implanted or embedded in the subject's upper chest. The pulse generator will run on a battery, which can in some cases recharged through techniques such as inductive coupling. The DBS therapy system can be adjusted generally through communication between a programming module or unit and the impulse generator. Such a system, as an example, is described in U.S. patent application Ser. No. 12/818,819, filed on Jun. 18, 2010, which is hereby incorporated by reference. Other systems as known or later developed in the art can also be adjusted with the devices, method and systems of the present invention. Thus, because of the highly invasive nature of therapies such as DBS, requiring surgical implantation of the therapy device, the present invention provides the screening capabilities to ensure the subject would benefit from such a therapy before undergoing the costly and onerous surgical procedure.

The subject may utilize this device at home and during other normal life activities as well, or in a clinical setting, and during the performance of motor and cognitive tests, and while taking his or her prescribed medications for treatment and management of the movement disorder symptoms. The device monitors and records the occurrence of these symptoms and then analyzes the data by means of algorithm(s) designed to account for the multitude of variables including demographic information (age, gender, weight, blood pressure, physical activity, medication use, disease duration, Hoehn & Yahr, marital status/caregiver support, patient expectations, and the like), the type of anticipated DBS therapy (DBS target, unilateral/bilateral implant, constant current versus constant voltage, and the like), non-motor response (UPDRS parts I and II scores, cognition and quality of life assessments, neuropsychology tests, and the like), motor response (UPDRS parts III and IV scores), sensor recordings of symptom occurrence and severity, response to medication, and the like. Generally, subjects who respond favorably to typical medications tend to respond well to DBS therapy. The algorithm analyzes all of the data and makes a determination as to whether the patient is likely to be a good candidate for DBS therapy. The algorithm may optionally employ any one, or a combination of, statistical models currently known to those in the art, including, but not limited to linear and non-linear classification methods such as logistic regression, artificial neural networks, k-means clustering, and the like. The algorithm may output the determination in different ways. In some embodiments, the algorithm may provide a binary output indicating that the subject is or is not a good candidate for DBS, in other words a report or display that indicates a Go/No go result on whether the subject would or would not benefit from such therapy. In other embodiments, the algorithm may provide a percent likelihood of favorable DBS outcomes for the subject. Still other embodiments may present the multitude of data described above in a chart or graph format in order to allow a clinician, physician or technician to review the data and results and confirm or deny that the subject would benefit from such therapy. By way of non-limiting example, various embodiment may employ a visual display that depicts a pie chart wherein the pie chart is populated showing the measured and quantified occurrences of a symptom(s) such as tremor throughout the day. Any other method known to those skilled in the art for displaying the test results and optionally the input data, or combinations of both, can be envisioned for presenting the data to a clinician,

physician or technician for optional review of the screening viability determination. Additionally, the algorithm may provide suggested DBS lead placements in the subject's brain based at least in part on the symptoms and side effects the subject experiences, their severity, and the like. If the subject is determined to be a favorable candidate, the clinician then initiates the process for DBS therapy, which begins with the surgical implantation of at least one DBS lead into the subject's brain. The main purpose of the screening process is to provide a pre-surgical indication of whether the subject would benefit from DBS therapy. This minimizes the likelihood of needless surgery and risk for the subject, as well as time, cost, and resources utilized.

For initial DBS tuning, it may be preferable to perform a monopolar review, which is using a single DBS contact in the monopolar configuration. However, bipolar, or greater, review may be accomplished as well with the present invention. A single DBS lead has several contact points, or electrodes, which can be used to administer the electrical stimulation. Typically, a DBS lead has at least one ground contact, and at least 3 battery contacts for delivering the electrical stimulation, though fewer or more battery contacts may be included. Once the lead, or leads, is implanted into the subject's brain, the subject then undergoes an initial programming session. During the initial programming session, a clinician enters a set of initial test variables (as described above), and administers the electrical impulse to the subject's brain. Typically, the initial test parameters are set and fixed, and then in later iterations the amplitude (voltage or current), as well as other variables are gradually increased or otherwise changed. The results of the impulse may be recorded in a tuning map for optional review indicating the effect which the given set of parameters had on the subject's symptoms, feelings, and the like, and are simultaneously communicated to the tuning algorithm(s). The subject is generally awake for the programming sessions and gives feedback to the clinician regarding any sensations or effects that the subject experiences. The results may include any sort of measured, observed, or calculated response, or combinations thereof including, but not limited to, sensor recordings (for quantifying symptoms as described above), patient responses and perceptions, clinician observations, and clinician scores (e.g., UPDRS, MDS-UPDRS, and the like). In some embodiments, it is possible that the tuning process may be entirely automated such that the tuning map is populated entirely by, and/or the algorithm(s) is supplied with sensor recordings and/or measured and quantified motor symptom data of the subject's response to the DBS therapy and no human observation or calculation is required. Based on the results of the initial test parameters, the tuning map is populated, and the tuning algorithm(s) changes the parameters to more effectively address the subject's symptoms, and the process is repeated. Preferably, in such embodiments, the tuning map is populated simultaneously with supplying the measurements to the algorithm(s), and the tuning map remains a hidden or dormant feature that can optionally be accessed for review of the parameters and settings. Typically, the end result of the tuning process is to optimize the effectiveness of the therapy (i.e., decrease the severity and occurrence of symptoms as much as possible) while minimizing the volume of activated brain tissue, but the particular goals and needs of the subject will dictate exactly what the desired result is for each subject.

Many embodiments of the present invention employ an intelligent system programming capability that greatly decreases the amount of clinician "guess-work" involved in

selecting the iterations of DBS parameter values by providing an expert system that efficiently determines appropriate DBS settings. Similar to above, for the first postoperative programming session, the system performs an automated monopolar survey. The subject may wear a motion sensor unit comprising sensors for measuring movement, and performs motor assessments at various DBS settings. Stimulation is incrementally increased from zero at each contact until symptoms stop improving as measured by the motion sensor unit, perceptions, clinician observations or scores, or the like, or until side effects appear as measured by the motion sensor unit, the clinician, or the patient. In many embodiments, adjustments will be based on current rather than voltage since it is the amount of current delivered to a specific target that determines the functional response. For constant current IPGs, the current amplitude will be set directly and for constant voltage IPGs, the voltage amplitude will be set based on the required current and impedance measured on the electrode. Preferably, the system is capable of operating in either constant current or constant voltage modes, depending on the clinician's preference and the needs of the particular subject. The monopolar survey helps determine the functional anatomy around the DBS lead site and narrows the search space for determining an optimal set of programming parameters. A therapeutic window will be defined as the region in which a patient exhibits optimal symptomatic benefits without side effects. This therapeutic window will be valuable at the initial postoperative programming session as well as all future adjustment sessions for determining the current amplitude when side effects begin to occur on each contact. This therapeutic window is then used to define a side effect region. The system includes internal electric field modeling to determine how this side effect region can be avoided, possibly by shaping the electric field with a bi- or tripolar configuration or altering the pulse width. Bipolar or tripolar configuration refers to the simultaneous delivery of electrical impulses from two or three, respectively, contacts on the DBS lead to shape the electrical field delivered to the subject's brain. For monopolar stimulation, current falls off proportionally to the distance from the negative electrode contact. For bipolar stimulation, current decreases proportional to the square of the distance to the negative contact, but increases by the square of the distance between the negative and positive contacts. Efficient stimulation algorithms are used to find a set of parameters that optimize for efficacy while minimizing side effects and battery usage. The patient and/or clinician will be able to give higher weight to a given item, parameter, or symptom (e.g., tremor severity) that may be most important to him or her. Many clinicians are ignorant of the battery voltage of the IPG battery and therefore unaware that a slight increase of the stimulation amplitude above the battery voltage will activate voltage doubler or tripler circuits in the IPG, significantly increasing battery drain and shortening battery life by half. The algorithms will automatically avoid increasing voltage above the battery voltage unless necessary for finding a therapeutic window. After the automated monopolar survey is completed and a patient-specific functional map is developed during the initial postoperative programming visit, subsequent programming adjustments will be much simpler and faster.

Basing the algorithm(s) on functional, rather than structural, anatomy has several advantages. First, most DBS programmers are not imaging experts and may not have the wherewithal to correctly interpret complex anatomies. More importantly, the therapeutic mechanisms of DBS are largely unknown. The optimal stimulation location differs across patients and is based on functional rather than structural

anatomy. Therefore, the system will be individualized to each subject's response and be far superior to recently developed DBS programming aids, which rely on anatomical assumptions, imaging, and statistical modeling to estimate the electric field at various anatomical targets.

The intelligent system programming capability takes the results of the initial test parameters and automatically populates the tuning map while simultaneously providing these results to an algorithm(s). The system's tuning algorithm(s) analyze these results and provide the next iteration of DBS parameters. These provided parameters or settings may be reviewed and possibly edited by the clinician, physician or technician, or may be automatically entered into the subject's therapy device thus programming the therapy device to operate according to those settings. Effectively, the system provides optimized DBS parameters or settings which either are automatically implemented, or may act as a guide for the clinician in setting the IPG for the next iteration of testing (particularly for the initial programming session). This reduces the clinician's need to perform the analysis and determine which parameters to change and how much to change them. The clinician may have the option to edit or elect any one or combination of the system's suggested parameters for the next iteration. This intelligent programming system may be performed in-clinic during a traditional programming appointment, whereby the system provides the suggested DBS parameters or settings to the clinician and the clinician can review and edit the parameters or settings through the software which in turn adjusts the settings and parameters on the IPG. Alternatively, the intelligent programming may take place remotely whereby the system automatically and intelligently provides optimized parameters or settings and programs them into the subject's therapy device, or communicates any data and suggested DBS parameter settings to the clinician, physician or technician who is located some distance away while the subject remains at home, and the clinician then reviews and elects to approve or edit the suggested settings, which are sent to the IPG to update the parameters of the DBS administered to the subject, or still further the clinician may instruct the diagnostic device to perform another iteration of testing, rather than editing the suggested parameters or settings herself. Where the clinician, physician or technician does perform the option or periodic review, the data (e.g., parameters and settings, test results, and the like) may be displayed for his or her review in a tuning map as described. Because the clinician is not required to be physically present at the time of programming, the system instead may rely on system and user reports, which are sent to the clinician. These reports may be made by the system sending reports to the clinician, video and/or audio conferences between the subject and the clinician, the subject keeping a medication diary to report medication schedules and symptom occurrence and severity, transmitted tuning maps, and the like.

Some embodiments of the present invention provide the clinician, physician or technician the ability to manually make the determination as to what therapy parameters to use with the subject's therapy device based in part on the tuning map or data corresponding to the subject's measured and quantified motor symptoms or based on other data measured by the movement disorder diagnostic device from the subject. Again, this is solely an optional review designed for periodic analysis of the algorithm(s)' function and to ensure that the subject's needs are being met adequately and safely. The present invention further optionally allows the clinician, physician or technician the ability to review recommended second level therapy parameters before or after those

therapy parameters or settings are entered into the therapy device and to change those recommended settings.

Additionally, other embodiments of the present invention may include intelligent remote programming methods and algorithms utilizing a database, which may be cloud-based in some embodiments, allowing for remote DBS adjustments being possible without the subject even leaving his or her home, and without clinician involvement. The system may provide automated, objective scoring and tuning algorithms to take the programming expertise out of the hands of a clinician, and enable high-quality programming to all DBS recipients, regardless of proximity to or availability of expert programmers. Ideally, the subject would not have to travel to the clinic or facility for programming unless a problem was detected requiring personal medical care. Such embodiments necessarily include the integrated system programming capability whereby the software directly communicates with the hardware to set the DBS parameters according to the values entered into the software in order to enable the periodic or optional review by the clinician, physician or technician. For such optional or periodic review, rather than in-clinic programming sessions, the software would communicate with the DBS hardware, which is located remotely, implanted into the subject. In these embodiments, the implanted device would perform the intelligent system analysis as above, creating a suggested set of DBS parameters for the particular subject, and then securely communicates all the data and suggested parameters to a centralized or cloud-based database, which analyzes all the information and then sends programming commands to the subject's IPG to change the DBS settings. This database and intelligent remote system allow for continued or repeated DBS tuning without requiring the patient to travel to a clinician, and without requiring a clinician to spend the time analyzing subject data. The benefits of such a system go beyond the convenience of minimizing travel time and access to clinicians and include the ability to deliver such reports at virtually any time (the subject and clinician are not tied to a particular appointment time and window, and the clinician can review any data at any time if desired), continuous and repeated monitoring of the subject and system's status, and delayed monitoring whereby results of changed parameters can be monitored later, which is particularly useful for symptoms that may not react to changes rapidly (i.e., during the normal clinical appointment time period).

Further, the system preferably includes the capability to perform hardware diagnostic tests remotely of any and all of the individual units or modules used with the present invention, including the subject's therapy device (e.g., implanted pulse generator), movement disorder diagnostic device, programmer unit, and the like. The hardware in such embodiments is able to monitor and/or periodically interrogate the system to detect changes in system conditions such as battery status, electrode impedance, and the like. The system then sends the results of these diagnostic tests back to the clinician who can monitor them to determine if a problem arises requiring the subject to return to the clinic for adjustments, repairs, or other such purposes. Further still, such embodiments using remote programming and control may include medication delivery systems as well. Such delivery systems include a drug reservoir for holding and storing medication, and infusion pump for delivering said medication from the reservoir to the subject. Such embodiments may determine based on recorded signals that the subject's symptoms are particularly severe or occurring more frequently.

Some embodiments may optionally include a closed-loop or semi-closed loop drug titration system. In such embodiments, when the subject's prescribed medication is initially taken or delivered, the system then monitors the subject's symptoms. The system continues to monitor the symptoms until and after it detects that the subject's symptoms increase, maintain, or only very slightly decrease in severity and rate of occurrence. In a semi-closed loop system, a report, warning, alert, or some other signal would then be sent to the subject or to the subject's clinician. In such case, the subject could take more medication, or the clinician could send a command for an integrated drug delivery pump to administer another dose. In a close-loop system, upon detection of the above indicators, the system would automatically administer an additional dose of medication through an integrated medication delivery pump. In either case, the system is capable of substantially continuous symptom monitoring to determine when the subject is experiencing an increase in symptom activity, ineffective medication delivery, or a wearing off of medication in order to administer additional medication to control the subject's symptoms. Further, such embodiments must also be able to monitor and detect the occurrence of side effects arising from the medication, and to stop administering medication when such side effects begin to manifest.

The above systems, devices, and methods are further contemplated for use in treating various mental health disorders, particularly major depression, bipolar disorder, and obsessive compulsive disorder. In particular treatment of mental health disorders would benefit from the patient screening system and method for determining if the patient would benefit from DBS therapy in dealing with his or her disorder, as well as the integrated and intelligent programming systems and methods for both in-clinic and remote programming of the DBS device once implanted.

The Applicants herein incorporate the following U.S. Pat. Application Nos. by reference: Ser. No. 11/082,668 filed Mar. 17, 2005, and Ser. No. 11/432,583 filed May 11, 2006.

FIG. 1 illustrates the therapeutic device programming (or "tuning," or "parameter settings adjustment") process with one embodiment of the invention. A subject 1 has a therapy device (not shown), which in the illustrated case is a therapy device for the treatment of a movement disorder, such as an implanted DBS device. Subject 1 wears a movement disorder diagnostic device comprising a sensor unit 2 and a command module 3. The sensor unit 2 comprises at least one sensor(s), preferably a physiological or movement sensor(s), such as accelerometers and/or gyroscopes (both not shown), or other similar sensors, as well as a transmission system (not shown). In one preferred embodiment, the sensor unit 2 comprises three orthogonal accelerometers and three orthogonal gyroscopes, or more preferably at least one 3-axis accelerometer and at least one 3-axis gyroscope. Preferably, where the at least one sensor is an accelerometer and/or a gyroscope, these are micro-electrical-mechanical (MEMS) accelerometers or gyroscopes. In a preferred embodiment, a single chip containing both a 3-axis accelerometer and a 3-axis gyroscope is used, rather than using separate sensors. An example of such a combined sensor chip is the Invensense MPU-6000. The transmission system may be wired or wireless, and may communicate via any medium and any transmission protocol known to those skilled in the art. In the illustrated embodiment, the sensor unit 2 communicates sensor readings to a command module 3 over a small flexible transmission cable 4, though this transmission could also be conducted wirelessly. In the more preferred embodiment where both the sensor unit 2 and

command module 3 are combined in a single housing constituting the entire movement disorder diagnostic device, the two modules may be integrated into the same electronics thus eliminating the need for wired or wireless communication between separate modules. In the illustrated embodiment, the sensor unit 2 is worn on the middle phalange of the middle finger and the command module 3 is worn on the wrist using a wristband, though the placement of the sensor unit 2 and command module 3 may vary depending upon the symptoms of the movement disorder. Alternate placements could include other fingers, the ankle, foot, shoulder, or elsewhere on the trunk of the body or on any part of any extremity. While the illustrated embodiment shows the sensor unit 2 and the command module 3 as having separate enclosures, permitting for a lighter-weight sensor unit 2 that is easily worn on the finger, in alternate embodiments the sensor unit 2 and command module 3 may be integrated into a single enclosure. In such embodiments where the sensor unit and command module are combined into a single enclosure forming the movement disorder diagnostic device, all components of each unit are enclosed or attached to the single enclosure, including, but not limited to the units and modules themselves, any power supply and communication electronics required for operation.

The command module 3 may provide numerous functions including, but not limited to supplying power to the sensor unit 2, storing data in memory, transmitting data. Preferably, it is controlled by firmware in processor, for example an Analog Devices ADuC7020 processor. The data acquisition (DAQ) section samples finger sensor unit data at 128 Hz for each of the six channels. Optional onboard memory preferably provides at least 12 hours of data storage. Some embodiments do not contain internal storage, but rather transmit the data substantially in real-time to a receiver unit 5, a centralized database (not shown) or to a cloud-based database (not shown). Still other embodiments utilize onboard, temporary data storage as well as substantially real-time data transmission to a receiver unit 5, centralized database (not shown) or a cloud-based database (not shown). A lithium-based battery provides at least 12 hours of continuous use and is rechargeable by a computer through a LEMO, or similar connector, to USB connector cable. The command module 3 also integrates a membrane switch label (not shown) with LED indicators for power and charging (not shown). Three membrane switches inside the label (not shown) provide on/off control and two subject diary inputs. The command module 3 may perform rudimentary signal processing, such as filtering and analog-to-digital conversion, on the movement signals received from the sensor unit 2 before transmitting the movement signals to a receiver unit 5. The receiver unit 5 may be of any type known to those skilled in the art, and useful for receiving data from the sensor unit and making it available to a clinician, physician or technician on computer device 6. The computer device 6 will be referred to as a tablet (or tablet computer), but it is meant to be understood that it may be any such similar device, including, but not limited to desktop computers, laptop computers, tablet computers, personal digital assistants (PDAs), "smart" cellular telephones, or the like. This transmission may be wired, but is preferably wireless, advantageously providing the subject the greater comfort and convenience of being untethered as well as endowing the system with enhanced safety and portability. The wireless link frees subject motion, which allows unimpeded and accurate assessment of subject symptoms. In an operating room, a small untethered system has the added benefits of reducing further subject discomfort and not impeding clini-

cal traffic. A wireless system, which is not directly connected to any source of AC power, has the added benefit of reducing or eliminating risk of electrical shock. Preferably, the wireless transmission is robust and operates in a frequency band designated for hospital or clinical-setting use. Preferably, the wireless transmission radio is a Bluetooth radio operating in the 2.4 GHz band. More preferably, radio transmission occurs over the Wireless Medical Telemetry Service (WMTS), dedicated by the FCC to wireless medical equipment used in hospitals, which comprises the frequencies 608 to 614 MHz, 1395 to 1400 MHz and 1429 to 1432 MHz. Preferably, radio communication is accomplished using a mix of traditional heterodyning techniques along with newer software radio techniques. For example, receiver structure consists of a band select function of either 608-614 MHz or 1395-1432 MHz, followed by a heterodyning operation. The lower frequency band undergoes one frequency translation while the upper undergoes two frequency translations. For the low band (608-614 MHz) the signal is translated to 44 MHz where it is then sampled by an A/D converter and demodulated in the "sampled" domain. The high band is translated first to the lower frequency band (608-614 MHz) and processed in the same fashion. The software radio demodulation approach accommodates many different data rates and modulation formats and advantageously allows future radio upgrades to be implemented simply by changing the signal processing program opposed to necessitating an entire analog hardware redesign. The low band transmit signal is a simple frequency source modulated with appropriate information. For the high band transmit signal, the same signal used for the low band transmit signal is mixed with a high frequency signal to produce the desired output. For transmitter operation, the signal processing hardware generates the modulating signal for all different signal formats and data rates. The signal processing hardware outputs a modulating signal input to an oscillator circuit that creates the modulated transmit signal. The modulated signal, for the high band, uses the low band modulator and translates that signal to the proper operating frequency. Since the modulator is the same for both low and high bands it ensures the same signal quality regardless of operation band. Since the radio is a transceiver (two-way link), the design can serve as a master or slave; thus the same design can be employed in the command module 3 as well as in the receiver unit 5.

Data may also be collected in an on-board memory contained within the command module 3. Such onboard or internal memory may be used for temporary storage so that the data may be saved and then downloaded to the tablet computer 6 later, advantageously allowing the subject to wear the movement disorder diagnostic device comprising sensor unit 2 and command module 3 for more prolonged symptom monitoring. Additionally, or in the alternative, the onboard memory may be used to temporarily store the movement data and provide a backup in the event of halted, corrupted, or otherwise incorrect transmission of the data from the movement disorder diagnostic device comprising sensor unit 2 and command unit 3 to the receiving unit 5.

The receiver unit 5 may be, and is preferably integrated into some larger system—for example, it may consist of a wireless receiver, such as a Bluetooth receiver, integrated into a device such as a laptop or tablet computer, a cellular phone, etc.—or it may be a separate device built into an enclosure. However, in the illustrated embodiment, the receiver unit 5 is connected to a tablet computer 6 via one of the USB ports (not shown, in a dongle-style connection that advantageously eliminates a cable), is powered thereby,

and comprises a radio frequency transceiver capable of 2-way radio frequency communication with the command module 3. Power regulation and USB-based data transmission protocols may be among any known in the art. The receiver unit 5 may be, in some embodiments, an off-the-shelf Bluetooth USB adapter dongle.

The tablet computer 6 is used to collect data transmitted from the control module 3, allow user inputs to store and track motor performance and therapy device parameter settings, and provide clinicians with real-time symptom quantification feedback. The tablet computer 6 of the illustrated embodiment may be any computing device with a user interface 7, including a smart phone, PDA, laptop computer, desktop computer, iPhone, iPad, or the like. Preferably, the tablet computer 6 is lightweight and portable, allowing for its easy transport within an operating room, and includes a touch screen. In some embodiments, the tablet computer 6 may be equipped with a clip or hanger (not shown) for easy mounting to, for example, an operating room pole.

The user interface 7 may be visual, preferably comprising a touch screen, or it may be an audio interface that accepts and transmits spoken commands. In addition, or alternatively, the user interface may be used to provide an automated testing protocol to the subject 1 by providing instructions to the subject 1 on which movement disorder test(s) to perform, and how to perform them. In other embodiments, the subject may be instructed on which tests to perform and/or how to perform them on a separate display, not on the tablet's user interface. The user interface 7 preferably provides several key components and an overall software wrapper. First, it preferably provides a main menu (not shown) to access all software features including a subject database (not shown), the tuning assistant software, which runs the therapy device parameter settings tuning algorithm, and software for automatically generating clinical reports following tuning sessions. Next, it preferably provides a module to view real-time motion data transmitted by the movement disorder diagnostic device comprising sensor unit 2 and command module 3, helping ensure proper setup and communication prior to clinical therapy device programming. The user interface 7 also preferably communicates with the system registry to store system parameters and clinician preferred settings. Finally, a help menu (not shown) with troubleshooting guides and frequently asked questions is preferably included.

Subject data management is an important aspect of clinically-used embodiments of the present invention. Preferably, the format of the software used with the system is designed for a high volume subject database. Any database known in the art may be used but is preferably one which scales well to accommodate thousands or tens of thousands of subjects. Preferably, the database has fields for subject history, including the subject's surgery dates, a running list of the subject's clinical sessions (past and/or future scheduled), the subject's primary physician, neurologist, medication dosage, etc. Preferably, the subject is also programmed with the ability to import e-mails and other documents into the subject history, and to export a standardized patient information sheet (reporting). Preferably, the database is programmed so as to permit all stored subject information to conform HIPAA guidelines for patient privacy and confidentiality.

A separate programmer device 8 is used in some embodiments by the clinician, physician or technician to program the subject's therapy device, that is, to adjust the therapy device's parameter settings. The separate programmer device 8 may be a separate device from the tablet 6, but more

preferably the tablet is capable of providing both functions. Whether the separate programmer device **8** is separate or integrated, it communicates with the subject's therapy device (e.g., DBS device), and transmits the desired therapy parameters to the subject's therapy device such that the therapy device operates under the transmitted parameters. Where the programmer device or unit is separate from the tablet, preferably communication between them is by wireless methods as described above. In all embodiments, the programmer device or unit preferably communicates wirelessly with the subject's therapy device.

In addition, or alternatively, the movement disorder diagnostic device comprising the sensor unit **2** and command module **3** may transmit to a server or group of servers constituting a centralized database, such as with cloud computing whereby the data resides on such server or group of servers and can be accessed at the point of testing or some remote location for review by a clinician, physician or technician. Further, the tablet and/or programmer device or unit may also communicate with and transmit data to a centralized database or cloud-based database in order to store the preferred therapy parameters for the particular subject, as well as information regarding the testing and tuning protocols used to arrive at the desired parameters. All the data that is transmitted and stored on a centralized database or cloud-based database is intended to be made accessible to the clinician, physician or technician for later review, for reference when the subject requires additional treatment or tuning, and to be readily available to other clinicians, technicians or physicians who the subject may receive treatment of any variety from and who might need to access the data in order to properly and safely attend to the subject. For example, a subject may reside in one state through the summer months and receive tuning of the therapy device and treatment there, but then may travel to another state for winter months and require similar attention there. The database storage of data allows clinicians, technicians or physicians in both states to readily obtain access to the subject's data and to provide the appropriate care to the subject. In all exchange of information that occurs in the above example and in all other embodiments of the present invention, it is important that information be exchanged securely and in ways that do not improperly disclose a subject's identity. Because of this, in certain preferred embodiments, all personal information of a subject is stored securely at a remote database and is accessible only through a secure network connection wherein both the database and connection protocol are compliant with standards required by the health insurance portability and accountability act (HIPAA). Often, this will require encryption of the data to eliminate the possibility that the data can be read by a third party and many preferred embodiments of the present invention include the use of data encryption.

As indicated in the above example, various embodiments of the present invention involve sending a movement disorder monitoring device home or to another remote location with a subject to be used for movement disorder testing away from a physician's or clinician's place of practice. This step occurs after programming of the display unit as described above. Once the subject arrives home, the movement disorder monitoring device is placed in the subject's home where it may be powered by either a single or multiple on-board batteries or by another power source in the subject's home such as a standard 120 volt alternating current outlet. Once in the home the display unit will, at intermittent times selected by the programming physician or clinician, alert the subject of the need to perform certain movement

disorder evaluation tasks. At these times, the display unit may produce a sound, provide a visual alert on its display screen, or a combination of both as a way to alert the subject. In response to the alert the subject will place at least one sensor on his or her extremity (ies) as instructed by the display unit and will proceed to follow other instructions provided regarding how to properly complete certain tasks used to evaluate the severity of the subject's movement disorder symptoms. In certain embodiments, the subject may be video recorded by the camera of the display unit so that a physician can at a later time verify that the tasks were indeed correctly completed. Preferably, the subject will also answer other questions at this time regarding a subject's self-assessment of his or her symptoms and the subject's adherence to and use of treatments prescribed by the subject's physician or another clinician. Such questions may consist of inquiries related to the subject's perception of the present severity of the subject's symptoms, the subject's most recent dose of pharmaceutically-based treatment, the subject's activity level throughout the day, and other similar pertinent information that is desired to be known by the physician to help better understand a subject's symptoms. As noted above, however, in certain other embodiments, the display unit may not be programmed to alert a subject, but instead may simply be left available for a subject to input data regarding his or her symptoms or to select movement disorder assessment tasks to perform from among various options according to the subject's personal preferences and schedule as well as the subject's own subjective view of the severity of his or her symptoms.

By way of a more specific example of the above situation, a physician or other clinician may see a subject for treatment of PD and the subject may indicate to the physician that his or her symptoms associated with PD vary greatly throughout the day. To better understand the diurnal fluctuations of the subject's symptoms, the physician may program a display unit to intermittently alarm over a certain duration of time and to instruct the subject to, for example, wear the sensor on the subject's right hand while performing hand grasping exercises, finger tapping exercises and to simply wear the sensor for a period of time while resting to examine the severity of a subject's rest tremor. The programmed display unit is then sent home with the subject with at least one accompanying sensor and sensor dock. After arrival at home, the subject follows the intermittent instructions provided by the display unit (or records his or her own symptoms according to personal preference if the display unit is not programmed) and then returns the display unit along with its collected data to the physician's office, allowing the physician to then better understand the subject's symptoms and to more effectively treat these symptoms by taking steps to more accurately minimize daily symptom fluctuations.

In still other embodiments, a physician or other clinician may meet with a subject who exhibits symptoms of a movement disorder and, before beginning treatment of the subject, send a programmed movement disorder monitoring device home or to another remote location with the subject where the subject uses the device to record his or her symptoms before returning the device and accompanying data to the physician or other clinician. By examining a subject's symptoms before beginning treatment, a physician or other clinician can establish a "baseline" against which to monitor changes in the severity of a subject's symptoms as treatment methods are changed and/or as time passes and movement disorder symptoms worsen or improve.

It will further be recognized that in certain preferred embodiments, it may be undesirable for a subject to first

meet with a physician or other clinician prior to undergoing movement disorder monitoring using the movement disorder monitoring device of the present invention at a remote location. Instead, it may be preferable to directly ship or deliver a movement disorder monitoring device to a subject at a remote location and instruct the subject to return the movement disorder monitoring device upon completion of the assigned monitoring period. In this way, both the subject and the physician can avoid the cost and inconvenience associated with a preliminary appointment if it is desirable that a symptom baseline first be established for a subject prior to a physician's meeting with the subject or if such monitoring is desired to be conducted to assess, for example, the efficacy of ongoing treatment.

The duration of time during which the movement disorder monitoring device may remain in a subject's home or other remote location with the subject and intermittently provides instructions to the subject and/or allows the subject to "journal" his or her symptoms can vary depending on the nature of the subject's movement disorder and the specific data desired by the programming physician or clinician. In certain preferred embodiments, the movement disorder monitoring device may be in a subject's home for a relatively short period of time such as 8 or even 6 hours. In still other embodiments it may be desirable that the movement disorder monitoring device remain in a subject's home or other remote location for a period of days. In yet other embodiments, it may be preferable to have the movement disorder monitoring device remain in a subject's home for a number of weeks. It will further be appreciated by those of ordinary skill in the art that keeping the movement disorder monitoring device of the present invention in a subject's home for even longer periods of time, such as months or even years, may be desired as a way to better understand a subject's symptoms over a greater duration of time and would provide powerful ways to examine trending in a subject's symptoms over a period of months, years, or even decades.

Certain embodiments of the present invention in which longer movement disorder monitoring times are especially valuable include embodiments in which the movement disorder monitoring device is used to monitor a subject's response to treatment directed at stopping or slowing the onset of a movement disorder. Continuous monitoring over times ranging from months to years to even decades is advantageous with the use of treatment directed at stopping or slowing the onset of a movement disorder because movement disorder symptoms can be very subtle when a movement disorder is in its early stages of development and the objective information provided by the movement disorder monitoring device of the present invention allows a physician or other interested individual to accurately and objectively monitor small changes in symptom severity over time. Thus, by allowing monitoring over a longer period of time, a physician or other clinician or even researcher could use the movement disorder monitoring device of the present invention to collect objective data regarding a subject's disease progression and, hence, the efficacy of a given treatment at stopping or slowing a subject's disease progression. It will be further noted that use of the device and method of the present invention in combination with treatment directed at stopping or slowing the progression or onset of a movement disorder is intended to optionally include the use of the movement disorder monitoring device with a broad scope of pharmaceutical agents and/or other treatments directed at stopping or slowing the progression or onset of a movement disorder. Neuroprotective drugs pro-

vide one specific example of a compound that can be used to stop or slow the progression or onset of a movement disorder. Briefly stated, neuroprotective drugs include a broad set of compounds that serve to eliminate or reduce neuronal death in the central and/or peripheral nervous systems, hence eliminating certain movement disorder symptoms that can follow neuronal death and stopping progression or onset of a movement disorder disease. By way of specific example, in the case of PD certain drugs have been and are being examined and may be found to be effective at eliminating or reducing death of a subject's dopamine producing neurons, and the efficacy of such drugs over extended periods of time could be objectively monitored using the device and method of the present invention as a means to collect and review movement disorder symptom data over extended periods of time. By way of example, neuroprotective drugs that have been and are being examined for their potential in stopping or slowing the progression of movement disorders such as PD include drugs such as selegiline, riluzole and lazabemide. It is to be understood that the scope of the present invention is intended to cover the use, with the device and as part of the method of the present invention, of these drugs as well as other neuroprotective drugs that may yet be discovered or are currently under investigation.

It may further be appreciated that just as the device and method of the present invention may be used in combination with treatments designed to slow or stop movement disorder onset or progression, the device and method of the present invention may also be used to monitor the efficacy of certain restorative treatments. By restorative treatments it is meant treatments that are directed at restoring the natural function, or partially restoring the natural function, of the part or parts of a subject's body, the failure of which acts as the source of the subject's symptoms. This differs from the other traditional treatments discussed above, such as administration of levodopa to a subject with PD or giving pramipexole to a subject with essential tremor, in that the goal of a restorative treatment is to eliminate dependence on external treatment of symptoms and instead focus on addressing the source of the problem itself. For example, in many instances such treatment may consist of gene therapy directed at restoring the function of certain cells critical to the development and/or symptoms of a subject's movement disorder. In other instances such treatment could consist of implantation of encapsulated cells from an external source, with the encapsulated architecture designed to provide immunoprotection to the encapsulated cells and allow the encapsulated cells to fulfill the role of the subject's native cells that no longer function correctly. By way of specific example but not limitation, such treatment may comprise the replacement of dopamine producing cells in the substantia nigra region of a subject's brain in a subject diagnosed with PD, using dopamine producing cells encapsulated to prevent a subject's immune response to the implantation. In all of the above, it is to be understood that the scope of the use of the device and method of the present invention in combination with treatments directed at preventing or slowing movement disorder onset or progression or at restoring function of native tissue extends beyond the basic examples provided above and includes all uses and approaches included within the general areas of treatment mentioned.

Various embodiments of the present invention also involve the subject returning the movement disorder monitoring device to a physician's office or another location after use. The method of return can vary depending on a subject's preferences and/or circumstances. In certain instances, the

movement disorder monitoring device may be returned in person. In other instances it may be preferable to return the device by mail or courier service. It will be apparent to those of ordinary skill in the art that the movement disorder monitoring device of the present invention is further amenable to return by any other method commonly used to deliver, ship, or transfer items between parties.

Upon return, or prior thereto, to the physician's or clinician's office or other desired location, the data collected from the subject using the movement disorder monitoring device is preferably transferred from the device to a different location for further analysis and review. Transfer of the data from the movement disorder monitoring device can occur in a number of ways including wireless transfer (e.g. IEEE 802.11 or Bluetooth as mentioned above) from the device, removal of the device memory module and subsequent upload to a separate location, and the use of various data exchange and transfer utilities including the internet or other communication system such as a local area network. Portable data storage devices such as USB flash drives and compact discs can also be used to transfer data from the movement disorder monitoring device for further analysis and review. It will further be noted that it is often the case that combination of the above-noted data transfer methods will be most advantageous.

By way of example, in certain preferred embodiments of the present invention, a USB flash drive may be used to initially transfer data from the display unit of the movement disorder monitoring device. The USB flash drive can then be plugged into a personal computer or other device that has a connection to the internet. The internet connection can then be used to transfer data to a remote database where further analysis and review of data can be performed. In still other preferred embodiments, data transfer may be accomplished by directly connecting the display unit of the movement disorder monitoring device to the internet such as using a Wi-Fi connection or a category 5 data cable connection to allow direct transfer of data from the display unit to a remote database. It will be clear to those of ordinary skill in the art that various other methods beyond the specific examples just provided exist through which movement disorder symptom data can be transferred from a device such as the display unit of the present invention and it is intended that these other methods be included within the scope of the present invention.

In the embodiment illustrated in FIG. 1, the subject 1 performs a movement disorder test according to instructions. In an in-clinic setting, a clinician, physician or technician may orally or visually instruct the subject on which test(s) or task(s) to perform, and/or how they are to be performed. Optionally, and more preferably, these instructions may be provided by an instructional video clip displayed on the user interface 7 of the tablet computer 6, or on a separate display device (not shown) advantageously providing the subject with a standardized visual aid to mirror while a test is conducted and data is collected. Such a system implemented in software and provided through user interface 7 ensures the same clinical examination protocol is used during subsequent test or programming sessions either in-clinic or in a non-clinical setting, advantageously allowing clinicians to more repeatedly and objectively track symptoms and assuring inter-subject data correspondence. In one embodiment, testing includes (or may be limited to) three types of tremor tasks (resting, postural, and kinetic) and three types of bradykinesia tasks (finger tapping, hand grasps, and pronation/supination). Either alternatively or in addition, testing may include various gait/balance tasks as well. The move-

ment disorder diagnostic device, or more specifically the sensor unit 2 of the diagnostic device, collects data, which is sent to command module 3 for transmission via radio link to a receiver unit 5. The processor of the tablet computer 6 processes the movement data to extract kinematic features, which are then fed into a trained scoring algorithm implemented as a software algorithm in the tablet computer 6. The trained scoring algorithm may output a score, which may then optionally be displayed on the user interface 7. A tuning algorithm of tablet computer 6 then computes suggested therapy device parameters or settings based at least in part upon the current therapy device parameter settings and the collected movement data and/or the quantified score computed therefrom. One example of a tuning algorithm for computing the suggested therapy device parameter settings is illustrated in FIG. 3. In many embodiments, the various algorithms utilized are able to analyze the measured and quantified movement data in correlation to the therapy parameters or settings being provided, determine if those parameters or settings or causing side effects to occur, and be able to anticipate similar parameters or settings that might cause the same side effect to occur. In such embodiments, the algorithm would then know to avoid the parameters or setting that are likely to cause the side effect, and thus avoid including them in the provided set or group of parameters and settings.

Once suggested parameter settings adjustments are computed by the tuning algorithm(s), the adjustments or new parameters or settings may optionally be displayed on the user interface 7. The tablet computer 6 may communicate directly with the programmer device or unit 8, wired or wirelessly, to adjust the parameter settings. The therapy device may be reprogrammed wired or wirelessly, and typical implanted therapy devices are enabled with means of wireless transcatheter reprogramming.

The alternate embodiment of the invention depicted in FIG. 2 advantageously combines the receiver unit 5, the tablet computer 6, the user interface 7, and the separate programming device 8 into a programmer unit 9 having improved user interface 10, which is preferably a touch-screen interface. The command module 3 of the movement disorder diagnostic device transmits movement data acquired by the sensor unit 2 of the movement disorder diagnostic device, as described above, to the programmer unit 9, where the movement data is analyzed and parameter settings adjustments are computed. The parameter settings may then automatically updated, with the programmer unit 9 interfacing with the therapy device (not shown) directly to reprogram the therapy device's parameter settings.

Preferably, the touch screens of tablet computer 6 and programmer unit 9 permit a user to interact with the user interface 7 or the improved user interface 10 using a large sterile stylus (not shown), or the user's finger.

In alternate embodiments of the invention, quantification of movement disorder symptoms may be performed using a different form of movement measuring apparatus. In one such example, a webcam built into the tablet computer 6 or programmer unit 9, or a video camera or set of multiple cameras connected thereto, view the subject 1 performing the motion disorder test and feed video data into the tablet computer 6 or programmer unit 9 where, for example, machine vision algorithms measure the motion of the limbs of the subject with respect to time according to any method known in the art. Such a method may consist, for example, in determining marker points along the limb of the subject in order to gauge relative motion, and such a method may be assisted by applying more visible markers (not shown) on

various points on the limb of a subject 1, such as is common with motion capture technology. In such case, the need for the movement disorder diagnostic device comprising sensor unit 2, with its accelerometers and gyroscopes, and command module 3, may be obviated.

In one embodiment, an initial programming session will be carried out when the device is first provided to the subject according to a protocol comprising the following steps. The clinician will assess all motor task baseline scores. The clinician will check electrode impedance for wire damage. The clinician will record medication dosages, which information preferably includes information relevant to the subject's present level of medication, such as time and dosage of last medication administration. The clinician will select programming motor tasks, and in conjunction with the programmer device or unit 8, 9, the subject will repeat the series of motor tasks for each stimulation setting. The DBS settings or parameters and corresponding scores for each chosen motor task will then be entered preferably automatically by the system, and where the clinician has the ability to switch between tasks for entering data and selecting which DBS parameters are fixed: frequency, current, pulse width, waveform type, contact setup (mono, bi, tripolar), and the like. Finally, the tuning algorithm(s) will assess all motor tasks post-programming, and the clinician will have the option of reviewing, and possibly editing the resultant parameters or settings. Preferably, for the clinician's optional review and editing, the system will provide the ability to enter DBS settings or parameters and scores completely either with a finger or stylus on a touch screen, and/or with a mouse and/or with a keypad or keyboard using the tab key to switch between data input fields. Preferably, the system provides three data input modes for clinician review and editing: (1) enter stimulation and score information, click update, enter next measurement; (2) enter information and display updated tuning map; (3) use stylus/finger/mouse to click on the tuning map for the appropriate measurement, with a new input box appearing to enter score/side effects/notes. Optionally, the initial programming session may be performed without direct interaction with the tuning map, but rather where the algorithm(s) populate the tuning map, and the clinician, physician, or technician engages the optional review function to view the tuning map and/or the parameters or settings chosen by the algorithm(s) in order to approve or deny them based on the subject's needs and the results from those therapy settings.

It is advantageous in some embodiments for the system to provide, or to permit the clinician performing the programming session to enter a large number of variables in order to provide a complete assessment of the DBS tuning. The following is a representative list of information that may be entered in each programming session: (1) general subject information, including patient ID, whether the subject's DBS implant is unilateral or bilateral, implant electrode site location and side; (2) motor evaluations performed during tuning, including (a) tremor: rest, postural, kinetic, (b) bradykinesia: finger tap, hand grasp, pronate/supinate, (c) rigidity: elbow/knee, head/neck, (d) leg agility: heel tapping, (e) rising from chair: with arms crossed, (f) posture, (g) gait: walking quality, (h) postural stability: pull back; (3) motor scores, in the form of integer scoring from 0 (no severity) to 4 (extremely debilitating); (4) DBS settings or parameters, including, but not limited to (a) contact: cathode/anode, monopolar, bipolar, tripolar, multiple channels with fractionalized control, waveforms, current steering, different waveforms, interleaving multiple waveforms, etc., (b) stimulation parameters including amplitude (in volts), fre-

quency (in Hz), current (in amps), pulse width (in microseconds), the type of waveform of the stimulation impulse, (c) side effects and/or capsule effects, including (i) motor effects, such as worsening of symptoms, dyskinesias, facial pulling, (ii) non-motor effects, such as blurry vision, soft or slurred speech, sweating, headache, tingling (transient/non-transient), fatigue, sense of euphoria, paresthesia, and/or (iii) new or atypical side effects and update list of notable effects.

FIG. 3 shows merely one optional example of a tuning algorithm used for computing suggested parameter settings adjustments. This basic tuning algorithm utilizes symptom severity data, detected stimulation induced dyskinesias (SID), and clinical inputs such as clinician defined improvement percentage (CDI %) to compute suggested stimulation parameter settings. Based on the typical clinical description, several constraints reduce the number of degrees of freedom in the tuning algorithm. During DBS programming, the clinician or the algorithm may utilize any subset of the motor task mentioned previously to evaluate motor performance. The average tremor score (ATS) is computed for the set of tremor tasks and the average bradykinesia score (ABS) is computed for the set of bradykinesia tasks utilized for a given iteration. This reduces the number of symptom severity outputs from a maximum of three to one for each symptom. Dyskinesia is either "on" or "off."

Recording symptom severity before the therapeutic device is turned on obtains baseline. In the case of DBS adjustment, the best monopolar electrode contact is determined by finding the contact that provides the largest therapeutic width, i.e., the largest change in supplied voltage from when a clinical benefit is noticed to when side effects occur. This is accomplished by fixing stimulation pulse width to initial settings, for example 60 μ s, frequency to 130 Hz, selecting one contact, and then stepping the voltage amplitude in small increments of approximately 0.2 V. The procedure is repeated for each contact. The contact that provides the largest therapeutic width is selected. With the pulse width (60 μ s) and frequency (130 Hz) set to typical values, the clinician or the algorithm then sets the amplitude to the lowest voltage that provides a significant decrease in symptoms. If a satisfactory result is not achieved, pulse width or frequency may also be increased. This can be a time consuming iterative process that must be completed several times over the first few months as microlesioning heals and requires a compensatory increase in stimulation amplitude to maintain clinical benefit. In various embodiments, the invention includes a sensitive tool, implemented in software and accessed through user interface 7 or improved user interface 10 to detect the instant of clinical benefit as voltage amplitude is increased and the instant any stimulation induced dyskinesias are detected. Use of the invention as a sensitive measure of clinical benefit onset and side effect occurrence advantageously ensures the contact with the greatest therapeutic width is selected.

Once the contact width is selected, the initial parameter settings adjustment iteration may be completed with literature-defined settings of 60 μ s and 130 Hz stimulation. Amplitude is set to 0.2 V initially, and then modified by the clinician or algorithm in subsequent iterations. After each stimulation parameter change, the clinician may use the user interface 7 or the improved user interface 10 to guide the subject through motor tasks, or, in some preferred embodiments, the device will automatically provide instructions to the subject via a display to provide such guidance. The tuning algorithm output provides a suggested parameter direction output after each motor task evaluation by utilizing the movement disorder quantification algorithm. The inven-

tion thereby maximizes clinical benefit by minimizing tremor and bradykinesia, minimizes adverse effects of stimulation-induced dyskinesias, and minimizes current consumption to maximize battery life. Thus, one objective function is to minimize the sum of average tremor score (ATS) and average bradykinesia score (ABS), known as the summed motor score (SMS) 11. This objective is achieved in the tuning algorithm by continuing to increase stimulation in the same direction as long as SMS is decreasing. A higher SMS corresponds to worse motor symptoms. A second constraint is that stimulation induced dyskinesias (SID) should not occur. If they are detected, the direction of the parameter change is reversed. Another system constraint is the minimization current consumption. This is accomplished by allowing a clinician defined improvement percentage (CDI %) 12 and considering any changes of less than 5% in SMS to be insignificant. When these conditions are met, the current parameter level is maintained 13 due to the SMS goal being achieved and with consideration given to diminishing clinical returns, in order to maximize battery life. Once optimized amplitude has been achieved or reaches 3.6 V, the clinician may adjust pulse width or frequency utilizing the same algorithm. The chances of the feedback system settling into local minimums are reduced by ensuring several of the settings are set at clinically accepted levels for the initial iteration and making only moderate adjustments as required.

While DBS programming frequently entails stepping through small incremental changes, this process can be wastefully time-consuming if the motor symptom response of the subject indicates larger changes are required. The implementation of an artificial neural network 14 to output suggested stimulation parameters minimizes programming iterations to reduce surgical and outpatient tuning session time.

Preferably, the artificial neural network 14 used in the tuning algorithm used by some embodiments of the present invention is trained with recorded clinician-made stimulator parameter changes in response to motor symptom severity changes during stimulator programming to minimize required iterations while still utilizing objective symptom severity measures to optimize performance. In this way, the algorithm takes clinician experience into account. Experienced clinicians are generally successful in quickly reducing the number of potentially successful parameter settings for tuning DBS systems. An expert clinician is capable of recognizing severe motor symptoms and modifying a parameter by a larger magnitude, then when the symptom is less and only fine-tuning is required. The present invention is therefore capable of quantitatively detecting motor symptom severity and suggesting a parameter change that approximates or mirrors the parameter change that would be made by an expert clinician.

Artificial neural network 14 may be implemented, for example, with the MATLAB Neural Network Toolbox offline using resilient backpropagation batch training. Inputs to the neural network may include current and previous stimulation settings 15 and motor responses.

FIG. 4 illustrates a two-layer network structure consisting of one hidden layer 17 with four neurons using “tansig” transfer functions and one output layer 18. As neural networks may fall into local minimums when being trained, each network is preferably trained three times with randomized initial weights and biases and the best training results are selected. Additionally, early stopping improves network generalization. Data is separated into training and generalization sets to ensure trained networks produce accurate

results both when the training set is reapplied and also when it is generalized to new data. Preferably, training data is collected from multiple patients. To ensure that the system generalizes to new patients, network generalization can be tested by training the system using a jackknife “one left out” method. Using such a method, the neural network is trained using data from, for example, only nine of ten subjects. Data from the nine subjects in the training set is then reapplied to the trained network to ensure good correlations while data from the “left out” subject is used to test generalization. The method is repeated, leaving out each subject one time. For each training and generalization set, both the mean squared error (MSE) and R-squared values between the clinician-made stimulator parameter changes and those output by the system for each stimulation parameter are calculated. The MSE and R-squared for all training and generalization sets are averaged. Preferably, the system achieves normalized MSE values of less than 10% and R-squared values of greater than 0.8 to show substantial agreement between system-suggested stimulation parameter changes and clinician-made stimulation parameter changes.

Preferably, separate data sets and acquired, and separate neural networks are trained, for the surgical and outpatient scenarios. Preferably, the data used to train the algorithm averages the experience of multiple expert clinician programmers.

Preferably, the tuning algorithm comprises a neural network as illustrated in FIG. 3, but it might instead or in addition comprise one or more of adaptive continuous learning algorithms, linear quadratic Gaussian control, Kalman filtering, and model predictive control.

When the tablet computer 6 is connected to the Internet or similar communications network, wired or wirelessly, it may therefore transmit subject data to remote systems, allowing general practitioners to conduct DBS programming remotely, minimizing travel for a subject 1 who lives far from a DBS implantation center or suitable programming clinic, so long as the subject 1 is equipped with the movement disorder diagnostic device comprising sensor unit 2 and command module 3 and means of programming and/or making parameter settings adjustments to his or her therapy device (including DBS implant).

FIG. 5 depicts a series specific display pages corresponding to reporting score provided for optional or periodic review by a clinician, physician or technician in various embodiments of the present invention. These examples of methods of reporting scores with visual displays associated with each display stage of the test process are merely exemplary, and many variations of the display method, as well as the labeling of the display, are envisioned. One example of displaying a score for optional or periodic review is the expandable menu view 19, where the user (i.e., clinician, physician, or subject performing self-testing away from the clinician) is presented with a list of the different types of movement disorders or movement disorder symptoms, which may or may not have been measured in a given test. In the portrayed example of this expandable menu view 19, the movement disorder symptoms that may be selected include tremor, bradykinesia, rigidity, gait and/or balance disturbances, and the like. The user is then given the option of expanding the results for each of those movement disorders or movement disorder symptoms through a series of levels (i.e., hand then to left or right), in order to view the score that was determined for each particular disorder or symptom in the indicated portion of the subject’s body. By way of clarification and example, the subject shown in menu view 19 received a score of 1 during rest for the symptom

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of tremor in the left hand, and a 2 for the pronation/supination task for bradykinesia in the left hand.

Another optional review display method, which may be independent or used in conjunction with the expandable menu view is the tuning map **20**. A tuning map **20** is generated for each task that the subject is directed to perform, symptom, or side effect and depicts the severity of the symptoms measured in each sensor that is used for the given task. Each task, symptom or side effect that is performed may be represented on a different tab (i.e., tab A, tab B, tab C, where the lettered tabs in FIG. 5 either correspond to a task, symptom or side effect). Alternatively, the tab letters may be replaced by other labels or indicators, or even the name, or abbreviation thereof, of the task, symptom or side effect. Further alternatively or in addition, the different tabs may represent combinations of tasks, symptoms (e.g., averaged results of multiple symptoms), and/or combinations of side effects. In this particular embodiment, the amplitude **21** at which the test was performed is measured in volts and indicated on one vertical axis of the tuning map **20**. Also in this particular embodiment, the contact being used to provide therapy or stimulation is indicated on the horizontal axis **23**. However, in many embodiments, the axes may represent any other test parameter or setting used, and in some preferred embodiments, one of the axes (typically the horizontal) may represent different groupings (see FIG. 6) of test therapy settings or parameters. In such embodiments, for example, the axis would provide a representation of a group (e.g., 1 representing group 1), which would comprise a predetermined set of therapy parameters or settings being tested, where the grouping may include variations of any of the above described therapy parameters or settings, such as contact, current, frequency, waveform, polarity, pulse width, and the like, as well as combinations thereof. Some embodiments further allow the condition to use varying or comparative settings within the groupings. For example, it may be decided to combine the settings in such a manner to provide weighted scores where one symptom, task, or side effect is given a greater weight than another, but they are combined to create a single weighted map. Similarly, therapy settings or parameters may be gradually increased or decreased as symptoms are continuously measured, rather than providing measurements or assessments at discrete amplitude levels. This allows a greater degree of freedom and versatility in defining the test settings instead of being limited to one or two test parameters per test. Such groupings decrease the amount of testing time required for tuning, and thus reduce cost to the subject or insurance company, as well as opportunity cost of the clinician's time spent with a single subject. The calculated or estimated score **22** is depicted on a vertical axis of the tuning map **20** as well, and is indicated as various cross-hatch patterns. Alternatively, the score may be represented by colors, shapes, or any other indicator. Each individual box that is shown represents a test performed **24**. Preferably, the tuning map **20** is shown on a color display (not shown) for optional review and the severity of the symptom is indicated by color. In this drawing, the colors are represented by different types of shading or cross-hatching rather than by the preferred color. Each column in the tuning map **20** represents a different contact on the DBS probe. Therefore, each individual test box **24** depicts the results of performing a task while administering DBS at a prescribed voltage amplitude **21** and provides both a severity of the symptom that was detected or measured by virtue of the color (represented by the cross-hatching), which also correlates to a given motor score. Additionally, each individual test box **24**

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may be selected, for example by pressing it on a touch-screen device, as representing by the test boxes **24** which are outlined in black. When a test box **24** is selected, the user is able to see a detailed view (not shown) of the statistics and parameters of the test corresponding to that box. In many embodiments, the clinician, physician or technician may be able to add notations to the different parameter or setting groupings, or even to the individual test scores. Such notations may include commentary or other notes regarding the efficacy of the given parameter grouping or score, of side effects that occur, or any other notations the clinician, physician or technician deem necessary.

A variable window **25** may display on the unit as well that allows the user to input various conditions that have an effect on the test and test results. These variables are calculated into the test results and help to give a more accurate calculated symptom score.

Other methods of displaying data corresponding to test therapy settings or parameters and results may also be envisioned and are considered for use with the present invention.

FIG. 6 portrays one example of the optional or periodic review tuning maps **20** or other visual display tool or method for a particular embodiment with amplitude **21** on the vertical axis and groupings of parameters or settings **23** on the horizontal axis, in greater detail. Each task performed, symptom or side effect is represented again by a separate tab with its own tuning map **20**. Though the tabs are labeled as A, B, C and A+B in the figure, in many preferred embodiments the tab may be labeled with a number, the name of the task, symptom or side effect it represents, an abbreviation thereof, or some other label indicating to the clinician, physician or technician what information is represented in the given tab. The amplitude **21** of the voltage at which the test was performed is tracked along one vertical axis of the map **20** for each grouping of parameters or settings **23** on the DBS lead (for this particular depicted embodiment), while the severity of the symptom detected or measured is displayed as a score **22** and correlated to an indicator (e.g., cross-hatching pattern, color, or the like) of each individual test result box **24**. Again, in many preferred embodiments, rather than the labeling the groupings of parameters or settings **23** used to provide stimulation with letters (e.g., W, X, Y, Z in the figure), they may instead be labeled by a grouping number, grouping name, or any other labeling scheme or plan, which indicates to the clinician, physician, or technician which grouping of settings or parameters is represented. Preferably, the groupings are cross-referenced within the software and/or GUI such that a user, clinician, physician or technician may readily and easily be able to see what parameters or settings correspond to the chosen grouping label. The right side **26** of FIG. 6 portrays a new tab **27**, which represents the combination of tabs A and B. This combination tab **27** represents the combination of the tuning maps for tasks A and B, and the combination can be of any mathematical variety such as averaging, weighted averaging, or the like.

The combination **27** is a result of the user selecting those two tuning maps to be combined together and optimized in some mathematical way (e.g., averaging) in order to show the results of how the scores for each task combine in order to optimize the DBS level for treating the subject. In other words, the goal is to minimize the voltage at which the DBS is to be supplied while simultaneously minimizing the severity of the subject's symptoms and/or side effects. Combining the tuning maps for each task allows the user to see a resulting score and select the DBS test parameters,

which are as close to optimal as possible. In a preferred embodiment, the system would be designed to be a closed-loop system, (i.e., for an implanted home-diagnostic and therapeutic device), which would not require extensive, or any, user input, but would perform the optimization automatically.

FIG. 7 depicts the screening process for determining whether a subject is a good and viable candidate for DBS therapy. When a subject begins to experience side effects 35 from medication he or she is taking to treat the symptoms of a movement disorder, the subject or his or her clinician may begin to consider new treatment methods other than simply relying on medication. The clinician may then have the patient undergo a home screening assessment 36 to perform monitoring and recording of the subject's symptoms and test results (motor and cognitive tests) with a system for monitoring and recording those results. Next, preferably the system utilizes predictive tuning algorithms 37 to analyze the many variables and test results in order to make a determination as to whether the subject would benefit from DBS therapy 38. If the determination is that the subject would benefit, then the clinician and subject may decide to undergo DBS surgery and therapy 38. However, if DBS is not a viable option for the subject, then alternative therapies 39 should be considered.

FIG. 8 depicts a method of the present invention utilizing an automated and intelligent DBS (or other therapy) training system. As the subject 40 performs motor and cognitive tests, as instructed via a display 42 (automated), while wearing the movement disorder diagnostic device 41, the system records and analyzes the results of those tests in light of the many variables. The system then populates a tuning map 45, in the background, to show the subject's response to the tested DBS (or other therapy) parameters for each symptom while simultaneously entering the same data into a tuning algorithm(s). As noted herein, the axes 43 of the optional or periodic review tuning map 45 may represent a single test variable, or may represent a grouping of therapy settings or parameters that are used while the subject 40 conducts a movement disorder test(s). The tuning map 45 is populated by scores 47 representing the severity of the subject's 40 symptom or side effect, or some other metric being measured, and the same data is entered into the tuning algorithm(s). The tuning map 45 also helps to map the side effect regions, as well as therapeutic and non-therapeutic regions for optional or periodic review. The tuning algorithm then determines, based on the test results, a set of therapy settings or parameters that are then entered into the subject's therapy (e.g., DBS) device for further testing or for delivering treatment and therapy to the subject.

FIG. 9 illustrates the difference between (a) monopolar and (b) bipolar DBS lead configurations and the ability to activate desired brain areas while avoiding others by shaping the electrical stimulation field. While only mono- and bipolar configurations are depicted, other configurations are envisioned for use with the present invention as well. A monopolar (a) configuration provides electrical stimulation in all directions from the given contact, and that stimulation dissipates as it radiates out from the contact. Bipolar (b) configurations simultaneously provide electrical stimulation from 2 contacts, and depending on the parameters used, can thus alter the shape of the electrical stimulation field. FIG. 9(a) shows a monopolar configuration, which provides electrical stimulation in all directions. Thus, with a monopolar configuration, the stimulation provided activates the desired portion 56, which corresponds to the area of the subject's brain which causes occurrence of a movement

disorder symptom, but also activates a side effect region 55, which causes a side effect to occur. However, with a bipolar configuration, as in FIG. 9(b), the electrical field may be shaped in such a manner so as to activate the desired region 56 corresponding to a symptom and thus treat that symptom, while avoiding the side effect region 55 and thus avoid causing the side effect to occur.

FIG. 10 depicts an exemplary embodiment of a device that can be carried, attached to, or otherwise readily used by the subject. The device 70 can be any variety of interface devices, such as a cellular or smart telephone, PDA, tablet, or the like. In the particular embodiment depicted, the subject can view the device 70 and view, determine and select the type of activity, action or task 78 he or she is about to perform or is performing. The activity list 78 depicts any number of activities, actions or tasks 78, such as activities of daily living that the subject may perform at various points in his or her day. When, for example, the subject knows he or she is going to go for a walk, the subject can merely select the walking activity, and the device 70 would then communicate a set of predetermined therapy settings or parameters to program the subject's therapy device (such as an implanted DBS device). The therapy device would then operate under the newly programmed therapy settings or parameters. The predetermined settings or parameters may be determined and programmed into the device by a clinician, physician or technician before the device is provided to the subject. Alternatively, the system may automatically and intelligently determine preferred settings or parameters based on the activity, action or task being performed, and the measurements acquired while performing said activity, action or task 78. This automated setting or parameter determination may initially be based on past performances of the elected activity, action or task 78, but real-time measurements during each performance of the activity, action or task 78 may allow for real-time setting or parameter needs to be determined and communicated from the device 70 to the therapy device. Thus, whenever the subject selects a desired activity, action or task 78, the device would automatically reprogram the subject's therapy device to provide the appropriate settings or parameters to most effectively allow the subject to perform the elected activity, action or task 78.

Particularly in the intelligent, automated parameter or setting determination embodiments, the subject may be given the option to define a new activity, action or task 82. Such capability would allow the subject to perform a new activity, action or task that he or she has not done before, or to create a new level of a previously defined activity, action or task (such as creating a different activity, action or task 78 in the activity list 74 for mowing the lawn and gardening, as opposed to just yard work, as depicted by way of example in the figure). It may be preferable, for some subjects, to only allow a clinician, physician or technician to define new activities, actions or tasks in which case this feature would either be disabled or removed from the interface or device 70. In any event, the automated, intelligent programming systems would allow the device 70 to take real-time measurements of the subject's movements, analyze the movement data, and determine the best therapy parameters or settings for maximizing the subject's ability to perform the activity, action or task. This maximization of the subject's ability to perform the activity, action or task may correspond to a minimization of symptoms, a minimization of side effects, a combination thereof, or any other combination of possible desired results or outcomes described herein as they pertain to the particularly elected activity, action or task.

Optionally, the device **70** may provide the ability to view and review the settings for an elected activity, action or task **76**. The subject or a clinician, physician or technician may simply select a button **80** corresponding to the settings for a particular activity, action or task, and the therapy parameters or settings would be displayed for review. Depending on the embodiment or the person reviewing the settings, this option may also provide the ability to manually edit the prescribed settings for a particular activity, action or task. Preferably, the ability to edit the parameters or settings is available only to a clinician, physician or technician. The subject may be able to view the settings in order to take notes or to discuss the settings with the clinician, physician or technician while at a remote location such as at home. In light of the various features of the device that upload and store the parameters and settings in a database, either in real-time or on-demand, the activity settings are similarly reported and stored for remote (in time, location, or both) access.

FIG. **11** depicts a flow chart representing a method embodiment of the present invention for tuning the movement disorder diagnostic device with new therapy parameters or settings. This particular embodiment is envisioned to be performed in an automated fashion, by an algorithm(s), with little or no interaction required with a clinician, physician or technician. First, a movement disorder diagnostic device is provided to a subject **90** who has a therapy device, such as a DBS therapy device. The movement disorder diagnostic device is as described herein, but preferably comprises at least one physiological or movement sensor for measuring the subject's external body motion, or some other physiological signal of the subject, where the sensor(s) has a signal related to the subject's motion or other physiological signal. Preferably, the movement disorder diagnostic device is a single unit, though may be multiple units (e.g., separate sensor unit and command/transceiver module), and is adapted to be worn or attached to a portion of the subject's body such that the sensor(s) of the movement disorder diagnostic device are able to measure the movement of that particular portion of the subject's body.

Once the movement disorder diagnostic device has been provided **90** to the subject, and the subject has donned or attached the device, the next step is to instruct the subject **92** to perform at least one movement disorder test(s) while the subject is undergoing therapy from the therapy (e.g., DBS) device, or is under the effects of recently administered therapy therefrom. These instructions may include a list of which test(s) the subject is to perform, directions on how to perform the test, or a combination of both. The instructions to perform a test or tests may be given **92** in person (if in a clinical setting) by a clinician, physician or technician, or more preferably may be provided in an automated or electronic fashion. For example, the instruction to perform the test(s) may be provided automatically to the subject via a video display, or a notification or alert message provided via such a display or perhaps the subject's smart phone, or the like. The instructions are provided **92**, either audibly, visually, or a combination thereof. The subject's performance of the tasks is thus affected by the therapy being provided, or recently provided by the therapy device.

While the subject is performing the at least one movement disorder test(s) as instructed **92**, the step of measuring and quantifying motor symptoms **94** of the subject based at least in part on the signal of the at least one sensor(s). The movement disorder diagnostic device uses the signal of the at least one sensor(s) to provide an objective measurement and quantification of the severity of the subject's motor symptoms while the subject performs the at least one

movement disorder test. The measured and quantified motor symptoms may include specific movement disorder symptoms, side effects from medication and/or therapy, or combinations thereof. The trained scoring algorithms of the movement disorder diagnostic device perform various measurements and calculations to provide this objective quantification of the subject's motor symptoms.

Once the subject's motor symptoms have been measured and quantified **94**, data corresponding to these measured and quantified motor symptoms is entered into an algorithm(s) **96** for automated analysis. Typically, the data corresponding to the measured and quantified motor symptoms is an object score, as described herein, but may be represented in numerous ways and means. Preferably, the quantified movement data is entered **96** directly and automatically into the determination algorithm(s) without the need for any manual human intervention, such as keying in the data.

The determination algorithm(s) then analyze the quantified movement data and determine a second level of therapy parameters or settings **98**. This second level of parameters or settings preferably corresponds to a mode of therapy or treatment that addresses the subject's needs as determined based on the measured and quantified motor symptom data, as well as other data, goals, or objectives. In other words, if the system determines that the subject is experiencing a very strong tremor, such determination being made as a result of the measurement of the subject's movement and quantifying the severity of the tremor, the algorithm would provide a second level of parameters or settings **98** that would reduce or minimize the tremor the subject is experiencing. As noted, the determination of a second level of parameters or settings may be based on any number of constraints or desired results for the subject, not solely the immediate symptom or side effect the subject is experiencing. For example, if the subject's main concern is reducing or minimizing symptom occurrence and or severity, algorithms will take this desired goal into account when determining the second level of parameters or settings. Similarly, the determination of settings or parameters may be based on a desired reduction or minimization of side effects from medication or the therapy. Also, the determination may be made to balance multiple desired results, such as if a slightly higher rate of occurrence of symptoms is acceptable to the subject in exchange for a minimization of side effects. Other examples of desired results that may be used to determine the second level of therapy parameters or settings for all embodiments include, but are not limited to, a therapeutic window (in terms of time or some other factor) in which the subject most positively responds to therapy, battery life, and other such constraints that might be considered in terms of optimizing the therapy parameters or settings. In any determination, the subject and the clinician, physician or technician decide what the initial desired result is, and these desired results, goals, or constraints are programmed into the algorithm for the decision making process. In some embodiments, the desired result, goal or constraints may be edited, either by a clinician, physician or technician, or by the subject, in order to allow the algorithm to most accurately analyze the data in light of the optimal treatment for the subject.

Once this second level of parameters or settings has been determined or provided by the algorithm(s) **98**, the parameters or settings are entered into the subject's therapy device **100**. The parameters or settings are preferably entered automatically via wireless communication, as described herein, between the diagnostic device or computer and the subject's therapy device. As such, the new, second level of parameters or settings is programmed into the subject's

therapy device, and the device then operates according to those new parameters or settings **102** and provides that newly determined course of therapy or treatment to the subject. Thus, the system measures and quantifies the subject's symptoms **94**, automatically determines a second level of therapy parameters or settings that would address those symptoms **98**, the second level of parameters or settings is automatically and wirelessly programmed into the subject's therapy device **100**, and the therapy device implements the second level of parameters or setting to in fact address the subject's symptoms **102**.

FIG. **12** depicts a flow chart representing a method embodiment of the present invention for tuning the movement disorder diagnostic device with new therapy parameters or settings. This particular embodiment is envisioned to be performed in an automated fashion, by an algorithm(s), with little or no interaction required with a clinician, physician or technician. First, a movement disorder diagnostic device is provided to a subject **90** who has a therapy device, such as a DBS therapy device. The movement disorder diagnostic device is as described herein, but preferably comprises at least one physiological or movement sensor for measuring the subject's external body motion, or some other physiological signal of the subject, where the sensor(s) has a signal related to the subject's motion or other physiological signal. Preferably, the movement disorder diagnostic device is a single unit, though may be multiple units (e.g., separate sensor unit and command/transceiver module), and is adapted to be worn or attached to a portion of the subject's body such that the sensor(s) of the movement disorder diagnostic device are able to measure the movement of that particular portion of the subject's body.

Once the movement disorder diagnostic device has been provided **90** to the subject, and the subject has donned or attached the device, the next step is to instruct the subject **92** to perform at least one movement disorder test(s) while the subject is undergoing therapy from the therapy (e.g., DBS) device, or is under the effects of recently administered therapy therefrom. These instructions may include a list of which test(s) the subject is to perform, directions on how to perform the test, or a combination of both. The instructions to perform a test or tests may be given **92** in person (if in a clinical setting) by a clinician, physician or technician, or more preferably may be provided in an automated or electronic fashion. For example, the instruction to perform the test(s) may be provided automatically to the subject via a video display, or a notification or alert message provided via such a display or perhaps the subject's smart phone, or the like. The instructions are provided **92**, either audibly, visually, or a combination thereof. The subject's performance of the tasks is thus affected by the therapy being provided, or recently provided by the therapy device.

While the subject is performing the at least one movement disorder test(s) as instructed **92**, the step of measuring and quantifying motor symptoms **94** of the subject based at least in part on the signal of the at least one sensor(s). The movement disorder diagnostic device uses the signal of the at least one sensor(s) to provide an objective measurement and quantification of the severity of the subject's motor symptoms while the subject performs the at least one movement disorder test. The measured and quantified motor symptoms may include specific movement disorder symptoms, side effects from medication and/or therapy, or combinations thereof. The trained scoring algorithms of the movement disorder diagnostic device perform various measurements and calculations to provide this objective quantification of the subject's motor symptoms.

Once the subject's motor symptoms have been measured and quantified **94**, data corresponding to these measured and quantified motor symptoms is entered into an algorithm(s) **96** for automated analysis. Typically, the data corresponding to the measured and quantified motor symptoms is an object score, as described herein, but may be represented in numerous ways and means. Preferably, the quantified movement data is entered **96** directly and automatically into the determination algorithm(s) without the need for any manual human intervention, such as keying in the data.

In this particular embodiment, the determination algorithm(s) then analyze the quantified movement data and determine at least two optional groups of therapy parameters or settings **110**. These optional groups of parameters or settings each preferably correspond to a different or separate mode of therapy or treatment that addresses the subject's needs as determined based on the measured and quantified motor symptom data, as well as other data, goals, or objectives, and each specifically addressing a separate need, goal, desired outcome, or constraint. For example, the system may detect that the subject is experiencing a high degree of rigidity along with speech problems caused by the electrical stimulation. In this particular embodiment, the system would provide two optional groups of parameters or settings **110**, one which more directly addresses the symptom of rigidity, and one that more directly addresses the side effect of speech problems. Similar to the second level of parameters or settings discussed above, the determination of these optional groups of parameters or settings may be based on any number of constraints or desired results for the subject, not solely the immediate symptom or side effect the subject is experiencing. In any determination, the subject and the clinician, physician or technician decide what the initial desired result is, and these desired results, goals, or constraints are programmed into the algorithm for the decision making process. In some embodiments, the desired result, goal or constraints may be edited, either by a clinician, physician or technician, or by the subject, in order to allow the algorithm to most accurately analyze the data in light of the optimal treatment for the subject.

Once the algorithm has determined and provided the at least two optional groups of parameters or settings **110**, the subject may then select which of the provided groups **112** use. This allows the user to determine on an as-needed basis which symptoms, side effects, or other constraints are the most relevant and important to the subject. When the subject makes his or her selection **112**, the parameters or settings of the selected group are entered into the subject's therapy device **112**. The parameters or settings are preferably entered automatically via wireless communication, as described herein, between the diagnostic device or computer and the subject's therapy device. As such, the new, selected group of parameters or settings is programmed into the subject's therapy device, and the device then operates according to those new parameters or settings **116** and provides that newly determined course of therapy or treatment to the subject. Thus, the system measures and quantifies the subject's symptoms **94**, automatically determines at least two optional groups of therapy parameters or settings that would address those symptoms or other constraints **110**, the selected group of parameters or settings is automatically and wirelessly programmed into the subject's therapy device **114**, and the therapy device implements the second level of parameters or setting to in fact address the subject's symptoms **116**.

FIG. **13** depicts a flow chart representing a method embodiment of the present invention for tuning the move-

ment disorder diagnostic device with new therapy parameters or settings. This particular embodiment is envisioned to be performed in an automated fashion, by an algorithm(s), with little or no interaction required with a clinician, physician or technician. First, a movement disorder diagnostic device is provided to a subject **90** who has a therapy device, such as a DBS therapy device. The movement disorder diagnostic device is as described herein, but preferably comprises at least one physiological or movement sensor for measuring the subject's external body motion, or some other physiological signal of the subject, where the sensor(s) has a signal related to the subject's motion or other physiological signal. Preferably, the movement disorder diagnostic device is a single unit, though may be multiple units (e.g., separate sensor unit and command/transceiver module), and is adapted to be worn or attached to a portion of the subject's body such that the sensor(s) of the movement disorder diagnostic device are able to measure the movement of that particular portion of the subject's body.

Once the movement disorder diagnostic device has been provided **90** to the subject, and the subject has donned or attached the device, the next step is to instruct the subject **92** to perform at least one movement disorder test(s) while the subject is undergoing therapy from the therapy (e.g., DBS) device, or is under the effects of recently administered therapy therefrom. These instructions may include a list of which test(s) the subject is to perform, directions on how to perform the test, or a combination of both. The instructions to perform a test or tests may be given **92** in person (if in a clinical setting) by a clinician, physician or technician, or more preferably may be provided in an automated or electronic fashion. For example, the instruction to perform the test(s) may be provided automatically to the subject via a video display, or a notification or alert message provided via such a display or perhaps the subject's smart phone, or the like. The instructions are provided **92**, either audibly, visually, or a combination thereof. The subject's performance of the tasks is thus affected by the therapy being provided, or recently provided by the therapy device.

While the subject is performing the at least one movement disorder test(s) as instructed **92**, the step of measuring and quantifying motor symptoms **94** of the subject based at least in part on the signal of the at least one sensor(s). The movement disorder diagnostic device uses the signal of the at least one sensor(s) to provide an objective measurement and quantification of the severity of the subject's motor symptoms while the subject performs the at least one movement disorder test. The measured and quantified motor symptoms may include specific movement disorder symptoms, side effects from medication and/or therapy, or combinations thereof. The trained scoring algorithms of the movement disorder diagnostic device perform various measurements and calculations to provide this objective quantification of the subject's motor symptoms.

Once the subject's motor symptoms have been measured and quantified **94**, data corresponding to these measured and quantified motor symptoms is entered into an algorithm(s) **96** for automated analysis. Typically, the data corresponding to the measured and quantified motor symptoms is an object score, as described herein, but may be represented in numerous ways and means. Preferably, the quantified movement data is entered **96** directly and automatically into the determination algorithm(s) without the need for any manual human intervention, such as keying in the data.

In this particular embodiment, the determination algorithm(s) then analyze the quantified movement data and determine at least two optional groups of therapy parameters

or settings **110**. These optional groups of parameters or settings each preferably correspond to a different or separate mode of therapy or treatment that addresses the subject's needs as determined based on the measured and quantified motor symptom data, as well as other data, goals, or objectives, and each specifically addressing a separate need, goal, desired outcome, or constraint. For example, the system may detect that the subject is experiencing a high degree of rigidity along with speech problems caused by the electrical stimulation. In this particular embodiment, the system would provide two optional groups of parameters or settings **110**, one which more directly addresses the symptom of rigidity, and one that more directly addresses the side effect of speech problems. Similar to the second level of parameters or settings discussed above, the determination of these optional groups of parameters or settings may be based on any number of constraints or desired results for the subject, not solely the immediate symptom or side effect the subject is experiencing. In any determination, the subject and the clinician, physician or technician decide what the initial desired result is, and these desired results, goals, or constraints are programmed into the algorithm for the decision making process. In some embodiments, the desired result, goal or constraints may be edited, either by a clinician, physician or technician, or by the subject, in order to allow the algorithm to most accurately analyze the data in light of the optimal treatment for the subject.

Once the algorithm has determined and provided the at least two optional groups of parameters or settings **110**, the subject may then select more than one of the provided groups **120** use for further therapy. This allows the user to determine on an as-needed basis which combination symptoms, side effects, or other constraints are the most relevant and important to the subject. When the subject makes his or her selection **120**, the parameters or settings of the selected group are then combined into a single set of parameters or algorithms **122** via an algorithm. The algorithm may combine the selected optional groups via many optimization methods, such as described herein, in order to provide the optimal therapy parameters or settings for addressing each of the subject's selected desired results, goals, or selections to the greatest degree possible. Essentially, this embodiment allows the subject to not simply take a binary approach and choose one symptom, side effect or constraint over the others, but rather to make minor concessions in several categories in order to address more needs, but to a lesser degree. Once the subject makes his or her selection of multiple groups, and the algorithm has performed the necessary optimization and combination of the selected parameters or settings into a single group, this new group of selected and combined parameters or settings is then entered into the subject's therapy device **124**. The selected and combined parameters or settings are preferably entered automatically via wireless communication, as described herein, between the diagnostic device or computer and the subject's therapy device. As such, the new, selected group of parameters or settings is programmed into the subject's therapy device, and the device then operates according to those new parameters or settings **126** and provides that newly determined course of therapy or treatment to the subject. Thus, the system measures and quantifies the subject's symptoms **94**, automatically determines at least two optional groups of therapy parameters or settings that would address those symptoms or other constraints **110**, the selected group of parameters or settings are combined and optimized **122** to address multiple needs of the subject, the selected and combined group of parameters or settings is

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automatically and wirelessly programmed into the subject's therapy device **124**, and the therapy device implements the second level of parameters or settings to in fact address the subject's symptoms **126**.

FIG. **14** is a flow chart depicting one embodiment of a process by which the tuning algorithm(s) is used to determine therapy parameters or settings and program those settings into a subject's therapy device. Like the embodiments described above, data corresponding to measured and quantified motor symptoms is entered into a tuning algorithm(s) **130** for automated analysis. Typically, the data corresponding to the measured and quantified motor symptoms is an object score, as described herein, but may be represented in numerous ways and means. Preferably, the quantified movement data is entered directly and automatically into the determination algorithm(s) **130** without the need for any manual human intervention, such as keying in the data.

The determination algorithm(s) then analyze the quantified movement data and determine a second level of therapy parameters or settings **132**. This second level of parameters or settings preferably corresponds to a mode of therapy or treatment that addresses the subject's needs as determined based on the measured and quantified motor symptom data, as well as other data, goals, or objectives. In other words, if the system determines that the subject is experiencing a very strong tremor, such determination being made as a result of the measurement of the subject's movement and quantifying the severity of the tremor, the algorithm would provide a second level of parameters or settings **132** that would reduce or minimize the tremor the subject is experiencing. As noted, the determination of a second level of parameters or settings may be based on any number of constraints or desired results for the subject, not solely the immediate symptom or side effect the subject is experiencing. Also, the determination may be made to balance multiple desired results, such as if a slightly higher rate of occurrence of symptoms is acceptable to the subject in exchange for a minimization of side effects. Other examples of desired results that may be used to determine the second level of therapy parameters or settings for all embodiments include, but are not limited to, a therapeutic window (in terms of time or some other factor) in which the subject most positively responds to therapy, battery life, and other such constraints that might be considered in terms of optimizing the therapy parameters or settings. In any determination, the subject and the clinician, physician or technician decide what the initial desired result is, and these desired results, goals, or constraints are programmed into the algorithm for the decision making process. In some embodiments, the desired result, goal or constraints may be edited, either by a clinician, physician or technician, or by the subject, in order to allow the algorithm to most accurately analyze the data in light of the optimal treatment for the subject.

In this embodiment, the second level of therapy parameters or settings is then output or displayed for review **134** prior to implementation. This is particularly useful for initial programming of the subject's therapy device, but is also very useful in subsequent or real-time programming as well. The second level (or selected group, or selected and combined group) of parameters or settings is transmitted from the subject's diagnostic device and displayed our output to a clinician, physician or technician **134** who can analyze the algorithm-generated settings to ensure they are within acceptable safety limits, as well as ensuring that they do in fact address the needs of the subject. Preferably, the transmission of the second settings is performed wirelessly and

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securely, particularly where the subject is located remotely from the clinician, physician or technician. The parameters or settings are preferably displayed visually to the clinician, physician or technician, such as on a monitor or display device. The parameters or settings may be displayed in any format known to those skilled in the art, including, but not limited to a textual report such as in an email or text document sent to the care provider, transmitted to a software package that displays the parameters or settings in a particular graphical user interface, or transmitted to a centralized or cloud-based database where the clinician, physician or technician can access them for review. Where the parameters or settings are viewed in a software package, the present invention is intended to be adaptable or formattable such that the parameter or settings report or transmission may be used in any commercially available package, or the like.

When the clinician, physician or technician reviews the parameters or settings, he or she then has the option of approving or denying the parameters or settings **136**. The decision to approve or deny is preferably based on the likelihood that the algorithm(s) provided parameters or settings will achieve the subject's desired outcome or goal, and any other constraints that have been elected either by the subject, or by the clinician, physician or technician in conjunction with the subject. If the clinician, physician or technician determines that the parameters or settings will not meet the desired results, goals or constraints, then the settings or parameters are denied **138**, and the determination algorithm(s) repeats the process of using the measured and quantified motor symptom data to provide another set of therapy parameters or settings **132**. This process may be repeated iteratively until acceptable parameters or settings are achieved. Optionally, the clinician, physician or technician may be given the ability to suggest starting parameters for the algorithm to begin with, thus circumventing the need for further motor symptom measurement and quantification. When the algorithm(s) are triggered to provide another iteration of parameters or settings, the above steps are repeated until the parameters or settings meet or achieve the desired end result for therapy.

When the caregiver determines that the parameters or settings are, in fact, likely to meet those goals, then he or she approves the parameters or settings **140**, and the second level, selected group or combined selected group of parameters or settings is entered automatically and wirelessly into the subject's therapy device **142** in the same or similar manner as described herein. As such, the new, selected group of parameters or settings is programmed into the subject's therapy device, and the device then operates according to those new parameters or settings **144** and provides that newly determined course of therapy or treatment to the subject.

FIG. **15** depicts an operating embodiment of the present invention beginning with the initial occurrence of movement disorder symptoms and ending with continuing monitoring and treatment or therapy. The depicted embodiment describes a system involving continuous home monitoring such that the device operates virtually full-time, monitoring, measuring and analyzing the subject's movement data continually; however, alternative embodiments exist that operate similarly, but with part-time or on-demand monitoring, measuring and analysis. In either embodiment, the subject initially exhibits symptoms of a movement disorder and attends an appointment with a clinician **160**. At the appointment, the clinician examines the subject **162**, and upon analysis, diagnoses the subject as having a movement dis-

order, and subsequently orders a course of treatment or therapy **164**. At a follow-up visit, the subject may report that symptoms fluctuate greatly throughout the day **166**, even while on the prescribed treatment or therapy plan. The clinician then decides to perform in-home continual testing **168** to better determine the severity of the subject's symptoms. Continuous home monitoring **170** begins with the initial programming **172** of any movement monitoring devices or automated treatment delivery devices for the subject, or instructing the subject how to do so. This initial programming may be performed in-clinic or remotely, and may be performed manually or integrated with the review of a clinician, physician or technician, or may be performed intelligently in an automated or semi-automated process utilizing trained tuning algorithms. The device, containing at least one sensor, preferably an accelerometer and/or gyroscope of at least three axes, but optionally another sensor capable of measuring motion, such as an EMG, continually records the subject's movement during activities of daily living **174**. In addition, the device can include two or more types of sensors, preferably accelerometers and gyroscopes. Activities of daily living may include folding laundry, handwriting, eating, dressing, self-care, walking, running, and the like. Optionally, the clinician may order the subject to perform clinical tasks such as finger tapping, nose touching, or the like, as defined by standardized scales such as the UPDRS, TRS, and the like, at regularly scheduled periods. Such movement data would also be continually recorded.

A trained tuning algorithm, preferably incorporated by at least one computer processor, analyzes the recorded movement data **176**. This data analysis may be performed at a scheduled time where the data is uploaded to the algorithm and analyzed, or, more preferably, may be performed in real time. The algorithm and processor function to distinguish voluntary motion of activities of daily living or clinician ordered tasks from movement disorder symptoms and quantify their severity. Preferably the trained algorithms and computer processor are also in two-way communication **178** with a central database **180** or multiple databases made up of previous patient movement data, disorder histories, treatment histories, and the like. Preferably two or more databases are used for reading. Such a database **180** would preferably retain information from the current subject for use with future subjects **182** (maintaining subject privacy and confidentiality at all times) and work with the trained algorithms and processor to determine a recommended treatment for the current subject based on the previous patient data. This database **180** could optionally be used as a real-time gateway for providing updates to the subject's clinician **184** regarding the subject's status.

If the trained algorithms and processor determine **176** that the patient still suffers from movement disorder symptoms **186**, a new course of treatment or therapy is determined **188** either based solely on the subject's measured and quantified symptoms, in conjunction with the central database **180** as previously described, or a combination of the two. If the subject has an automated treatment delivery device, it is preferably reprogrammed according to the new treatment protocol **190**. Preferably, this reprogramming of the subject's device is performed remotely, and in an automated, intelligent fashion requiring little, or more preferably no interaction from a clinician, physician or technician. The movement measuring device then continues to record new movement data and the process repeats. If the trained algorithms and processor determine **176** movement disorder

symptoms no longer persist **192** then no new treatment is needed, no changes are made **194**, and the device continues recording movement data.

FIG. **16** is a flow chart depicting one example of a tuning algorithm used to determine the therapy parameters or settings that can be programmed into a subject's therapy device for another iteration of testing or for providing therapy according to those parameters or settings. This particularly depicted embodiment is merely an example, and many other combinations of parameters or settings, number of variables and/or steps are envisioned. First, a set of initial parameters **200** is programmed into the subject's therapy device. These initial parameters may be pre-programmed, programmed manually or as part of an integrated system, or automatically programmed via an intelligent algorithm. Further, the initial parameters may define any number of variables or parameters or settings as disclosed herein to be set while others are altered or modified through the tuning process. In the depicted embodiment, some initial parameters or settings include a pulse width (PW) of 90 μ s, frequency (F) of 130 Hz, and pulse amplitude (PA) of 0 mA. An amplitude of 0 mA essentially means that no therapeutic current is being provided, or the therapy is turned off; but is provided as an initial setting to being the testing process from which the amplitude is increased gradually throughout the testing iterations. Once these parameters are programmed into the subject's therapy device such that the device can provide therapy according to these initial parameters or settings, the tuning process may begin by initially increasing the pulse amplitude **202** to a first test level provided to the subject, in the figure shown to be an increase of 0.25 mA. As the therapy is provided to the subject according to the initial parameters and the first level of test parameters, the movement disorder diagnostic device (not shown) measures and quantifies the subject's movement (not shown) in order to determine the level of symptoms, side effects, and the like that the subject experiences while receiving therapy according to the provided parameters or settings. Based on the measured and quantified movement data, the depicted algorithm first determines whether side effects are occurring **204**. If side effects occur under the provided therapy parameters or settings, then the algorithm may determine that using the presently tested contact may result in persistent side effects, and move on to the next contact **218** to test and determine whether therapy can be provided in an alternative pattern from the different contacts to avoid the side effects.

If side effects do not occur or do not persist from one iteration of the test to the next, then the algorithm then determines whether the subject's symptoms have improved **206** as a result of the tested therapy parameters or settings. This determination, too, is based on the measured and quantified movement data provided by the movement disorder diagnostic device. If the algorithm determines that the symptoms have not improved based on the provided therapy parameters or settings, it may next perform a query to see how many individual parameters or settings have been tested (i.e., changed individually to determine the particular parameter's effect on the subject's side effects and/or symptoms), or how many groups of parameters or settings have been tested **214**. In the depicted embodiment, the threshold value is eight parameters/settings or groups thereof. Therefore, in the depicted embodiment, if eight parameters/settings or groups thereof have been tested through the depicted process, the algorithm determines whether there has been any improvement in the last four changes in parameters/settings or groups **216**. If there has been no such improve-

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ment, then again, the algorithm may move on to the next contact **218** in order to try and achieve a therapeutic response from another contact that may address the subject's symptoms and/or side effects and other constraints.

If, however, the depicted algorithm determines that a) the subject's symptoms did improve under the provided therapy parameters or settings **206**; b) symptoms did not improve and fewer than eight parameters/settings or groups thereof have been tested **214**; or c) at least eight parameters/settings or group have been tested, but there has been some improvement within the last four iterations **216**, then the algorithm next determines whether the therapy parameters or settings are being provided at a maximum value for one of those parameters or settings **208**—in the depicted case pulse amplitude. Pre-determined parameter or settings limits may be defined to keep the therapy device operating within safe limits in order to protect the subject. If the specific parameter or setting that is being tested (in the depicted case, pulse amplitude) has not reached its maximum value, then the algorithm may again increase the value of that parameter or setting **202** and repeat the process. If the maximum value has already been reached for the particular parameter or setting, then that variable cannot be increased any further, and the algorithm determines whether another individual parameter or setting is at its maximum level **210**—in the depicted case frequency. If the frequency has not reached its maximum value yet, then the algorithm reduces the amplitude to zero, increases the frequency **220** and then again begins the iterative process by initially increasing the amplitude **202** to provide therapy to the subject at the new levels of parameters or settings.

If, however, symptoms have shown improvement as a result in the change in therapy parameters or settings, and both of the first two parameters or settings have reached their maximum levels, then the algorithm next determines if yet another parameter or setting (in the depicted case, pulse width) has yet reached its maximum value **212**. If all of the parameters or settings desired to be tested have reached their maximums, then the algorithm determines that the tested contact, though close, may not provide the best course of therapy, and moves on to the next contact **218** to begin the process over. If, however, the next parameter or setting **212** has not reached its maximum value, the algorithm again reduces the amplitude to zero, maintains the maximum frequency, and increases the pulse width **222** (again, these are exemplary parameters or settings, and other combinations are envisioned). Once the updated parameters or settings have been determined, the algorithm again begins the process over by slightly increasing the amplitude **202** to provide a therapeutic current according to the new settings, and repeats the testing process. This decision process is repeated by the intelligent algorithm until it determines that the best combination of contact(s) and parameters or settings has been achieved, resulting in an optimized therapy that takes into account the subject's side effects, symptoms, and/or other constraints.

It will be apparent to those skilled in the art that various modifications and variations can be made to the present invention without departing from the spirit and scope of the invention. Thus, it is intended that the present invention cover the modifications and variations of this invention provided they come within the scope of the appended claims and their equivalents.

What is claimed:

1. A method of optimizing movement disorder therapy comprising steps of:

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providing or accessing a database at a remote location with a processor in communication with the database, the database comprising reference data of the current subject and/or of other subjects, the reference data comprising:

- a) movement data collected with a system or device adapted to objectively detect and quantify severity data of at least one movement disorder symptom from external body motion data, and
- b) one or more of:
 - i) subject diary data related to medications, medication schedules, symptom occurrence, and/or symptom severity,
 - ii) clinician data including at least measurements and/or observations related to symptoms, symptom quantification scores, and/or treatment or therapy prescriptions, and
 - iii) initial and adjusted treatment or therapy parameter data;

objectively detecting the external body motion data with at least one sensor of the system or device used to quantify the severity data of the at least one symptom from the external body motion data of the current subject;

quantifying the severity data of the at least one movement disorder symptom of the current subject over a time period with a processor based at least in part on the external body movement data;

transmitting the severity data of the at least one movement disorder symptom over the time period a transceiver or transmitter to the database at the remote location; and outputting, with the processor or an additional processor, through an interface, an adjusted set of therapy or treatment parameters based in part on the severity data and reference data to apply to the subject.

2. The method of claim 1, wherein the system or device adapted to objectively detect the external body motion data is a wrist-worn device comprising one or more accelerometers and/or one or more gyroscopes.

3. The method of claim 2, wherein the at least one symptom is selected from the group consisting of tremor, dyskinesia, bradykinesia, rigidity, and gait or balance disturbances.

4. The method of claim 3, wherein the processor adapted to quantify the severity data and the transceiver or transmitter are comprised in the wrist-worn device and the processor comprises a first software stored on a non-transitory computer readable medium and executable by the processor, the first software when executed by the processor is adapted to quantify the severity data of the at least one movement disorder symptom of the current subject.

5. The method of claim 3, further comprising a smartphone adapted to receive the movement data from the wrist-worn device, wherein the smartphone comprises the processor adapted to quantify the severity data, and the processor comprises a first software stored on a non-transitory computer readable medium and executable by the processor, the first software when executed by the processor is adapted to quantify the severity data of the at least one movement disorder symptom of the current subject.

6. The method of claim 5, wherein the smartphone comprises the transceiver or transmitter and is adapted to transmit the movement data and/or the quantified severity data via one-way or two-way communication with the database, at least one remote device, and/or another database, the

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remote device or database(s) adapted to provide access to data of the system to a remote physician, clinician, technician, or other caregiver.

7. The method of claim 3, wherein the transceiver or transmitter are comprised in the wrist-worn device and the processor and/or the additional processor is comprised in the remote device or database and the wrist-worn device is adapted to transmit the movement data to the remote database and processor.

8. The method of claim 3, wherein the system is further adapted to determine subject activity and/or sleep data based on the movement data.

9. The method of claim 8, wherein the system is adapted to optimize therapy parameters based on patterns determined from the movement data, the diary data, and the activity and/or sleep data to increase efficacy of the therapy.

10. The method of claim 9, wherein the therapy or treatment parameters apply to one or more therapy modalities including deep brains stimulation (DBS) systems, drug delivery systems, and drug or medication regimens.

11. A system for optimizing movement disorder therapy comprising:

a database at a remote location comprising reference data of the current subject and/or of other subjects, the reference data comprising:

a) movement data collected with a system or device adapted to objectively detect and quantify severity data of at least one movement disorder symptom from external body motion data, and

b) one or more of:

i) subject diary data related to medications, medication schedules, symptom occurrence, and/or symptom severity,

ii) clinician data including at least measurements and/or observations related to symptoms, symptom quantification scores, and/or treatment or therapy prescriptions, and

iii) initial and adjusted treatment or therapy parameter data;

the system or device adapted to objectively detect the external body motion data with a sensor used to quantify the severity data of the at least one symptom from the external body motion data of the current subject;

a processor adapted to quantify the severity data of the at least one movement disorder symptom of the current subject over a time period based at least in part on the external body movement data;

a transceiver or transmitter for transmitting the severity data of the at least one movement disorder symptom over the time period to the database at the remote location; and

the processor further adapted to, or an additional processor adapted to, output through an interface an adjusted

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set of therapy or treatment parameters based in part on the severity data and reference data to apply to the subject.

12. The system of claim 11, wherein the system or device adapted to objectively detect the external body motion data is a wrist-worn device comprising one or more accelerometers and/or one or more gyroscopes.

13. The system of claim 12, wherein the at least one symptom is selected from the group consisting of tremor, dyskinesia, bradykinesia, rigidity, and gait or balance disturbances.

14. The system of claim 13, wherein the processor adapted to quantify the severity data and the transceiver or transmitter are comprised in the wrist-worn device and comprises a first software stored on a non-transitory computer readable medium and executable by the processor, the first software when executed by the processor is adapted to quantify the severity data of the at least one movement disorder symptom of the current subject.

15. The system of claim 13, further comprising a smartphone adapted to receive the movement data from the wrist-worn device, wherein the smartphone comprises the processor adapted to quantify the severity data, and the processor comprises a first software stored on a non-transitory computer readable medium and executable by the processor, the first software when executed by the processor is adapted to quantify the severity data of the at least one movement disorder symptom of the current subject.

16. The system of claim 15, wherein the smartphone comprises the transceiver or transmitter and is adapted to transmit the movement data and/or the quantified severity data via one-way or two-way communication with the database, at least one remote device, and/or another database, the remote device or database(s) adapted to provide access to data of the system to a remote physician, clinician, technician, or other caregiver.

17. The system of claim 13, wherein the transceiver or transmitter are comprised in the wrist-worn device and the processor and/or the additional processor is comprised in the remote device or database and the wrist-worn device is adapted to transmit the movement data to the remote database and processor.

18. The system of claim 13, wherein the system is further adapted to determine subject activity and/or sleep data based on the movement data.

19. The system of claim 18, wherein the system is adapted to optimize therapy parameters based on patterns determined from the movement data, the diary data, and the activity and/or sleep data to increase efficacy of the therapy.

20. The system of claim 19, wherein the therapy or treatment parameters apply to one or more therapy modalities including deep brains stimulation (DBS) systems, drug delivery systems, and drug or medication regimens.

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